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Master Thesis

Prediction of Earth Orientation Parameters using Machine Learning

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Declaration of Authorship

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Abstract

The complexities and non-linearities embedded in the Earth Orientation Parameter (EOP) - inclusive of polar motion, precision, nutation, and UT1 - have always made their accurate prediction a challenging endeavor in space engineering. EOPs hold a paramount role in representing the Earth's rotational variations against space-fixed reference frames. Such rotational shifts are driven by the complex interaction of the solid Earth, oceans, and atmospheric components, resulting in transfers of angular momentum. Given this backdrop, this Master's thesis proposes a novel approach to predict EOPs, harnessing the power of Machine Learning (ML), specifically Long Short Term Memory (LSTM) networks.

Initially, a comprehensive evaluation was carried out on a range of ML models such as Support Vector Regression (SVR), Random Forests (RF), and Multi Layer Perceptron (MLP), gauging their performance in forecasting the five crucial EOP components over multiple time horizons - 1, 5, 10, 20, 40, and 90 days. Following these preliminary analyses, the LSTM model was introduced into the research, renowned for its competency in handling temporal data, recognising intricate time-based patterns and preserving information over extensive sequences. The LSTM model was trained and tested using the extensive IERS EOP 14 C04 (IAU2000A) time-series dataset. The results revealed its superior performance over the initial models, demonstrating its exceptional capability in understanding and predicting the complex, non-linear behaviour of EOPs. The scope of this research encompasses the definition of various use-cases and constraints, bearing in mind mission types, satellite design requirements, and types of EOP performance simulations.

The validation procedure was undertaken in line with the European Space Agency (ESA) accuracy standards and utilised CONT14 data supplied by International Earth Rotation and Reference Systems Service (IERS). The success of the validation process confirms the proficiency of the LSTM model in generating precise simulation results, even at the nascent design stage. Moreover, the model furnishes accurate pointing performance outcomes such as position knowledge, attitude knowledge, pointing accuracy, and pointing stability, adhering to ESA's stringent requirements.

In conclusion, this Master's thesis signifies a contribution to the intersection of ML and space engineering, demonstrating the LSTM's effectiveness in accurately predicting EOPs. It lays a robust foundation for wider application and exploration of LSTM networks in intricate space engineering tasks, thereby setting the stage for advanced EOP prediction systems and fostering innovative research in this domain.

Acknowledgement

It has been a lifelong dream of mine to contribute significantly to the space industry, a dream that has come closer to realization with the completion of my Master of Science in Space Engineering. This journey towards fulfilling my childhood ambition would not have been possible without the invaluable support and guidance from many esteemed individuals.

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Contents

List of Figures **xiii**

List of Tables **xvii**

Acronyms **xix**

1 Introduction **1**
1.1 Background and Motivation 1
1.2 Objectives of the Thesis 3
1.3 State of the Art 4
1.4 Structure of the thesis 5

2 Geodesy Fundamentals and Related Work **7**
2.1 Motion of Earth 7
2.2 International Celestial Reference System (ICRS) 8
2.3 International Terrestrial Reference System (ITRS) 9
2.4 Earth Orientation Parameter (EOP) 9
2.4.1 Polar motion 10
2.4.2 Celestial Pole Offsets (CPO) 11
2.4.3 Universal Time 13
2.4.4 Length Of Day (LOD) 14

3 Machine Learning Fundamentals **15**
3.1 Long Short Term Memory (LSTM) 15
3.1.1 Core Components of LSTMs 16
3.2 Autoregressive (AR) Modeling 21
3.2.1 Autoregressive Integrated Moving Average (ARIMA) Modeling . 21
3.2.2 Seasonal ARIMA(SARIMA) Modeling 23
3.3 Random Forests (RF) Modeling 24
3.4 Support Vector Regression(SVR) Modelling 25
3.5 Multilayer Perceptron (MLP) Modelling 26
3.6 Evaluation Metrics and ESA Standards 27
3.6.1 Evaluation matrices 27
3.6.2 ESA Standards 28

4 Data **29**
4.1 Data Collection and Preparation 29
4.1.1 Dataset Overview 29

4.1.2 Dataset Integrity and Format . . . 30
4.1.3 Data Access and Utility . . . 30
4.1.4 CONT14 . . . 31
4.2 Data Preprocessing . . . 31
4.2.1 Initial Data Cleaning and Transformation . . . 31
4.2.2 Advanced Preprocessing with Singular Spectrum Analysis . . . 31
4.2.2.1 Embedding Step . . . 31
4.2.2.2 Decomposition Step . . . 32
4.2.2.3 Grouping Step . . . 33
4.2.2.4 Reconstruction Step . . . 34

5 Methodology **37**
5.1 Autocorrelation Analysis . . . 37
5.2 Model Selection and Evaluation Standards . . . 39
5.3 Model Comparative Analysis . . . 39
5.4 Approach for predicting EOPs . . . 41
5.4.1 LSTM Prediction . . . 41
5.4.2 SARIMA Time Series Forecasting . . . 41
5.5 Integration and Comparative Analysis . . . 41
5.5.1 SSA-Improved Long Short-Term Memory . . . 41
5.5.2 SSA-Improved SARIMA Model . . . 42
5.5.3 Evaluation of Performance . . . 42
5.5.4 Context and Applicability of the Model . . . 42
5.5.5 Methodological Adaptation . . . 42
5.6 Approach for Predicting UT1-UTC Using LOD and leap second . . . 42
5.6.1 Data Preparation and Transformation . . . 42
5.6.2 Calculation of UT1 . . . 43
5.6.3 Long Short-Term Memory (LSTM) Forecasting Model . . . 44
5.6.4 Adjustment of Prediction after Modeling . . . 44
5.6.5 Alternative Approach: Direct LOD Prediction for UT1-UTC Calculation . . . 44
5.6.5.1 LOD Prediction Modeling . . . 44
5.6.5.2 Dynamic Time Correction Calculation . . . 45
5.6.5.3 Final Conversion to UT1-UTC . . . 45
5.6.6 Evaluation and Verification of Model Performance . . . 45
5.7 Correlation Analysis Between Earth Orientation Parameters . . . 45
5.7.1 Correlation Analysis Results . . . 45
5.7.2 Discussion . . . 46

6 Results and Discussion **49**
6.1 Overview of Results . . . 49
6.2 Prediction of x_p . . . 50
6.2.1 1 Day . . . 50
6.2.2 5 Day . . . 52
6.2.3 10 Day . . . 54
6.2.4 20 Day . . . 56
6.2.5 40 Day . . . 58
6.2.6 90 Day . . . 59

6.3 Prediction of y_p . . . 62
6.3.1 1 Day . . . 62
6.3.2 5 Day . . . 63
6.3.3 10 Day . . . 64
6.3.4 20 Day . . . 66
6.3.5 40 Day . . . 67
6.3.6 90 Day . . . 68
6.4 UT1-UTC prediction . . . 69
6.4.1 1 Day . . . 69
6.4.2 5 Day . . . 70
6.4.3 10 Day . . . 70
6.4.4 20 Day . . . 71
6.4.5 40 Day . . . 71
6.4.6 90 Day . . . 72
6.5 CONT14 . . . 72
6.5.1 Comparison with ESA Benchmarks . . . 75
6.6 Summary of results . . . 78

7 Conclusion **81**
7.1 Future work . . . 82

A Appendix **83**
A.1 Python Code . . . 83
A.1.1 File Download Code . . . 83
A.1.2 Data Preparation Code . . . 83
A.1.3 LSTM Model for Time Series Forecasting . . . 83
A.1.4 Prepare Data for UT1-UTC Prediction and Use of Leap Seconds . 83
A.1.5 Converting Predicted UT1 to UT1-UTC and Model Evaluation . 84
A.2 Prediction of dX and dY . . . 85
A.2.1 Overview . . . 85
A.2.2 Forecasting Timeframes . . . 85

Bibliography **89**

List of Figures

1.1 Illustration of Earth Orientation Parameter. [Not to scale] by (Modiri (2021)) 1

1.2 Earth Motion Factors by NASA 2

2.1 International Celestial Reference Frame 3 S/X band by (Charlot et al. (2020)) 9

2.2 x_p by EOP 14 C04 (IAU2000A) from 01-01-1990 to 31-12-2023 11

2.3 y_p by EOP 14 C04 (IAU2000A) from 01-01-1990 to 31-12-2023 11

2.4 dX by EOP 14 C04 (IAU2000A) from 01-01-1990 to 31-12-2023 . . . 12

2.5 dY by EOP 14 C04 (IAU2000A) from 01-01-1990 to 31-12-2023 . . . 12

2.6 UT1-UTC by EOP 14 C04 (IAU2000A) from 01-01-1972 to 31-12-2023 13

2.7 LOD by EOP 14 C04 (IAU2000A) from 01-01-1990 to 31-12-2023 . . 14

3.1 Illustration of a RNN, (Olah (2015)) 15

3.2 Illustration of Forget Gate, (Olah (2015)) 16

3.3 Illustration of Input Gate, (Olah (2015)) 17

3.4 Illustration of Update Gate, (Olah (2015)) 19

3.5 Illustration of Output Gate, (Olah (2015)) 20

5.1 Autocorrelation analysis for x_p 38

5.2 Autocorrelation analysis for UT1-UTC 38

5.3 Initial Model selection process for different ML techniques 40

5.4 UT1-TAI vs UTC-TAI 43

5.5 History of recorded Leap Seconds(s) 43

5.6 UT1 44

5.7 Correlation Matrix for the Earth Orientation Parameters 46

6.1 1 day prediction using AR 50
6.2 1 Day prediction using SARIMA 51
6.3 1 Day prediction using LSTM 51
6.4 Comparison of 1 Day prediction for different models 52
6.5 5 Day Prediction using LSTM 52
6.6 5 Day prediction using SARIMA 53
6.7 Comparison of 5 Day prediction for different models 53
6.8 LSTM 10-day prediction 54
6.9 LSTM with SSA combined 10-day prediction 54
6.10 10 Day prediction using SARIMA 55
6.11 Comparison of 10 Day prediction for different models 55
6.12 20 Day prediction using LSTM 56
6.13 20 Day prediction using LSTM and SSA 56
6.14 20 Day prediction using SARIMA 57
6.15 Comparison of 20 Day prediction for different models 57
6.16 40 Day prediction using LSTM 58
6.17 40 Day prediction using LSTM and SSA 59
6.18 40 Day prediction using SARIMA 59
6.19 Comparison of 40 Day prediction for different models 60
6.20 90 Day prediction using LSTM 60
6.22 Comparison of 90 Day prediction for different models 61
6.21 90 Day prediction using LSTM and SSA 61
6.23 1 Day prediction using AR 62
6.24 1 Day Prediction using SARIMA 62
6.25 1 Day prediction using LSTM 63
6.26 Comparison of 1 Day prediction for different models 63
6.27 5 Day prediction using LSTM 64
6.28 5 Day prediction using SARIMA 64
6.29 10 Day prediction using LSTM 65
6.30 10 Day prediction using LSTM 65
6.31 10 Day prediction using SARIMA 66
6.32 20 Day prediction using LSTM 66

6.33 20 Day prediction using LSTM+SSA 67
6.34 40 Day prediction using LSTM vs LSTM + SSA 68
6.35 90 day prediction using LSTM and LSTM+SSA 68
6.36 Comparison of model for 90 Days 69
6.37 1 Day prediction using LSTM 70
6.38 5 day prediction using LSTM 70
6.39 5 day prediction using LSTM 71
6.40 20 day prediction using LSTM 71
6.41 5 day prediction using LSTM 72
6.42 90 day prediction using LSTM 72
6.43 CONT14 Prediction using LSTM for x_p 73
6.44 CONT14 Prediction using LSTM for y_p 73
6.46 CONT14 Prediction using LSTM and LOD for UT1-UTC 74
6.45 CONT14 Prediction using LSTM for UT1-UTC 74
6.47 Model performance vs ESA requirements for x_p 76
6.48 Model performance vs ESA requirements for y_p 77
6.49 Model performance vs ESA requirements for UT1-UTC 77

A.1 10 Day Prediction of dX using LSTM 85
A.2 10 Day Prediction of dY using LSTM 86
A.3 90 Day Prediction of dX using LSTM 86
A.4 90 Day Prediction of dY using LSTM 87

List of Tables

1.1 Earth Orientation Parameters (EOP) 3

5.1 Original vs predicted x_p values for different ML techniques 40

5.2 MSE and RMSE for each method 40

Acronyms

AR Autoregressive.

ARIMA Autoregressive Integrated Moving Average.

CIP Celestial Intermediate Pole.

CNES Centre National D'Etudes Spatiales.

CPO Celestial Polar Offsets.

DORIS Doppler Orbitography and Radiopositioning Integrated by Satellite.

DUT1 Difference of UT1 and UTC.

EOC Earth Orientation Center.

EOP Earth Orientation Parameter.

ESA European Space Agency.

FCN Free Core Nutation.

GCRS Geocentric Celestial Reference System.

GLONASS Global'naya Navigatsionnaya Sputnikovaya Sistema.

GNSS Global Navigation Satellite System.

GPS Global Positioning System.

IAU International Astronomical Union.

ICRS International Celestial Reference System.

IERS International Earth Rotation and Reference Systems Service.

ITRF International Terrestrial Reference Frame.

ITRS International Terrestrial Reference System.

ITT Invitation To Tender.

LLR Lunar Laser Ranging.

LOD Length Of Day.

LSTM Long Short Term Memory.

MAE Mean Absolute Error.

MJD Modified Julian Date.

ML Machine Learning.

MLP Multi Layer Perceptron.

MSE Mean Squared Error.

PACF Partial Autocorrelation Function.

PM Polar Motion.

RBF Radial Basis Function.

RF Random Forests.

RMSE Root Mean Square Error.

RNN Recurring Neural Network.

SARIMA Seasonal Autoregressive Integrated Moving Average.

SLR Satellite Laser Ranging.

SSA Singular Spectrum Analysis.

STLS Structural Total Least Squares.

SVD Singular Value Decomposition.

SVM Support Vector Machine.

SVR Support Vector Regression.

TAI Atomic Time.

UT1 Universal Time 1.

UTC Coordinated Universal Time.

VLBI Very Long Baseline Interferometry.

1. Introduction

1.1 Background and Motivation

The Earth Orientation Parameters (EOP) are a set of five angles: Polar Motion (PM) represented by x_p and y_p, phase of rotation (UT1-UTC), and Celestial Polar Offsets (CPO) denoted as dX and dY. The PM, which has an annual motion and Chandlerian-free wobble with a period of 433 days, is the motion of the Earth's pole relative to the International Terrestrial Reference System (ITRS). The phase of rotation, known as UT1-UTC, represents the difference between the fundamental form of Universal Time 1 (UT1) and Coordinated Universal Time (UTC). (See Figure 1.1)

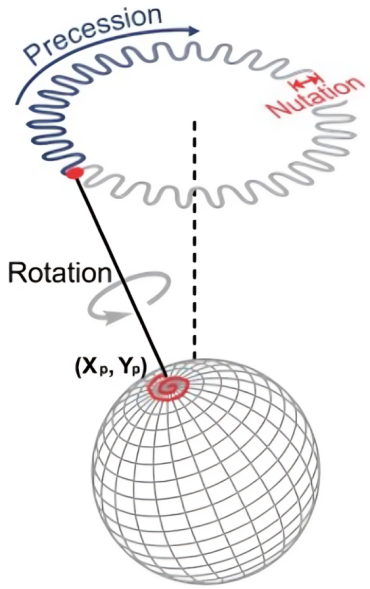


Figure 1.1: Illustration of Earth Orientation Parameter. [Not to scale] by (Modiri (2021))

Figure 1.2 demonstrates how the Earth's rotation and orientation are influenced by both gravitational torques exerted by external masses, such as the moon and the sun, and by geophysical processes like mantle convection in Earth's interior, mass redistribution in the atmosphere, oceans, and hydrosphere, and angular momentum exchange within the Earth-moon system.

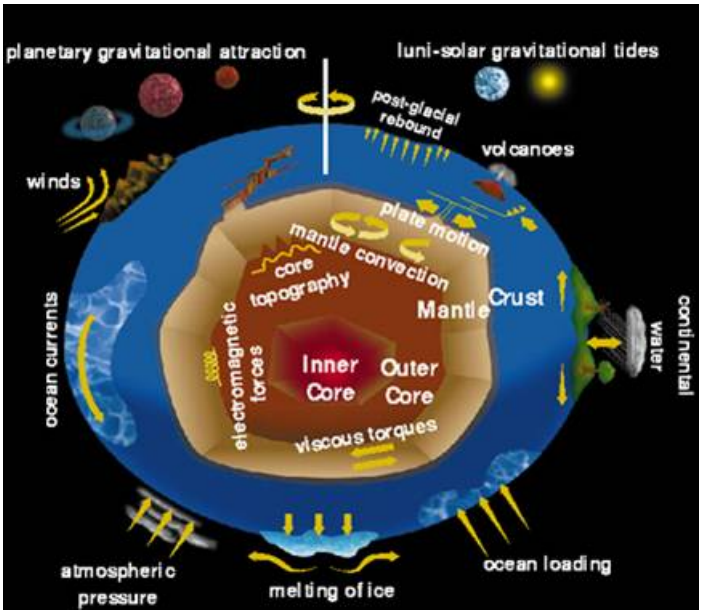


Figure 1.2: Earth Motion Factors by NASA

Traditional geodetic surveillance of these variations hinges on sophisticated techniques. Very Long Baseline Interferometry (VLBI) (Schuh and Böhm (2010)) gauges the Earth's rotation by measuring the time difference between the arrival of radio waves from distant quasars to telescopes located thousands of kilometers apart (Schuh and Behrend (2012)). Global Navigation Satellite System (GNSS), including systems like Global Positioning System (GPS) (Kaplan and Hegarty (2017)), Global'naya Navigatsionnaya Sputnikovaya Sistema (GLONASS), Galileo, and BeiDou, continuously monitor the variable positions of GNSS satellites relative to the Earth to deduce orientation parameters. Doppler Orbitography and Radiopositioning Integrated by Satellite (DORIS) provides satellite orbit data by capturing the frequency shift by the movement of the satellites which is governed by the principle of Doppler's effect which was designed and developed by Centre National D'Etudes Spatiales (Saunier (2023)). Satellite Laser Ranging (SLR) (Combrinck (2010)) involves sending laser pulses to satellites and measuring their round-trip time to determine the satellites' orbits and the Earth's gravitational shape, contributing data to deduce the rotation parameters. These established methods deliver vital geodetic information enabling the definition and maintenance of terrestrial and celestial reference frames.

Abbreviation	Designation	Unit	Narrative Explanation
x_p, y_p	Polar Motion Components	Arcseconds (")	Measures the shift of the Earth's rotation axis relative to its surface, affecting latitude.
UT1-UTC	Universal Time Correction	Seconds (s)	Difference between UT1 (Earth rotation angle) and Coordinated Universal Time (UTC), impacting timekeeping.
LOD	Length of Day	Seconds (s)	Variation in Earth's rotational speed, influencing the exact length of a day.
dX, dY	Celestial Pole Offset	Arcseconds (")	Displacements of the celestial poles from their mean positions, affecting star positions.

Table 1.1: Earth Orientation Parameters (EOP)

EOP comprises as shown in Table 1.1 of two polar motion components x_p and y_p expressing the shifting of the poles on the Earth's surface, UT1, which correlates with the rotation angle of the Earth and is critical for timekeeping, and Figure 1.1 depicts the precession-nutation of the celestial pole (Luzum (2010)), which is essential for aligning terrestrial coordinates with celestial coordinates.

1.2 Objectives of the Thesis

The overarching aim of this thesis is to dissect and harness the predictive power of Machine Learning (ML) for Earth Orientation Parameter (EOP) forecasting (Schuh et al. (2002)), with a particular focus on benchmarking against established geodetic prediction models (Kiani Shahvandi et al. (2023)). This encompasses a multi-faceted technical exploration that advances from data acquisition to model optimization and predictive validation (Luzum (2010)). The specific technical objectives are as follows:

Development of a Comprehensive ML Model
The thesis sets out to architect and fine-tune ML model complex enough to encapsulate the intricacies embedded in historical EOP datasets. This task involves:

- Curating a rich dataset from reputable EOP data repositories, ensuring a representative compilation of variability in Earth's rotation and orientation.
- Preprocessing and feature engineering to structure the data effectively for ML consumption, involving noise reduction, normalization, and trend decomposition.
- Exploring various ML architectures, including but not limited to neural networks, support vector machines, and ensemble methods, to assess their suitability for temporal and non-linear pattern recognition within EOP data.
- Tuning hyperparameters and regularizing the ML models to enhance their predictive accuracy and prevent overfitting to the historical data.

Validation of Model Predictions
To establish the ML model as a reliable tool for EOP forecasting, rigorous validation mechanisms will be employed:

- Implementing cross-validation techniques to evaluate the model's performance on unseen data segments, thereby assuring the model's robustness and repeatability.

- Comparing ML-predicted EOP changes against the outcomes recorded by conventional space-geodetic techniques to quantify improvements in prediction fidelity.
- Utilizing statistical metrics such as Root Mean Square Error (RMSE), Mean Absolute Error (MAE), and correlation coefficients to measure prediction accuracy.

Generalizability and Predictive Analysis under Varying Geophysical Conditions
A central thrust of this research is to interrogate the ML model's extrapolation prowess across a spectrum of geophysical scenarios:

- Assessing model performance across different temporal scales, from daily up to 3 months predictions, to understand the temporal limits of ML applicability in EOP forecasting.
- Testing the model under various historical database feeds to gauge its resilience and adaptability.
- Employing feature importance analysis to identify which aspects of the EOP dataset most heavily influence the model's predictions, guiding future enhancements in data collection and preprocessing.

The pursuit of these technical objectives is expected to yield a ML framework that not only challenges the status quo of EOP prediction but also provides an empirical foundation for the adoption of ML in geodesy and related scientific disciplines. The resulting comparative analysis between machine-learned and traditional methods will underscore the transformative potential that contemporary ML techniques bear on the field of Earth science.

1.3 State of the Art

In the past, the prediction of Earth Orientation Parameters (EOP) has depended on the utilization of VLBI (Schuh and Böhm (2010)), SLR (Combrinck (2010)), GNSS (Schuh and Böhm (2013)), and other space geodetic techniques. These techniques enable accurate measurements of the Earth's rotation and orientation by monitoring quasars and satellites, allowing for direct observations of variations in EOP (Charlot et al. (2020)), (Schuh and Behrend (2012)). The IERS has a crucial role in collecting and distributing these observations, providing a standardized reference for EOP predictions (Luzum (2010)).

Phenomena like the exchange of angular momentum between the atmosphere and oceans, as well as variations in the water cycle, can cause disturbances in Earth's rotational motion. By integrating these geophysical insights into machine learning models, either as characteristics or by modifying the model's structure, it is possible to enhance the accuracy and resilience of predictions. The works cited are (Kiani Shahvandi et al. (2023)) and (Luzum (2010)).The Earth Orientation Center (EOC) of the International Earth Rotation and Reference Systems Service (IERS) publishes daily Earth Orientation Parameters (EOP) values in the C04 series. These values are important for a range of applications, such as astronomy, geodesy, and timekeeping, extending beyond the scope of natural disaster mitigation. Although

various prediction techniques have been developed, accurately predicting short-term Earth Orientation Parameters (EOP) is still a difficulty. This highlights the need for methodological enhancements to meet the needs of the Global Geodetic Observing System (GGOS) (Schuh et al. (2002)), (Kalarus et al. (2010)) and (Dick and Thaller (2020)).

The progression of machine learning (ML) has brought out new approaches in EOP prediction, which provide a challenge to conventional models by utilizing data-driven methodologies capable of capturing intricate, non-linear patterns that are inherent in the EOP time series. Neural networks, specifically LSTM networks, have demonstrated potential in accurately representing time-based sequences with extended dependencies, rendering them appropriate for challenges involving the prediction of EOPs (Dhar et al. (2022)), (Wang et al. (2019)), (Kumar Dubey et al. (2021)), (Huang et al. (2022)), (Sirisha et al. (2022)) and (Khaira et al. (2020)). The capacity of machine learning algorithms to acquire knowledge from past data and apply it to new, unfamiliar data provides a notable benefit in forecasting variations in EOP across different geophysical conditions (Gou et al. (2023)) and (Albeladi et al. (2023)).

A notable advancement in the analytical techniques for EOP prediction is the application of SSA. SSA has been instrumental in decomposing time series into constituent components, facilitating a deeper understanding of the underlying dynamics. This method's integration into EOP forecasting has enriched the predictive models with enhanced analytical depth, enabling more nuanced and accurate predictions of EOP variations (Modiri (2021)).

1.4 Structure of the thesis

This master thesis is organized into eight chapters, each delving into a specific aspect of predicting (EOPs) using (ML) techniques, with an emphasis on LSTM networks integrated with SSA.

Chapter 2 reviews the theoretical underpinnings of the Earth's rotation and orientation parameters, provides an exposition on the dynamics of these parameters, and delves into the current state-of-the-art techniques for monitoring EOPs, including space-geodetic techniques like VLBI. It also surveys the role and evolution of Machine Learning in the context of EOP prediction, setting the stage for subsequent analyses.

Chapter 3 articulates the research methodology in detail, delineating every step from data acquisition from the IAU2000A C04 dataset to preprocessing, ML model selection, parameter tuning, and establishing benchmarks for evaluation. It elucidates the integration of SSA with LSTM for decomposing and reconstructing the time series data for enhanced prediction performance.

Chapter 4 presents the experimental setup, describes the training and testing processes of the chosen ML models, and discusses the results from the initial set of ML methods compared to the enhanced LSTM-SSA combination. It meticulously analyzes predictions across different time horizons and contrasts them against the established ESA accuracy standards.

Chapter 5 offers a critical evaluation of the LSTM model's validation process, using CONT14 data to affirm the model's prediction accuracy. This chapter provides a detailed account of how the model satisfies ESA's requirements for short and long-term predictions for EOP.

Chapter 6 expands on the implications of the findings in the field of space engineering, with a particular focus on the impacts of prediction accuracy on mission planning and satellite design. It discusses the utility and adaptability of the ML models developed within this research and their compliance with ESA's strict standards.

Chapter 7 synthesizes the overall study, reflecting on the encountered challenges, limitations, and the practical applications of the results. It also includes a discussion of the research's contributions to the broader scientific community's understanding of EOP prediction.

Chapter 8 casts a vision for future research directions prompted by this thesis. It outlines potential enhancements for prediction models, suggestions for further integrating ML in space engineering, and possible expansions into new data sources and prediction methodologies.

2. Geodesy Fundamentals and Related Work

2.1 Motion of Earth

The Earth's motion, both in its orbit around the Sun and its rotation on its axis, is a foundational aspect of our planet's dynamics, deeply influencing various terrestrial and celestial phenomena. As the third planet in the solar system, Earth follows an elliptical orbit around the Sun, completing one revolution annually. This revolution, coupled with the Earth's axial tilt of 23.5 degrees relative to its orbital plane, leads to the cyclical nature of seasons.

Simultaneously, the Earth rotates on its axis, an imaginary line connecting the North and South Poles, once every 24 hours. This rotation is responsible for the daily cycle of day and night and contributes to the apparent westward motion of the Sun, Moon, and stars across the sky. The tilt of the Earth's axis also means that the Sun's path, as observed from Earth, appears to traverse a tilted trajectory relative to this axis. This phenomenon is what gives us the varying lengths of day and night throughout the year.

The Earth's orbital path projects onto the celestial sphere as the ecliptic. The points where this path intersects the celestial equator are known as the equinoxes. Conversely, the solstices – the summer solstice around June 21st and the winter solstice around December 21st – mark the extremes of the Sun's apparent path on the celestial sphere.

A critical aspect of Earth's motion is precession, the gradual shift in the direction of the Earth's axis of rotation. Influenced primarily by gravitational forces from the Sun and the Moon, this slow movement alters the orientation of the Earth's axis over a cycle of approximately 26,000 years. This precession movement means that the celestial poles and equators are not fixed in location but gradually shift, affecting the celestial coordinates of astronomical objects. As a result, the reference frame for celestial coordinates, traditionally tied to the position of the vernal equinox. The

commonly used reference frame today is the Equinox J2000.0, which is based on the equinox and mean equator positions as of January 1, 2000 12:00:00.

Understanding and monitoring Earth's rotation and orbital movement is crucial for various scientific and practical applications. It is foundational for timekeeping, navigation, and the establishment of global reference frames. The continuous observation of these motions provides insights into the interactions between the Earth's components, such as the atmosphere, oceans, and solid Earth, and the internal dynamics involving the Earth's core and mantle (Chao et al. (1996)). Today, precise space geodetic techniques like VLBI (Schuh and Behrend (2012)), SLR/ LLR, GNSS, and ring laser gyroscopes are employed to measure these movements with high accuracy, contributing to a comprehensive understanding of Earth's intricate motions within the solar system.

2.2 International Celestial Reference System (ICRS)

The International Celestial Reference System (ICRS) as described in IERS Technical notes by (Petit and Luzum (2010)) is a highly precise and stable celestial coordinate system that serves as a foundation for modern astronomical observations and data. It is based on the positions of distant extragalactic objects, primarily quasars, which are considered to have negligible proper motions, making them ideal for defining a fixed frame of reference. The ICRS is realized through the ICRF, which provides a set of precise coordinates for these objects.

The ICRF's establishment and ongoing refinement are crucial for maintaining the accuracy of the ICRS. The first realization of the ICRF (ICRF1) (Ma et al. (1998)) was constructed in 1995 using Very Long Baseline Interferometry (VLBI) to measure the positions of 212 defining extragalactic radio sources. The second realization, ICRF2 (Fey et al. (2015)) introduced in 2009, marked a significant enhancement in the frame's precision and stability. It expanded the number of sources to over 3,000, significantly reducing the formal uncertainties in position measurements and increasing the frame's overall stability by January 1, 2010.

The International Astronomical Union (IAU) adapted to ICRF3 and has further enhanced its accuracy by containing 4,536 objects at S and X bands as of January 1, 2019 as shown in the Figure 2.1

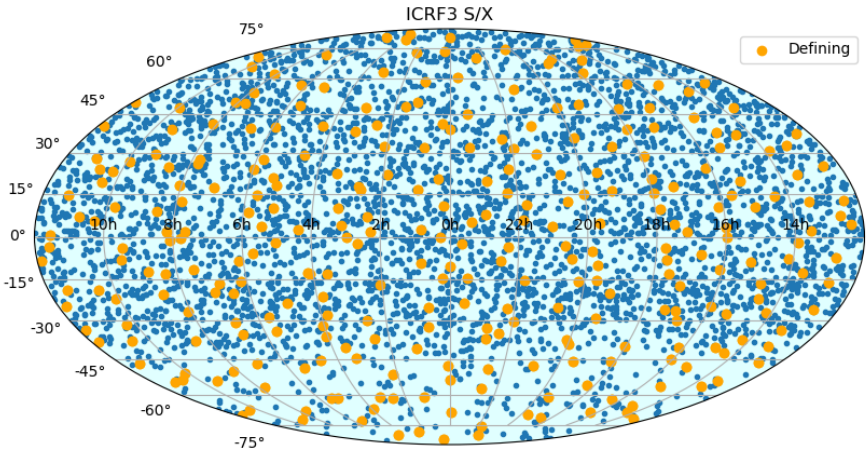


Figure 2.1: International Celestial Reference Frame 3 S/X band by (Charlot et al. (2020))

2.3 International Terrestrial Reference System (ITRS)

According to (Petit and Luzum (2010)) the ITRS comprises a global framework designed to define the positions and motions of objects on and near the Earth's surface with high precision. It is operationalized through the International Terrestrial Reference Frame (ITRF), which combines careful determined three-dimensional coordinates and velocities of points distributed across the Earth's crust. These points are anchored at geodetic observatories equipped with sophisticated space geodesy techniques such as VLBI, SLR, GNSS including GPS, and DORIS. The ITRF's construction entails a sophisticated combination of data from these diverse geodetic techniques, necessitating the establishment of highly precise local ties at co-location sites to integrate the disparate data streams into a singular, cohesive reference frame.

Over the years, the ITRF has undergone numerous revisions, each iteration marked by advancements in data quality, network density, and analytical methodologies. Successive versions, including ITRF88, ITRF2000, ITRF2005, and ITRF2008, have systematically enhanced the frame's accuracy and stability. These enhancements reflect the cumulative improvements in geodetic instrumentation, data processing techniques, and the expanding global network of geodetic observatories. The evolution of the ITRF, coordinated by the IERS, underscores a commitment to maintaining the highest standards of precision and consistency in global terrestrial referencing, pivotal for a myriad of scientific research and practical applications ranging from plate tectonics and Earth dynamics studies to precise navigation and environmental monitoring.

2.4 Earth Orientation Parameter (EOP)

EOP play a crucial role in the precise alignment of terrestrial and celestial reference frames, essential for high-precision geodetic and astronomical observations. These

parameters facilitate the transformation between the ITRS and the Geocentric Celestial Reference System (GCRS), accounting for the dynamic and complex motion of the Earth. The EOP consist of five key components: two polar motion coordinates (x_p, y_p), the difference in Universal Time (UT1-UTC), and celestial pole offsets (dX, dY). These components reflect the Earth's irregular rotation and orientation in space, influenced by various geophysical and astronomical factors.

The polar motion (PM), x_p and y_p, describe the shifting of the Earth's rotation axis relative to its surface, encompassing the Chandler wobble with a period of about 433 days and an annual wobble. The UT1-UTC component, a measure of the Earth's rotation angle, is vital for timekeeping and navigation, correlating with the length of the day and affected by factors such as tidal forces and changes in the Earth's moment of inertia (Capitaine and Wallace (2006)). CPO, dX and dY, represent the corrections to the IAU precession-nutation model, accounting for the small but significant discrepancies between the observed and modeled motion of the celestial pole, influenced by phenomena such as the Free Core Nutation (FCN) (Dehant and Mathews (2015)).

The transformation matrix, incorporating these EOP, is mathematically represented as

$$[\text{GCRS}] = Q(t) \cdot R(t) \cdot W(t) \cdot [\text{ITRS}] \tag{2.1}$$

where Q(t) includes the celestial pole offsets, R(t) represents Earth's rotation, and W(t) describes the terrestrial pole offsets (Schuh and Böhm (2013)). The precise determination of these parameters is predominantly achieved through Very Long Baseline Interferometry (VLBI), along with contributions from other space geodetic techniques like GNSS and SLR. The IERS plays a pivotal role in monitoring and providing updated EOP values, essential for a broad spectrum of scientific and practical applications, including satellite navigation, climate research, and global telecommunications (Petit and Luzum (2010)).

2.4.1 Polar motion

Polar motion, a key component of Earth Orientation Parameters (EOPs), refers to the movement of the Earth's rotational axis relative to its surface, specifically tracked in the ITRS and denoted by x_p and y_p as shown in Figure 2.2 and 2.3 respectively. This motion is crucial for understanding the dynamics of Earth's rotation and for accurately transforming coordinates from the ITRS to the GCRS.

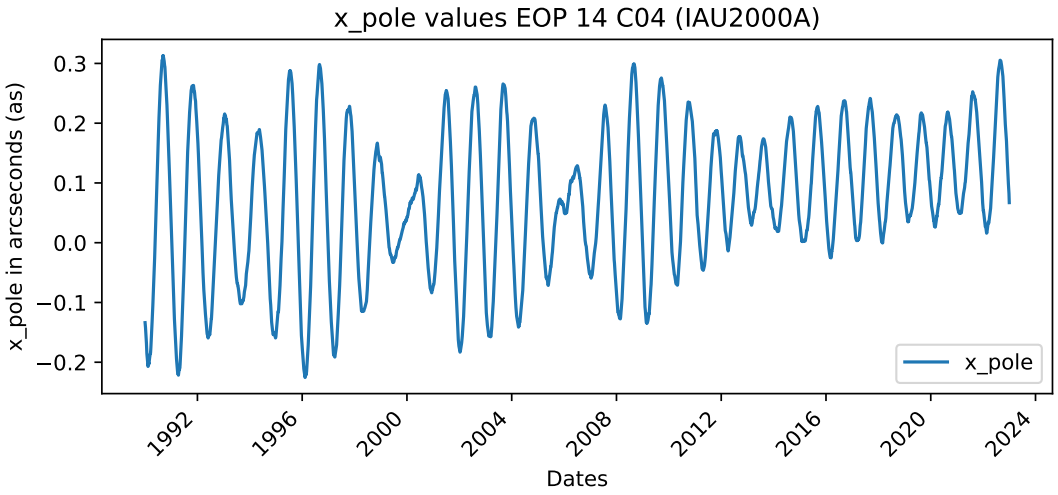


Figure 2.2: x_p by EOP 14 C04 (IAU2000A) from 01-01-1990 to 31-12-2023

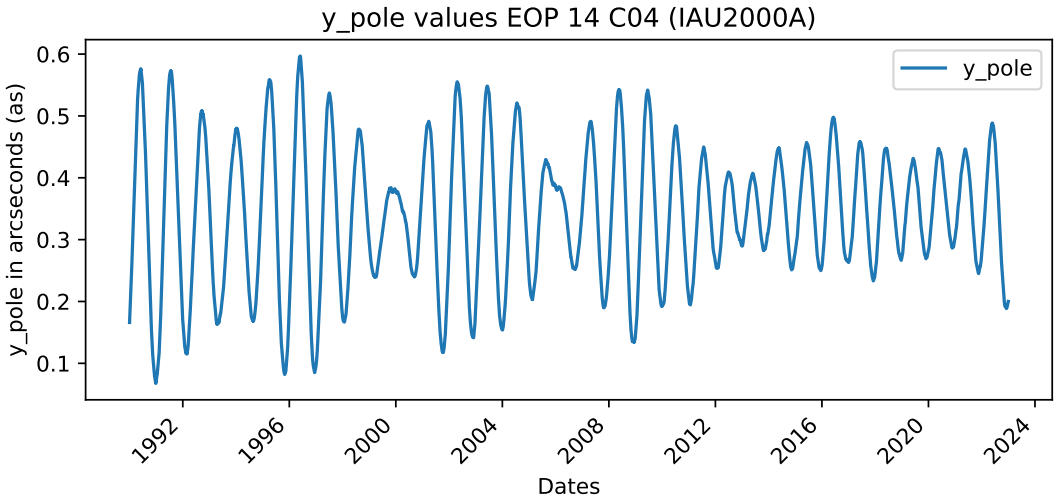


Figure 2.3: y_p by EOP 14 C04 (IAU2000A) from 01-01-1990 to 31-12-2023

Polar motion is mainly characterized by the Chandler Wobble (Bizouard et al. (2011)), with a period of about 433 days (Mathews et al. (2002)), and an annual motion. It is important to note that these movements are almost entirely in the high-frequency domain, with significant components observed below -1.5 and above -0.5 cycles per sidereal day in the ITRS. This characteristic makes polar motion distinct from celestial motions (precession and nutation) that are primarily observed in the celestial frame at low frequencies between -1.5 and -0.5 cpsd in ITRS.

2.4.2 Celestial Pole Offsets (CPO)

CPO are critical components in the area of EOPs, playing a significant role in the accurate transformation of coordinates between the ITRS and the GCRS. These offsets represent the time-dependent corrections needed to align the conventional celestial pole with the CIP.

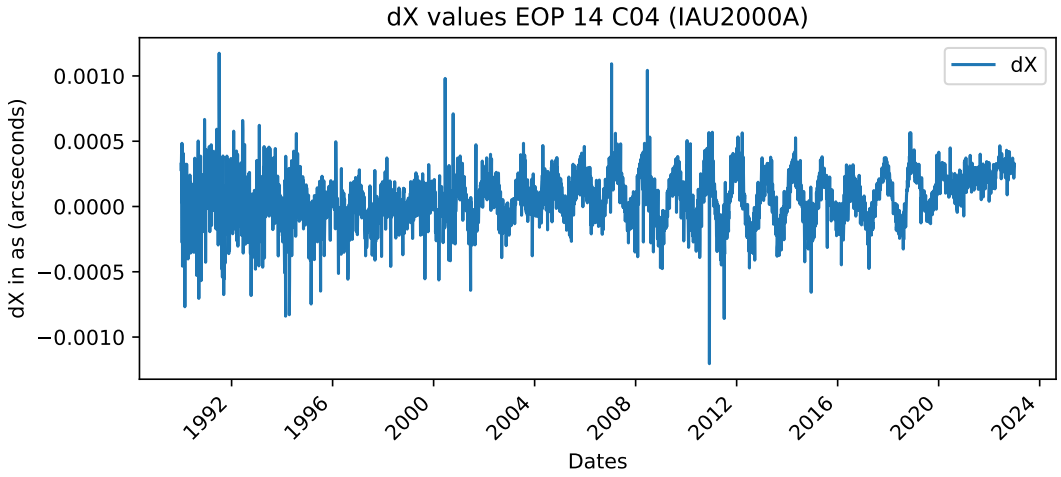


Figure 2.4: dX by EOP 14 C04 (IAU2000A) from 01-01-1990 to 31-12-2023

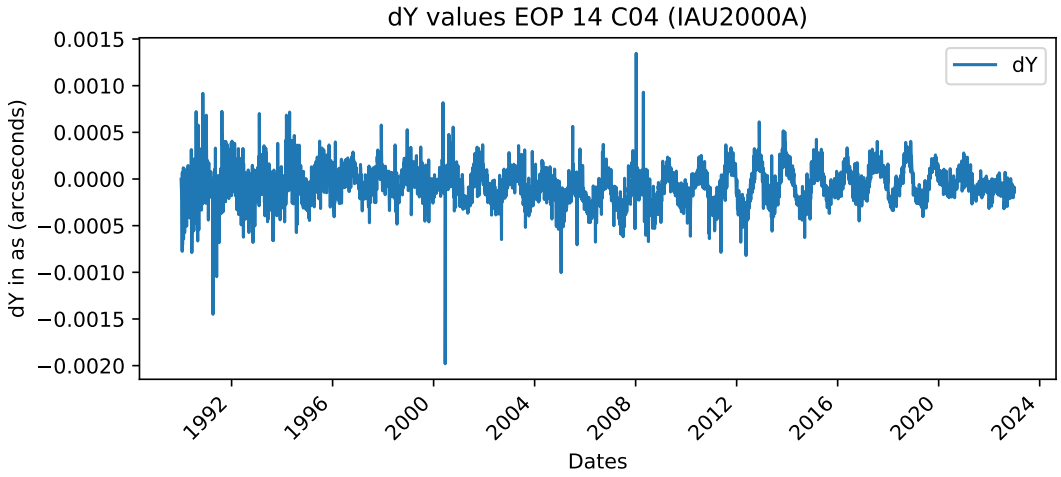


Figure 2.5: dY by EOP 14 C04 (IAU2000A) from 01-01-1990 to 31-12-2023

In the context of EOPs, CPOs are denoted as dX and dY. They are part of the precession/nutation matrix, $Q(t)$, in the transformation equation:

$$[\text{GCRS}] = Q(t) \cdot R(t) \cdot W(t) \cdot [\text{ITRS}]$$

Here, $Q(t)$ specifically accounts for the CPOs, alongside the other matrices $R(t)$ for Earth's rotation and $W(t)$ for polar motion (Schuh and Böhm (2013)).

The conventional celestial pole's position is defined by the IAU precession and nutation models. However, these models do not perfectly represent the actual motion of the CIP due to various geophysical factors. CPOs dX and dY are introduced to correct these discrepancies and are shown in Figure 2.4 and 2.5 respectively. These offsets are essential for accommodating residual errors in the a priori precession/nutation model and phenomena such as the FCN, which is a resonance effect caused by the misalignment of the rotation axes of the Earth's mantle and core.

The FCN manifests as a retrograde motion with a period of about 433 days in the GCRS and can have significant amplitudes, making precise measurement and correction through CPOs crucial for maintaining high positioning accuracy. The IERS provides these offsets, derived from ongoing observations and space geodetic techniques, such as VLBI.

2.4.3 Universal Time

DUT1 (Delta UT1) is a critical component of Earth Orientation Parameters (EOPs) and plays a significant role in precision timekeeping and astronomical observations. It represents the difference between UT1, which is based on the Earth's rotation, and UTC, which is a standardized atomic time scale.

UT1 is a form of Universal Time that corrects for variations in the Earth's rotation which is the mean solar time. The Earth's rotation is not uniform due to various geophysical and astronomical factors, including tidal forces, the movement of the Earth's core, and changes in atmospheric pressure. UT1 reflects these irregularities.

UTC, on the other hand, is maintained through a network of highly precise atomic clocks and is kept within 0.9 seconds of UT1 by the insertion of leap seconds (Nelson et al. (2001)). This adjustment ensures that UTC remains in close synchronization with the mean solar time.

DUT1 is defined as:

$$DUT1 = UT1 - UTC \tag{2.2}$$

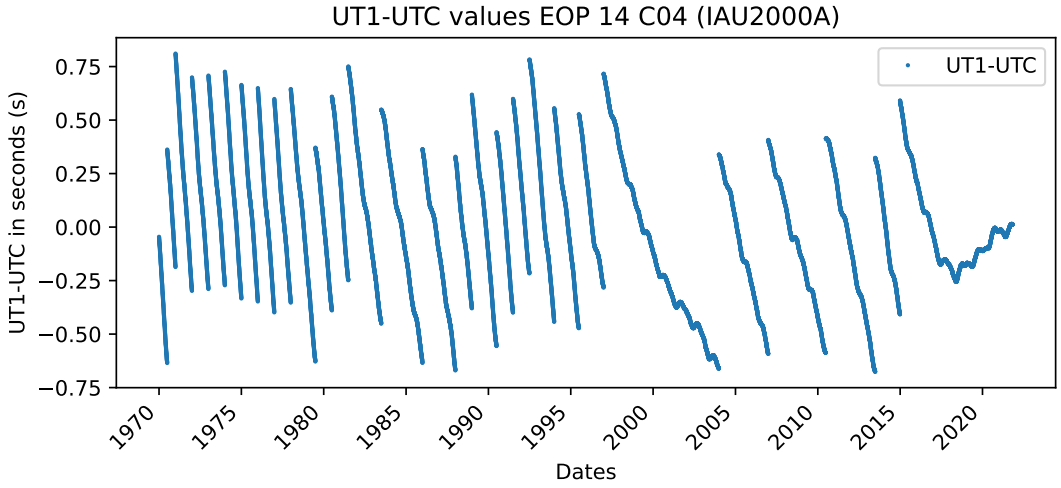


Figure 2.6: UT1-UTC by EOP 14 C04 (IAU2000A) from 01-01-1972 to 31-12-2023

This difference is important for applications that require a high degree of time precision. For example, in satellite navigation systems, small inaccuracies in timekeeping can lead to errors in determining a position. In radio astronomy and VLBI, accurate timekeeping is crucial for precisely locating and observing celestial objects.

DUT1 is also essential for aligning astronomical observations with the Earth's rotation. As the Earth rotates at an irregular rate, DUT1 provides the necessary correction to align the observatories' time with the actual rotational position of the Earth.

2.4.4 Length Of Day (LOD)

The LOD measures the actual time taken for one complete rotation of the Earth, which can vary due to geophysical forces like tidal interactions with the Moon and Sun and changes in atmospheric conditions. These variations in the Earth's rotation speed lead to changes in the length of a day, sometimes deviating from the standard 24 hours or 86,400 seconds.

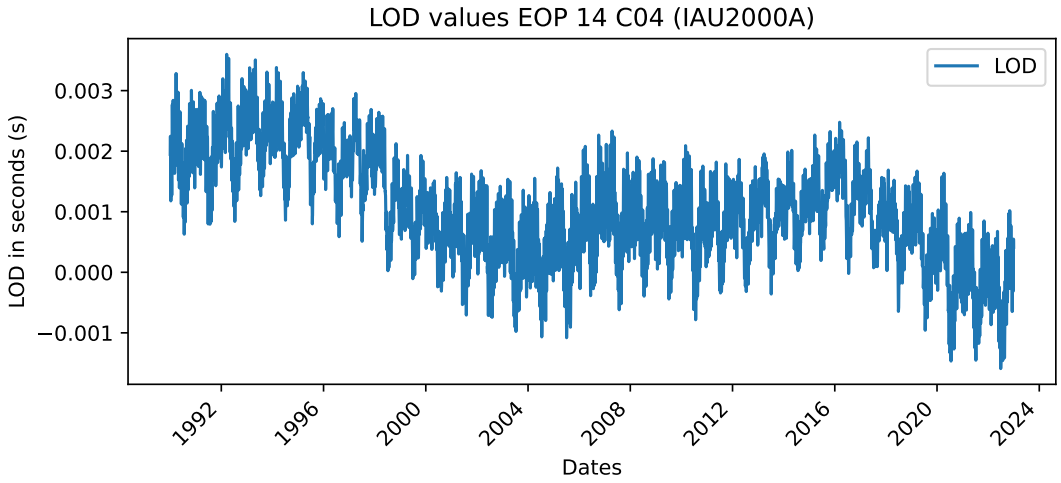


Figure 2.7: LOD by EOP 14 C04 (IAU2000A) from 01-01-1990 to 31-12-2023

The relationship between LOD and DUT1 as described in (Bizouard et al. (2019)). Before 1997, LOD is derived from UT1 values as

$$\mathrm{LOD} = -D\frac{d(\mathrm{UT1} - \mathrm{TAI})}{dt}$$

where D = 86,400 s is the nominal length of day.

This relationship highlights the critical role of DUT1 in aligning UT1, a time scale based on Earth's rotation and mean solar time, with the consistent atomic time standard of UTC. The IERS diligently monitors these parameters, ensuring that global timekeeping systems remain in sync with the natural, albeit irregular, rhythms of our planet's rotation.

3. Machine Learning Fundamentals

3.1 Long Short Term Memory (LSTM)

LSTM is a specialized variant of Recurring Neural Network (RNN) designed to effectively process sequences of data. Originating from the groundbreaking work in 1997 by Hochreiter and Schmidhuber (1997) and subsequently refined by numerous researchers, LSTMs have garnered widespread adoption due to their exceptional performance across a myriad of problems.

One of the critical challenges in sequential data processing is the need to remember information over long periods. Traditional RNNs often falter in this aspect due to a phenomenon known as ”vanishing gradients,” where the network loses the ability to learn connections between events separated by long time intervals.

LSTMs address this challenge with a unique architecture that includes components called ”gates.” These gates control the flow of information, allowing the network to retain or discard data as needed. This mechanism enables LSTMs to ”remember” relevant past information and ”forget” irrelevant data, making them exceptionally adept at understanding long-range dependencies in the data.

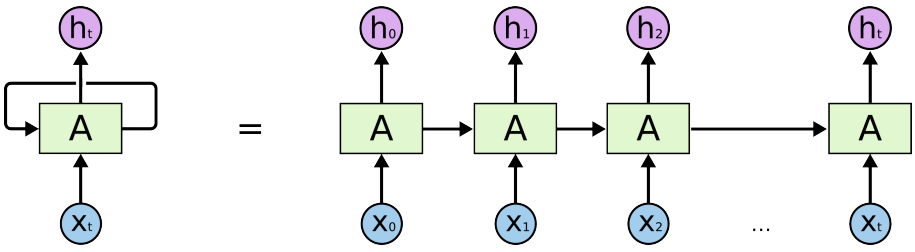


Figure 3.1: Illustration of a RNN, (Olah (2015))

All RNNs share a structure resembling a chain of repeating neural network modules shown in Figure 3.1 that in Recurring Neural Network (RNN), ”A” represents a single unit or cell within the network. Each cell takes an input x_t at a given timestep t

and produces an output h_t, which is also referred to as the hidden state. In Figure 3.1 pink circles are vector operations, yellow boxes are the trained neural network layers while the lines represent the flow of information or vector transfer. The hidden state h_t encapsulates information learned from previous timesteps, allowing the network to maintain a form of memory over sequential data. In the case of standard RNNs, this repeating module typically comprises a single layer, often employing a simple structure like a hyperbolic tangent (tanh) layer(to normalize the values to be between -1 and 1). However, LSTMs exhibit a similar chain-like arrangement but boast a distinct architecture. Within their repeating module, LSTMs incorporate four interacting layers, each playing a specialized role in information processing and retention. Further discussions can be found in (Deb et al. (2017)), and (Lindemann et al. (2021)).

3.1.1 Core Components of LSTMs

Cell State At the heart of LSTMs lies the cell state C_t, acting as a conveyor belt carrying information across different time steps. The cell state retains information over long sequences and allows selective updates through various gates.

Gates in LSTMs LSTMs employ gates to regulate the flow of information into and out of the cell state, consisting of four main gates:

- **Forget gate**: The weights linked to the gate are important for understanding how the forget gate makes its choice. The weights play an essential part when calculating the significance of the inputs (current input x_t and prior hidden state h_{t-1} in the decision-making process of retaining or discarding information as shown in Figure 3.2.

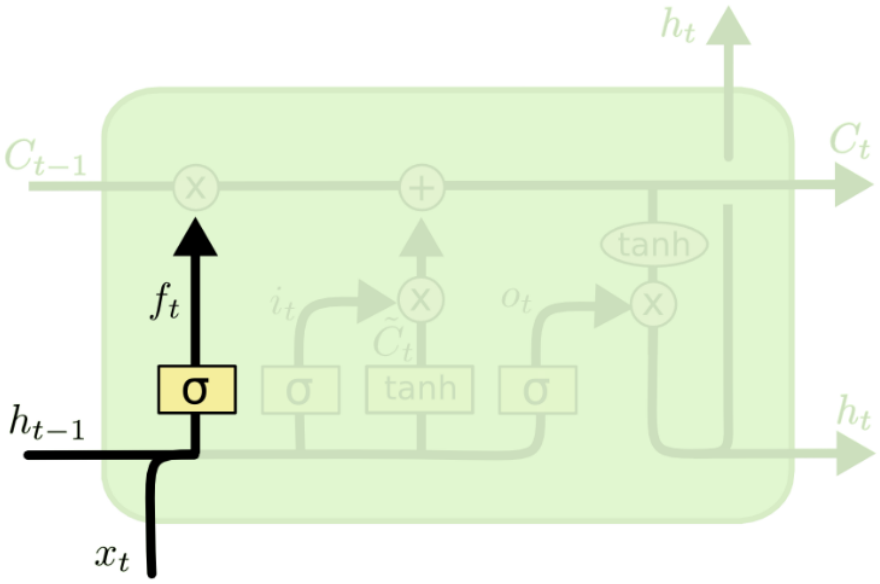


Figure 3.2: Illustration of Forget Gate, (Olah (2015))

The procedure can be defined mathematically as follows:

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f) \tag{3.1}$$

Where:

- f_t is the output of the forget gate.
- σ represents the sigmoid function, ensuring the output is between 0 and 1.
- W_f are the weights associated with the forget gate, applied to the inputs.
- $[h_{t-1}, x_t]$ is the matrix containing information of the previous hidden state and the current input.
- b_f is the bias term for the forget gate.

1. **Decision-Making in the Forget Gate**: The LSTM network adjusts the weights W_f during training using backpropagation to decide which components of the cell state are crucial to keep or discard for the task.

   A high weight assigned to a certain feature in the input or the previous concealed state indicates the significance of that feature in determining whether the corresponding information in the cell state should be retained. As a result, the sigmoid function will produce a value closer to 1 for that feature, indicating to the cell to save that information. If the weight is low or negative, it signifies that the feature is not significant for the current decision or job. This causes the sigmoid function to produce a value closer to 0, leading the cell to disregard that information.
2. **Adaptive Learning**: The weights are not fixed; they are learned and consistently adjusted when the LSTM model encounters additional input throughout the training phase. This enables the forget gate and the LSTM as a whole to adjust to the unique attributes of the data it is being trained on, hence enhancing its efficiency.

- Input gate

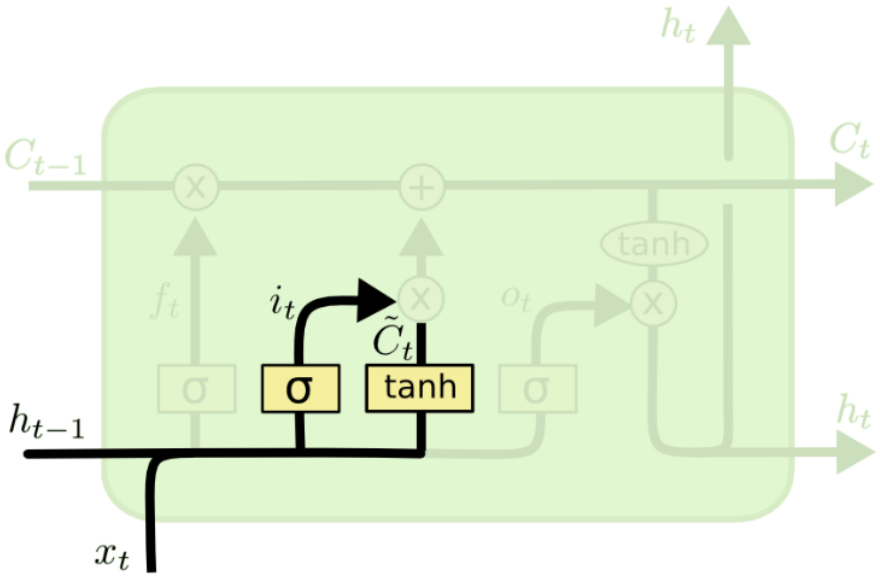


Figure 3.3: Illustration of Input Gate, (Olah (2015))

The input gate as shown in Figure 3.3 of an LSTM is important for integrating new information into the cell state and consists of two main operations:

1. **Value Selection via Sigmoid Function:** The sigmoid layer assesses both the current input x_t and the previous hidden state h_{t-1}, assigning a value between 0 and 1 to each element in the cell state, indicating the need for updating based on the new information's relevance. This process is given by:

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \tag{3.2}$$

   where i_t is the sigmoid layer's output, W_i represents the weights, and b_i is the bias. A value close to 1 implies high relevance for an update, while a value near 0 suggests irrelevance.

2. **Generation of Candidate Values via Tanh Function:** A tanh layer creates a vector of new candidate values $\tilde{C}_t$, normalized between -1 and 1, suitable for updating the cell state:

$$\tilde{C}_t = \tanh(W_C \cdot [h_{t-1}, x_t] + b_C) \tag{3.3}$$

   Here, W_C and b_C denote the weights and bias for this layer, respectively.

The cell state is updated by combining these outputs, where the sigmoid output i_t determines the addition extent of each candidate value $\tilde{C}_t$ to the cell state, post-modification by the forget gate:

$$C_t = f_t \odot C_{t-1} + i_t \odot \tilde{C}_t \tag{3.4}$$

In this equation, f_t is the forget gate's output, C_{t-1} is the previous cell state, and $\odot$ denotes element-wise multiplication.

This process enables the input gate to selectively update the cell state with pertinent new information, preserving the LSTM's capacity for managing long-term dependencies.

- Update of cell state

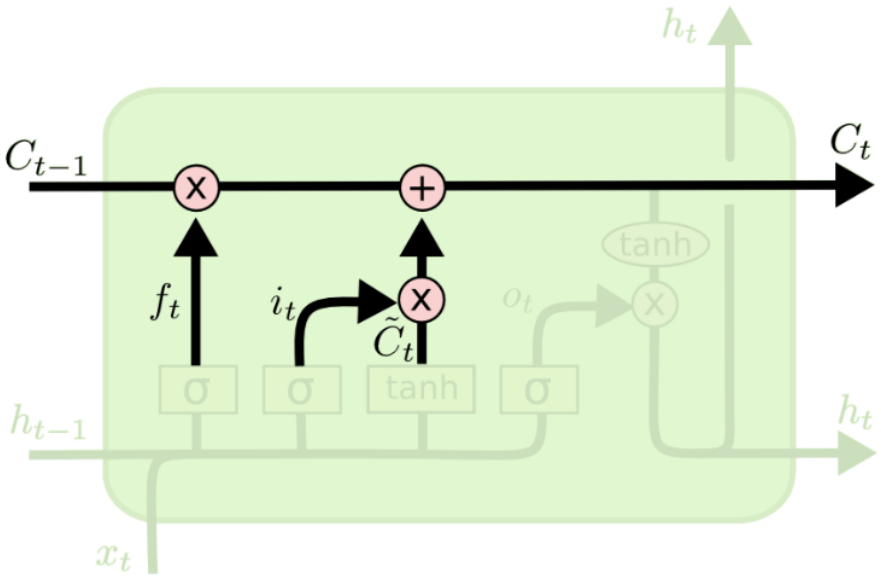


Figure 3.4: Illustration of Update Gate, (Olah (2015))

The cell state update shown in Figure 3.4 is a central aspect of the LSTM's functionality, allowing it to carry relevant information across time steps. This update process involves the following key steps:

1. **Updating the Cell State:** The final update to the cell state C_t combines the previous cell state's retained information (as decided by the forget gate) with the new information (as decided by the input gate):

$$C_t = f_t \odot C_{t-1} + i_t \odot \tilde{C}_t \tag{3.5}$$

This update mechanism ensures that the LSTM cell state is a dynamic representation of the information deemed relevant up to the current time step, enabling the LSTM to effectively manage long-term dependencies in sequential data.

- Output gate

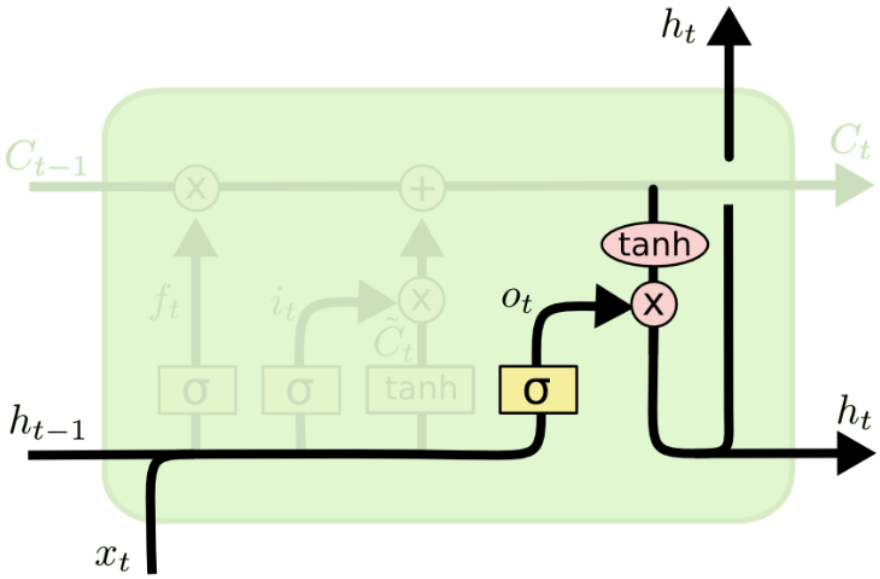


Figure 3.5: Illustration of Output Gate, (Olah (2015))

The output gate in Figure 3.5 in an LSTM network determines the final output at each time step by performing the following steps:

1. **Output Decision via Sigmoid Layer:** A sigmoid layer evaluates the current input x_t and the previous hidden state h_{t-1} to decide which parts of the cell state C_t are significant for the output. This decision is quantified as values between 0 and 1, represented by:

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o) \tag{3.6}$$

   where o_t is the output of the sigmoid layer, with W_o and b_o being the associated weights and bias.
2. **Cell State Filtering:** The updated cell state C_t is normalized through a tanh function, preparing it for the output by ensuring values are between -1 and 1:

$$\tilde{C}_t = \tanh(C_t) \tag{3.7}$$

3. **Final Output Generation:** The final output h_t at time step t is produced by element-wise multiplication of the sigmoid layer's output o_t with the normalized cell state $\tilde{C}_t$:

$$h_t = o_t \odot \tilde{C}_t \tag{3.8}$$

This process ensures that the LSTM selectively passes only the relevant information through the output gate, making the final output h_t a filtered version of the cell state. The output h_t serves as the hidden state for the next time step and can be utilized by the neural network for further processing or predictions, underpinning the LSTM's ability to manage long-term dependencies and make informed decisions based on sequential inputs.

3.2 Autoregressive (AR) Modeling

Autoregressive (AR) models are a fundamental class of models used in time series analysis, particularly effective in scenarios where successive values in the series are correlated. These models are especially relevant when predicting time-dependent phenomena such as EOPs. An AR model predicts future behavior based on past behavior. It assumes that past values have a linear influence on future values. The model is essentially a regression of the time series onto itself at previous times.

The general form of an AR model of order p, denoted as AR(p) is:

$$Y_t = c + \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \ldots + \phi_p Y_{t-p} + \varepsilon_t \tag{3.9}$$

Here, Y_t is the value of the series at time t, and $\phi_1, \phi_2, \ldots, \phi_p$ are the coefficients that measure the influence of each lagged value. c is a constant (intercept) term, and ε_t is white noise error. p indicates the number of lagged observations included in the model.

Dynamics of AR models: In an AR model, each term can be thought of as a weighted version of past observations. The weights (ϕ coefficients) determine how much influence each past observation has on the forecast. The order p of the model defines how far back in the past the model will go to predict the current value. High ϕ values indicate that the time series is influenced significantly by its past values.

Application in EOP forecasting: AR models are particularly useful for EOP forecasting if the data shows that past orientation parameters significantly influence future parameters. For instance, if EOP data demonstrates autocorrelation over a specific number of lags, an AR model of that order could effectively capture this dependence.

Selecting the order of AR Model: The order p is typically determined through analysis methods like the Partial Autocorrelation Function (PACF), which helps identify the extent of lagged influence. Overfitting is a risk with higher-order AR models, so it's crucial to choose p such that it sufficiently captures the autocorrelation without being overly complex.

Limitations: AR models assume linearity and may not perform well with non-linear data. They may not adequately handle time series data with trend and seasonal components unless appropriately modified or differenced. AR models can be sensitive to outliers, which can significantly influence the coefficient estimates.

3.2.1 Autoregressive Integrated Moving Average (ARIMA) Modeling

ARIMA models, standing for Autoregressive Integrated Moving Average, play an important role in forecasting EOPs, which are essentially time series data. These models uniquely blend the elements of autoregression, integration, and moving averages to analyze and predict time-dependent data, making them particularly suitable for EOPs.

ARIMA in time series analysis: The concept of stationarity is central to ARIMA's approach, which is crucial for accurate time series forecasting. A time series is

stationary if its properties like mean and variance do not change over time. ARIMA achieves this through integration, with differencing being used to stabilize the mean of the time series.

The ARIMA(p,d,q) Forecasting Equation: The ARIMA model is defined by its parameters: p (the number of autoregressive terms), d (the number of nonseasonal differences), and q (the number of moving average terms). The forecasting equation is a linear function that involves these parameters:

$$\begin{aligned}\text{Predicted } Y = \text{constant } &+ \sum(\text{weighted sum of recent values of } Y) \\ &+ \sum(\text{weighted sum of recent forecast errors})\end{aligned} \tag{3.10}$$

Various configurations of these parameters lead to different ARIMA models. For example:

- ARIMA(1,0,0): First-Order Autoregressive Model The ARIMA(1,0,0) model, classified as a first-order autoregressive model, is one of the simplest yet effective configurations in the ARIMA family, particularly useful for time series data like Earth Orientation Parameters (EOPs). The model is expressed through the equation:

  $$\hat{Y}_t = \mu + \phi_1 Y_{t-1} \tag{3.11}$$

  In this equation, $\hat{Y}_t$ represents the forecasted value of the series at time t, μ is a constant term, representing the mean level of the series, ϕ_1 is the coefficient of the first lag of Y, essentially quantifying the influence of the immediate past value of the series on its current value, and Y_{t-1} is the observed value at time $t-1$, the direct predecessor of the current period.

  Model Dynamics: The ARIMA(1,0,0) as suggested by (Khaira et al. (2020)) model posits that the current value of the time series can be primarily explained by its immediate past value. The coefficient ϕ_1 plays a crucial role. If ϕ_1 is close to 1, it suggests that the series has a strong momentum and the past value has a significant influence on the current value. If ϕ_1 is close to 0, it implies that the past value has little to no effect on the current value, reducing the model to essentially $\hat{Y}_t = \mu$. The constant μ adjusts the level around which the series oscillates. If the mean of the series is zero, then μ might not be included in the model.

  Applicability in EOPs: When applied to EOPs, this model is especially useful if the data shows a pattern where each value tends to be close to its preceding value, with slight variations. For instance, if EOP data demonstrates gradual changes rather than abrupt shifts, the ARIMA(1,0,0) model can effectively capture this behavior, assuming that each day's orientation parameter is a slight modification of the previous day's parameter.

  Limitations: While ARIMA(1,0,0) is powerful in capturing linear relationships in time series data, it may not be as effective in scenarios where the data exhibits non-linear patterns, significant seasonal effects, or trends. The model's simplicity can be a drawback if the EOP data has complex underlying structures that a single lag and a constant term cannot adequately capture.

- ARIMA(0,1,0): The ARIMA(0,1,0) model is described by the equation:

$$\hat{Y}_t = Y_{t-1} + \mu \tag{3.12}$$

In this model, $\hat{Y}_t$ is the forecasted value at time t. μ represents the mean change from one observation to the next. It's the constant term that indicates the average period-to-period change in the series.

Model Dynamics: This model implies that the best predictor for the next value in the series is the most recent observation, with an adjustment by the constant mean change. The absence of autoregressive and moving average components means this model does not account for patterns or trends beyond a simple shift by the mean change. A zero value of μ would suggest a pure random walk without drift, where the best forecast for tomorrow is simply today's value.

Applicability to EOPs: ARIMA(0,1,0) is suitable for EOP data that shows a stable trend over time with minimal fluctuations. If EOPs exhibit a consistent directional movement (either increasing or decreasing) without significant variations, this model can effectively capture that behavior.

Limitations: This model may not perform well with EOP data exhibiting seasonality, significant fluctuations, or where past values and trends significantly influence future values.

- ARIMA(1,1,0): Differenced First-Order Autoregressive Model The ARIMA(1,1,0) model can be represented as:

$$\hat{Y}_t = \mu + Y_{t-1} + \phi_1(Y_{t-1} - Y_{t-2}) \tag{3.13}$$

Here, ϕ_1 is the coefficient of the first lag of the differenced series. The term $(Y_{t-1} - Y_{t-2})$ represents the first difference of the series at time $t-1$.

Model dynamics The model combines differencing (to achieve stationarity) with autoregression. It predicts the current value based on the previous value and the change that occurred in the last period. This model is particularly useful when the series, once differenced, shows a strong correlation with its lagged values.

Applicability to EOPs This model is apt for EOP data that shows trends or patterns only becoming apparent or stationary after differencing. It's effective for EOPs where each value tends to deviate from the previous value by an amount that is dependent on the recent changes.

Limitations ARIMA(1,1,0) may not be ideal for EOP data with high seasonality or non-linear trends. The model assumes a linear relationship between past changes and future values, which might not always hold in complex time series data like EOPs.

3.2.2 Seasonal ARIMA(SARIMA) Modeling

Seasonal ARIMA (SARIMA) (Hyndman and Athanasopoulos (2018)) models are an extension of the basic ARIMA models, specifically designed to handle and forecast time series data with seasonal variations. This feature makes SARIMA particularly suitable for data like EOPs, which may exhibit seasonal patterns. SARIMA models

add a seasonal component to the standard ARIMA models, addressing both the trend and seasonality in a time series. The model can account for shifts or patterns that repeat over a fixed period.

SARIMA Model Equation: A SARIMA model is denoted as $SARIMA(p,d,q)(P,D,Q)s$, where p,d,q are the non-seasonal components (same as in ARIMA). P,D,Q represent the seasonal parts of the model. s is the length of the seasonal cycle. The full SARIMA model is expressed as:

$$Y'_t = c + \sum_{i=1}^{p} \phi_i Y'_{t-i} + \sum_{i=1}^{q} \theta_i \varepsilon_{t-i} + \sum_{i=1}^{P} \Phi_i Y'_{t-is} + \sum_{i=1}^{Q} \Theta_i \varepsilon_{t-is} + \varepsilon_t \tag{3.14}$$

Here, Y'_t represents the differenced and seasonally differenced series. The ϕ, θ, Φ, and Θ terms are coefficients for the autoregressive and moving average components for both non-seasonal and seasonal parts.

Dynamics of SARIMA Models: SARIMA models integrate both non-seasonal and seasonal factors in a multiplicative manner. They can handle complex structures in time series data by modeling the dependencies within the observations. The seasonal elements P, D, and Q capture the seasonality, allowing the model to account for patterns that repeat every s periods. The differencing terms d and D help in making the series stationary, both at the regular and seasonal levels.

Application in EOP Forecasting: SARIMA is particularly effective for EOP data if it exhibits seasonality – patterns repeating at regular intervals. For instance, if EOPs change in a similar way across the same months or seasons each year, SARIMA can capture this periodicity. The model's ability to integrate both the long-term trends and seasonal fluctuations makes it a comprehensive tool for forecasting EOPs.

Model selection and challenges: Selecting the appropriate values for p,d,q,P,D,Q, and s requires careful analysis, often involving autocorrelation and partial autocorrelation plots, along with seasonal decomposition techniques. Overfitting can be a concern, particularly with complex models having many parameters. A balance must be struck between adequately capturing the seasonal patterns and maintaining model parsimony.

Limitations: The complexity of SARIMA models means they require more parameters than non-seasonal models, which can lead to overfitting if not properly managed. They assume a stable seasonal pattern over time, which might not hold in cases where the seasonality evolves.

3.3 Random Forests (RF) Modeling

RF is a machine learning algorithm widely used for regression and classification tasks, including time series forecasting like EOPs. It's particularly effective in capturing complex, nonlinear relationships in data. RF models operate by the principle of constructing multiple decision trees during the training phases and output the mode of the classes (classification) or mean prediction (regression) of the individual trees. The general prediction equation for RF in a regression context is:

$$\hat{Y} = \frac{1}{N} \sum_{i=1}^{N} T_i(X) \tag{3.15}$$

Where $\hat{Y}$ is the predicted value, N is the number of trees, $T_i(X)$ is the prediction of the ith tree for input X.

In time series forecasting, the input features X are often constructed from lagged values of the time series. For example, for a time series $\{Y_t\}$, the feature set for prediction at time t could be $\{Y_{t-1}, Y_{t-2}, \ldots, Y_{t-p}\}$. To transform the time series data into a feature, it can be represented as:

$$X_t = [Y_{t-1}, Y_{t-2}, \ldots, Y_{t-p}] \tag{3.16}$$

Where p is the number of lags (past observations) used as predictors.

Training and Prediction process: The training process involves constructing each decision tree T_i using a random subset of the data and features. For regression problems, each tree provides a continuous output, and the final prediction is typically the average of all trees' outputs.

Benefits and challenges in EOP Forecasting:

- Handling Non-linearity: RF can model complex patterns in EOP data, which might be nonlinear or have interactions between predictors.
- Feature importance: RF can evaluate the importance of different lagged values in forecasting future EOPs.
- Robustness: It's less prone to overfitting, thanks to the averaging process across multiple trees.

Limitations: RF does not inherently account for seasonality or trend components in time series data. The ensemble nature of RF makes it less interpretable than simpler models like linear regression. Large numbers of trees and deep trees can make the model computationally intensive.

3.4 Support Vector Regression(SVR) Modelling

SVR is a powerful machine learning technique derived from Support Vector Machine (SVM)s, extensively used for regression tasks, including time series forecasting like EOPs. SVR is particularly beneficial for its ability to handle high-dimensional spaces and its effectiveness in modeling complex, non-linear relationships.

Core concept of SVR: SVR is based on the principle of finding a function that approximates the relationship between input variables and continuous output with a margin of tolerance (ε). The objective of SVR is to minimize this margin while penalizing data points that fall outside it. The basic form of SVR for a linear relationship is given by:

$$f(X) = \langle w, X \rangle + b \tag{3.17}$$

Where $f(X)$ is the predicted value for the input X, w is the weight vector, $\langle \cdot, \cdot \rangle$ denotes the dot product, and b is the bias term. For non-linear relationships, SVR uses kernel functions to map input features into a higher-dimensional space where

a linear relationship can be found. A common choice is the Radial Basis Function (RBF) kernel:

$$K(X_i, X_j) = \exp(-\gamma \|X_i - X_j\|^2) \tag{3.18}$$

Where γ is a parameter that determines the spread of the kernel.

SVR in Time Series Forecasting: In time series forecasting, SVR can be utilized by transforming the series into a supervised learning problem. The input features are typically lagged values of the series. For instance, for EOP forecasting, the input set X_t for prediction at time t could be $[Y_{t-1}, Y_{t-2}, \ldots, Y_{t-p}]$.

Advantages of Using SVR in EOP Forecasting Through the use of kernel functions, SVR can model complex, non-linear relationships in EOP data. SVR is robust to outliers as it focuses on the points near the margin. The use of different kernels allows for modeling various data patterns effectively.

3.5 Multilayer Perceptron (MLP) Modelling

MLP, a class of feedforward artificial neural networks, is widely used in various machine learning tasks, including time series forecasting like EOPs. MLP is particularly noted for its ability to capture complex, nonlinear relationships in data.

An MLP consists of at least three layers of nodes: an input layer, one or more hidden layers, and an output layer. Each node, except for the input nodes, is a neuron that uses a nonlinear activation function.

MLP Model Structure: The general structure of an MLP for regression tasks can be represented as:

- Input Layer: Receives the input features (e.g., lagged values in time series).
- Hidden Layers: Each layer computes a weighted sum of its inputs, followed by an activation function. Common activation functions include ReLU (Rectified Linear Unit), sigmoid, and tanh.
- Output Layer: Produces the final prediction, typically with a linear activation in regression tasks.

MLP in Time Series Forecasting: For time series forecasting, such as predicting EOPs, the MLP model can be structured to take in a series of past observations as inputs. If Y_t is the time series, the input features for MLP at time t might be $[Y_{t-1}, Y_{t-2}, ..., Y_{t-p}]$, where p is the number of lags used.

Training and Prediction with MLP: The MLP learns to map the input features to the output (future value of the series) by adjusting the weights and biases in each neuron (training). The typical loss function used for training an MLP in a regression setting is Mean Squared Error (MSE).

Advantages of MLP in EOP Forecasting: MLPs can model complex nonlinear relationships, making them suitable for EOP data which may not follow linear patterns. MLPs can learn to capture various temporal dynamics by adjusting the number and size of hidden layers and the activation functions.

3.6 Evaluation Metrics and ESA Standards

3.6.1 Evaluation matrices

Mean Squared Error (MSE) The MSE is calculated as the average of the squared differences between predicted ($\hat{y}_i$) and actual (y_i) values:

$$\text{MSE} = \frac{1}{n}\sum_{i=1}^{n}(\hat{y}_i - y_i)^2 \tag{3.19}$$

where:

- $\hat{y}_i$: Predicted EOP value.
- y_i: Actual EOP value.
- n: Total number of data points.

The MSE quantifies the average squared difference between predicted and actual values, assigning higher penalties to larger errors.

Root Mean Square Error (RMSE)

The RMSE is a metric that measures the magnitude of the difference between the predicted values ($\hat{y}_i$) by a model or estimator and the actual observed values (y_i). It is derived from the MSE, which represents the mean or average of the squared differences between predicted and actual values. The RMSE is defined as the square root of the MSE:

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(\hat{y}_i - y_i)^2} \tag{3.20}$$

where:

- $\hat{y}_i$ is the predicted value.
- y_i is the actual observed value.
- n is the total number of observations or data points.

The RMSE is particularly valuable as it gives more weight to larger errors, making it a useful metric in situations where large errors are especially undesirable. By extracting the square root of the MAE, the RMSE is expressed in the same units as the parameter being evaluated, facilitating a more intuitive interpretation. The RMSE's emphasis on larger errors makes it a more precise measure of model performance when significant deviations are particularly critical to address.

3.6.2 ESA Standards

ESA Standards for Prediction Product Accuracy The ESA has set strict accuracy requirements for EOP prediction products. These standards are important in assessing the reliability and precision of predictive models utilized in various scientific and navigational applications. The specified accuracy requirements across different prediction horizons are set according to the announcement by the Invitation To Tender (ITT) of ESA and are assumed to be reasonable and correct. The requirements are given as follows:

1. 1-Day Predictions:
   - x_{pole} 0.30 milliarcseconds (mas)
   - y_{pole} 0.23 milliarcseconds (mas)
   - UTC-UT1 0.06 milliseconds (ms)
2. 5-Day Predictions:
   - x_{pole} 1.8 mas
   - y_{pole} 1.3 mas
   - UTC-UT1 0.215 ms
3. 10-Day Predictions:
   - x_{pole} 3.5 mas
   - y_{pole} 1.99 mas
   - UTC-UT1 0.53 ms
4. 20-Day Predictions:
   - x_{pole} 6.8 mas
   - y_{pole} 2.7 mas
   - UTC-UT1 1.9 ms
5. 40-Day Predictions:
   - x_{pole} 12.8 mas
   - y_{pole} 4.1 mas
   - UTC-UT1 2.9 ms
6. 90-Day Predictions:
   - x_{pole} 23.8 mas
   - y_{pole} 16.5 mas
   - UTC-UT1 8.6 ms

These standards serve as crucial benchmarks for evaluating the accuracy of predictive models in estimating Polar Motion (x_p and y_p) as well as the difference between UTC and UT1. Adhering to these stringent accuracy requirements ensures the reliability of predictive models, facilitating their application in navigation systems, geodetic surveys, and various scientific endeavors reliant on precise Earth orientation information.

4. Data

4.1 Data Collection and Preparation

4.1.1 Dataset Overview

The EOP 14 C04 series, curated by the Earth Orientation Parameter (EOP) Product Center at the Paris Observatory under the auspices of the International Earth Rotation and Reference Systems Service (IERS), is a cornerstone dataset for Earth orientation studies. This dataset is meticulously aligned with the IAU 2000 Precession-Nutation Model and is consistent with the International Terrestrial Reference Frame 2020 (ITRF2020) Bizouard et al. (2019), offering unparalleled insights into Earth's dynamic orientation. The dataset is sampled bi-daily at 12-hour intervals UTC, providing a dense temporal resolution for detailed analysis.

Each data entry within the EOP 14 C04 series follows a precise format, defined by the format string, and can be understood as follows:

- **Date and Time:** The first four integer fields, each 4 characters wide, denote the Year (YR), Month (MM), Day (DD), and Hour (HH), respectively, providing the exact timestamp for each measurement.
- **Modified Julian Date (MJD):** A floating-point field, 10 characters wide with 2 decimal places, offering a continuous count of days that facilitates chronological analysis across the dataset.
- **Pole Coordinates (x_p, y_p):** Two floating-point fields, each 12 characters wide with 6 decimal places, represent the x and y components of Earth's polar motion in arcseconds, indicating the axial tilt and wander.
- **UT1-UTC (s):** A 12-character wide floating-point field with 7 decimal places, detailing the difference between Earth-rotation-based Universal Time 1 (UT1) and Coordinated Universal Time (UTC) in seconds.
- **Celestial Pole Offsets (dX, dY):** Adjustments in Earth's rotational axis orientation relative to the celestial frame are recorded in two 12-character wide floating-point fields, each with 6 decimal places.

- **Rates of Change (xrt", yrt"):** These fields, each 12 characters wide with 6 decimal places, indicate the temporal variations in the pole coordinates, reflecting dynamic changes in Earth's orientation.
- **Length of Day (LOD) Variation (s):** A 12-character wide floating-point field with 7 decimal places, capturing the variability in Earth's rotational period, measured in seconds.
- **Measurement Uncertainties:** The dataset also includes uncertainties for each measurement, denoted by fields such as x Er, y Er, UT1-UTC Er, dX Er, dY Er, xrt Er, yrt Er, and LOD Er. Each of these uncertainties is presented in a 12-character wide floating-point field with 7 decimal places, providing insights into the precision and reliability of the data.

4.1.2 Dataset Integrity and Format

The IERS EOP 14 C04 dataset undergoes rigorous annual data preparation and daily combination processes as explained by (Bizouard et al. (2019)) to ensure its accuracy, consistency, and alignment with international reference frames which are as follows:

Annual Data Preparation:

Each year begins with the selection or update of EOP series, incorporating measurements like UT1-UTC, LOD, and pole coordinates from various space geodetic techniques. This step ensures a diverse and comprehensive dataset reflective of Earth's orientation changes.

Formal uncertainties of EOP values are rescaled to rectify potential inconsistencies and normalize the dataset, making it more reliable. This involves adjusting the uncertainties to match the average standard deviation of paired differences within the selected series.

Additionally, the dataset is rigorously aligned with the current space reference frames, such as ICRF and ITRF, through a detailed characterization process. This alignment corrects for network effects and certifies the dataset's relevance and precision.

Daily Combination Procedure:

The daily integration of selected EOP series into the EOP 14 C04 dataset involves collecting EOP values with their a priori uncertainties and ensuring series homogeneity by addressing network instabilities and rescaling uncertainties. Differences with reference values are calculated and averaged to derive the combined EOP values, ensuring dataset integrity.

4.1.3 Data Access and Utility

The dataset, encompassing records from 1962 to the present, is publicly accessible via FTP from the Earth Orientation Centre, allowing for comprehensive temporal analyses of the EOPs. The dataset, EOP 14 C04 series for 1962-2024 (IAU2000) is bi-weekly updated and can be accessed at ftp://hpiers.obspm.fr/iers/eop/eopc04/

4.1.4 CONT14

The CONT14 campaign, part of a series of continuous VLBI sessions, was conducted in May 2014 to acquire high-precision EOP data over two weeks, starting from May 6, 2014 to May 20, 2014 (Haas et al. (2017)). This campaign aimed to demonstrate the utmost accuracy achievable by the current VLBI system, supporting Earth rotation studies, reference frame stability investigations, and the analysis of site motions. It involved a network of 16 stations located worldwide, to ensure geographical diversity and precision. The campaign facilitated studies of high-frequency EOP variations, ocean tide models, and atmospheric phenomena, contributing to ionospheric research. Additionally, it provided an opportunity to improve tropospheric parameter estimates and assess the repeatability of reference frames. The CONT14 data were instrumental in comparing site motions with external factors and necessary to enhance our understanding of Earth's rotational behavior and geodetic measurements. This campaign marked a significant step toward advancing VLBI techniques and precision in Earth sciences.

4.2 Data Preprocessing

4.2.1 Initial Data Cleaning and Transformation

The initial preprocessing of the raw IERS dataset involves several key activities aimed at refining the dataset for predictive modeling:

- **Data Cleaning:** Removal of extraneous columns not important to the core study objectives such as Type, Type.1, Type.2, xrt(”), yrt(”), x Er, y Er, UT1-UTC Er, dX Er, dY Er, xrt Er, yrt Er, LOD Er are removed. These columns either contain redundant information or data not pertinent to the study of .
- **Column Selection:** Focusing on key parameters and their uncertainties, directly relevant to the prediction of EOP like 'Year', 'Month', 'Day', 'MJD', x_p, y_p, UT1-UTC, dX, dY, LOD.
- **Date Handling:** Establishing a coherent date format through a new 'Date' column derived from the 'Year', 'Month', and 'Day' columns, enabling effective time series analysis.

4.2.2 Advanced Preprocessing with Singular Spectrum Analysis

The Singular Spectrum Analysis (SSA) method is a technique employed for trend extraction in time series analysis, comprising several fundamental steps described in prior works by (Golyandina and Zhigljavsky (2013)), Ghil et al. (2002) and Harvey and Peters (1990)).

4.2.2.1 Embedding Step

Embedding in Singular Spectrum Analysis (SSA) suggested by (Hyndman and Athanasopoulos (2018)), Kalarus et al. (2010), Rocco S (2013) and Hassani (2022)) is a fundamental step that transforms a one-dimensional time series $X = (x_1, \ldots, x_N)$

into a multidimensional representation. This process involves creating lagged vectors $X_1, \ldots, X_K$ from X, where each X_i is a vector containing L consecutive elements of the original series. The number of resulting vectors K is determined by the window length L and the length of the original series N such that $K = N - L + 1$.

The window length L is a crucial parameter in embedding. It defines the length of each lagged vector and influences the properties of the resulting trajectory matrix. The choice of L is essential; it needs to strike a balance. If L is too small, it might not capture essential characteristics of the time series. Conversely, if L is excessively large, it could create too many lagged vectors, losing the specific dynamics of the original series.

Different fields have varying approaches to selecting L. In dynamical systems analysis, for instance, smaller L values are used to calculate correlation dimensions, which relate to the fractal dimensions of the system's attractor. Here, L is small relative to a very large K ($K \to \infty$). Conversely, methods like Structural Total Least Squares (STLS) with Hankel matrix structure often select $L = r + 1$, where r represents the estimated rank of the approximation matrix.

In SSA, L should be chosen to be sufficiently large to encapsulate essential dynamics of the original series within each lagged vector. Larger L values offer the advantage of treating each lagged vector X_i as an independent series. This enables the exploration of specific characteristics and dynamics across this set of derived series.

The choice of L is a critical consideration in SSA, impacting how well the resulting lagged vectors represent the underlying dynamics of the original time series. It's essential to balance the trade-off between capturing essential information and avoiding excessive fragmentation of the data. Detailed discussions on optimal L selection can be found in specialized literature, allowing practitioners to tailor embedding for various types of time series analysis.

4.2.2.2 Decomposition Step

The Decomposition step entails expressing the original matrix X as a sum of its constituent components, typically obtained through the Singular Value Decomposition (SVD). This allows for the extraction of individual patterns or signals embedded in the time series data.

Singular Value Decomposition (SVD)

The SVD of an arbitrary non-zero $L \times K$ matrix $X = [X_1 : \ldots : X_K]$ is represented as:

$$X = \sum_{i=1}^{d} \sqrt{\lambda_i}\, U_i\, V_i^T \tag{4.1}$$

Here,

- λ_i (for $i = 1, \ldots, L$) are eigenvalues of the matrix $S = XX^T$ arranged in descending order.
- $d = \max\{i : \lambda_i > 0\}$ represents the rank of X.

- $U_1, \ldots, U_d$ form an orthonormal system of eigenvectors of S, and $V_i = X^T U_i / \sqrt{\lambda_i}$.
- The left singular vectors are U_i, the right singular vectors are V_i, and $\sqrt{\lambda_i}$ denotes the singular values.

Optimal Features of SVD in Decomposition

- The SVD offers an optimal approximation of X by matrices of rank $r < d$, where $\sum_{i=1}^{r} \lambda_i / ||X||_F^2$ characterizes the contribution of individual matrices X_i to the entire trajectory matrix X.
- Each matrix X_i contributes to X based on the ratio $\lambda_i / ||X||_F^2$, highlighting its role in the overall trajectory matrix.

Dual geometrical descriptions The SVD yields two geometric interpretations:

- $U_1, \ldots, U_d$ form an orthonormal basis in the d-dimensional trajectory space $X(L)$.
- $V_1, \ldots, V_d$ represent another orthogonal basis, providing an alternate viewpoint.

Principal directions and components Eigenvectors U_i determine principal directions, with V_i and $Z_i = \sqrt{\lambda_i} V_i$ termed factor vectors and principal components, respectively.

4.2.2.3 Grouping Step

Concept of Separability The primary goal of the grouping step is to segregate the matrix terms obtained from the SVD into distinct groups that correspond to different underlying components or patterns within the time series.

Separability: Consider a time series matrix X and its trajectory matrix $X(L)$, which consists of lagged vectors. The SVD yields eigenvalues, eigenvectors, and elementary matrices that are crucial for the grouping step.

Formalizing separability: Suppose X can be decomposed into two distinct series, $X(1)$ and $X(2)$, such that $X = X(1) + X(2)$. The series $X(1)$ is separable by a specific SVD decomposition if there exists a set of indices I such that $X(1) = \sum_{i \in I} X_i$ and $X(2) = \sum_{i \notin I} X_i$.

Grouping strategy The principles of grouping rely on analyzing the singular vectors U_i and V_i and the associated eigenvalues λ_i derived from the SVD decomposition (previously explained in SVD).

Matrix Components Arrangement: The grouping step involves organizing the elementary matrices X_i derived from the SVD into distinct groups $I_1, I_2, \ldots, I_m$ that demonstrate certain structural characteristics or patterns within the time series.

Grouping from a geometric Perspective

- Orthogonal Subspaces: By considering the trajectory space $X(L)$ spanned by the eigenvectors, the grouping corresponds to splitting this space into m orthogonal subspaces $X(L,1), X(L,2), \ldots, X(L,m)$ via the eigenspace decomposition.

- Representation via Subspaces: Each subspace $X(L, i)$ is defined as the span of specific eigenvectors, e.g., $X(L, i) = \text{span}(U_i \mid i \in I_i)$, signifying the contribution of certain eigenvectors to a particular subgroup.

Practical Considerations:

- Analysis of Eigenvectors: The process of determining the optimal grouping is based on insights derived from the eigenvectors, eigenvalues, and their relationships, as revealed through the SVD.
- Complexity in Formalization: While formal criteria can be established for separability, the complete formalization of the grouping step remains challenging due to the intricacies of identifying the SVD components and their inclusion in distinct groups.

4.2.2.4 Reconstruction Step

Trend Component

- Identification: The trend component in SSA is reconstructed by aggregating a subset of dominant eigenvalues and their corresponding eigenvectors. These eigenpairs are selected based on their representation of the long-term behavior within the time series.
- Reconstruction Process: To extract the trend component, consider r dominant eigenvalues λ_i and their associated eigenvectors U_i (where $i = 1, 2, \ldots, r$). The reconstructed trend component T can be obtained by summing the outer products of these selected eigenvectors:

$$T = \sum_{i=1}^{r} \lambda_i U_i V_i^T \tag{4.2}$$

  Here, λ_i represents eigenvalues, U_i is the corresponding left singular vector, and V_i is the right singular vector.

Seasonal Component

- Identification: Seasonal patterns are identified through the selection of eigenvalues and their associated eigenvectors that capture periodic behavior or cyclical variations within the time series.
- Reconstruction Process: To reconstruct the seasonal component S, consider s dominant eigenvalues λ_i and their corresponding eigenvectors U_i (where $i = 1, 2, \ldots, s$). The seasonal component S is derived by summing the outer products of these selected eigenvectors:

$$S = \sum_{i=1}^{s} \lambda_i U_i V_i^T \tag{4.3}$$

  Similar to the trend component, λ_i represents eigenvalues, U_i is the corresponding left singular vector, and V_i is the right singular vector.

Residual Component

- Identification: The residual component captures the unexplained variability in the time series after the trend and seasonal patterns have been extracted.
- Reconstruction Process: After obtaining the trend (T) and seasonal (S) components, the residual component (R) is calculated as the difference between the original time series and the sum of the trend and seasonal components:

$$R = [X - (T + S)] \tag{4.4}$$

Understanding the components

- Trend: The trend component T characterizes the overall long-term behavior or direction of the time series, providing insights into the overarching patterns or tendencies.
- Seasonality: The seasonal component S captures periodic variations or cycles within the data, showcasing repetitive patterns occurring at specific intervals.
- Residual: The residual component R represents the remaining signal after extracting the trend and seasonal patterns. It encompasses irregularities, noise, or unexplained variations present in the original data.

5. Methodology

5.1 Autocorrelation Analysis

Autocorrelation analysis is conducted on EOP 14 C04 (IAU2000A) time series to inform the parameter settings for LSTM models, providing insights into the time dependencies within the EOP data.

Following is a correlation test performed on the x_p data from the year 1990 to 2022. The x-axis in Figure 5.1 represents the lag, the lag measures how well the current value of the series is related to its past value. The graph spans from 0 to 30 time steps. The y-axis depicts the autocorrelation coefficient, which spans from -1 to 1. A value of 1 signifies complete positive autocorrelation, -1 signifies complete negative autocorrelation, and 0 signifies no autocorrelation.

The vertical lines depict the autocorrelation coefficient for each lag. The presence of bars positioned above zero, which progressively decrease as the lag grows, indicates a significant positive autocorrelation at earlier lags that decreases as time progresses.

The blue shaded area in the autocorrelation plot represents the confidence interval, usually at 95%, which helps to distinguish significant correlation values from those that could occur by chance. If the autocorrelation bar for a specific lag exceeds this blue area, it indicates that the correlation at that lag is statistically significant.

The gradual decrease of the autocorrelation indicates that the x_p variable is subject to prediction based on its previous values, so approaches such as SARIMA or LSTM can be effective for prediction. The autocorrelation is much stronger at shorter lags, which can be used to set parameters for time series forecasting models.

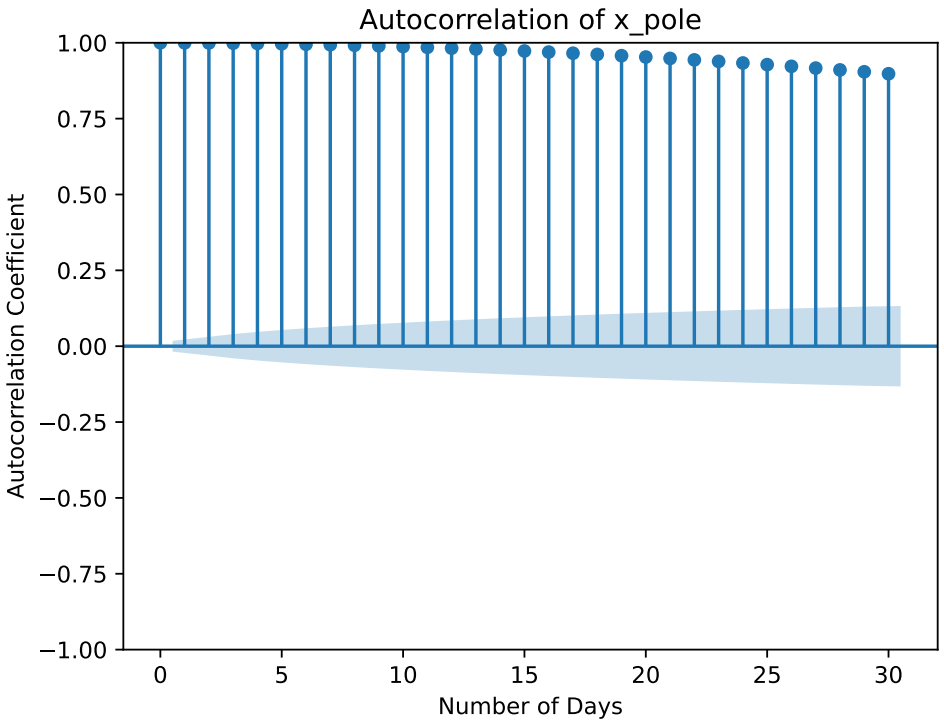


Figure 5.1: Autocorrelation analysis for x_p

Similarly, the same approach can be used for y_p, dX, dY, and a similar understanding of the correlation was gained.

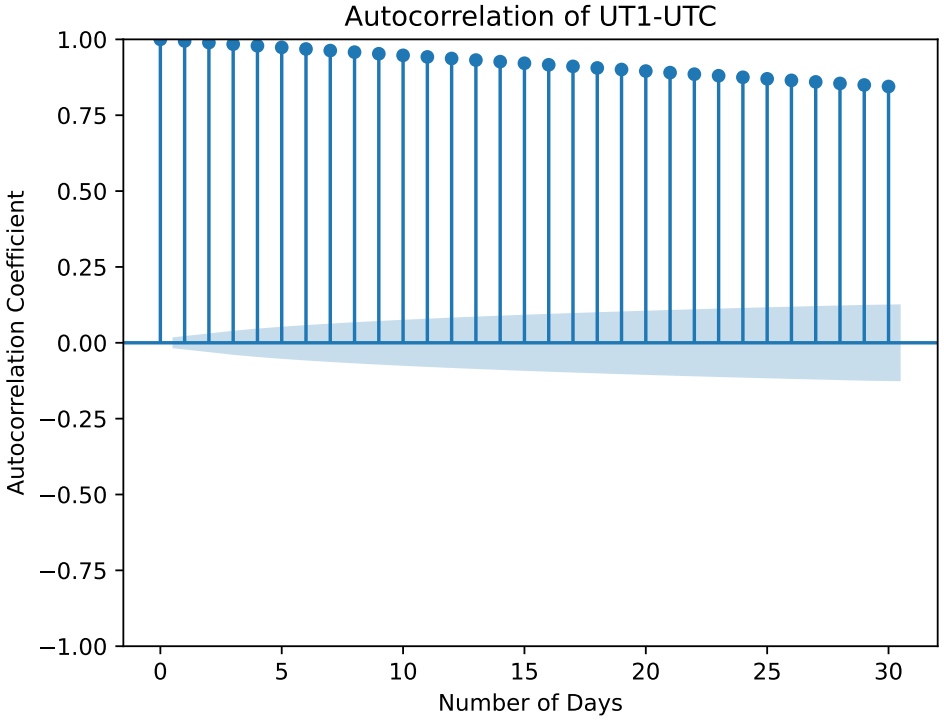


Figure 5.2: Autocorrelation analysis for UT1-UTC

For UT1-UTC we can perform another correlation analysis as shown in Figure 5.2, we can see unlike x_p which shows a high correlation for at least 10 lags, the UT1-UTC shows better correlation only for the initial 3 days and then it decreases rapidly.

5.2 Model Selection and Evaluation Standards

The selection of appropriate predictive models is crucial in the application of EOP prediction, As we have noticed for correlation tests that for all 4 EOPs we can use similar techniques so we test it for x_p so to understand how the models will work for y_p, dX and dY. A variety of models are used in this study, including as LSTM model and SARIMA model, which were selected based on how well they processed time series data. The selection criteria are centered on how well each model can capture the complex dynamics and patterns present in EOP data.

The initial analysis phase uses data from the 90 days, which are taken from the EOP 14 C04 (IAU2000A) time series, to analyze a ten-day forecast period. This time frame provides a reasonable but comprehensive scope for model evaluation. The MAE, a statistical measure that expresses the average size of errors in a model's predictions, is the primary performance parameter used (Kalarus et al. (2010)). For this study we use MAE as it provides an interpretable assessment of accuracy, making it an ideal metric for comparing different models.

5.3 Model Comparative Analysis

The analysis of predictions using several machine learning models between January 1st and January 10th, 2023, using a dataset consisting the previous 90 days i.e. from October 3, 2022. AR, ARIMA (Jia et al. (2017), Albeladi et al. (2023)), SARIMA (Sirisha et al. (2022)), RF, SVR, MLP, and LSTM networks (Soja et al. (2023)) are among the models that are being investigated. This wide variety of models makes it possible to thoroughly examine each one's effectiveness in forecasting x_p. The Original values from EOP 14 C04 (IAU2000A) time series are compared against the predicted values from different models are shown in Table 5.1

Table 5.1: Original vs predicted x_p values for different ML techniques

Date	Original x_p	Predicted x_p						
		AR	ARIMA	SARIMA	RF	SVR	MLP	LSTM
01-01-2023	0.062699	0.063593	0.062843	0.062843	0.093919	-0.057848	0.005179	0.067611
02-01-2023	0.059003	0.060561	0.058953	0.058953	0.092650	-0.059644	0.006080	0.065503
03-01-2023	0.055638	0.057364	0.055341	0.055341	0.092607	-0.061429	0.007109	0.062677
04-01-2023	0.052933	0.054409	0.051954	0.051954	0.091454	-0.063204	-0.004736	0.059736
05-01-2023	0.050795	0.048306	0.048748	0.048748	0.090805	-0.064979	0.003141	0.057724
06-01-2023	0.048795	0.045462	0.045690	0.045690	0.089491	-0.066745	0.021987	0.055803
07-01-2023	0.046853	0.042608	0.042752	0.042752	0.088658	-0.068497	0.029284	0.053650
08-01-2023	0.045146	0.042608	0.039911	0.039911	0.088411	-0.070243	0.013874	0.050506
09-01-2023	0.043370	0.039535	0.037148	0.037148	0.088194	-0.071982	-0.002396	0.049664
10-01-2023	0.041325	0.036535	0.034449	0.034449	0.087333	-0.073706	-0.005751	0.045608

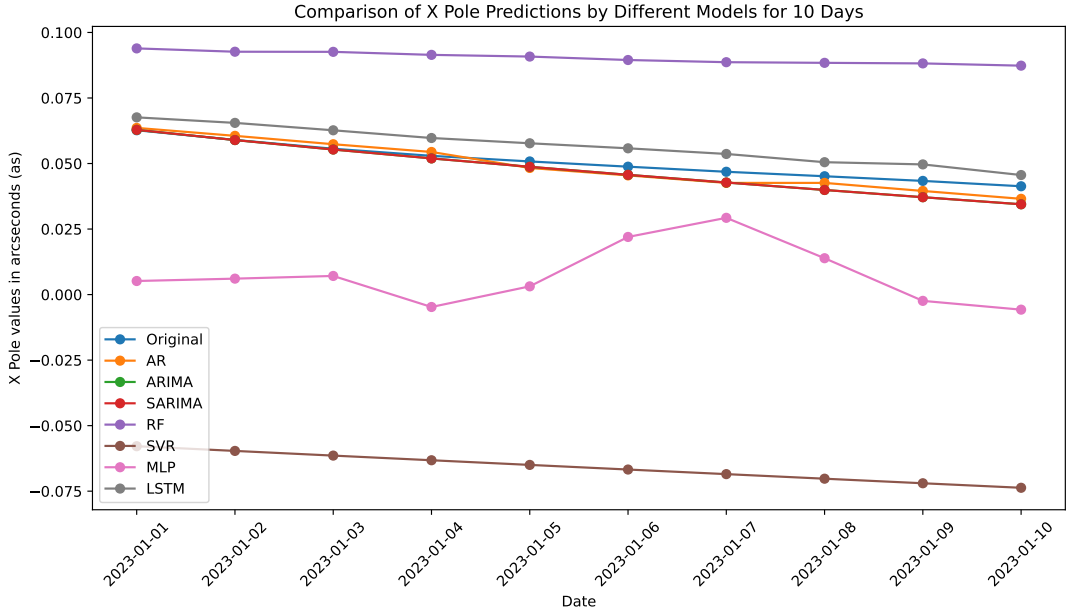


Figure 5.3: Initial Model selection process for different ML techniques

Table 5.2: MSE and RMSE for each method

Model	MAE(mas)	RMSE(mas)
AR	1.926	2.344
ARIMA	2.905	3.809
SARIMA	0.014	3.82
RF	39.69	39.95
SVR	116.48	116.49
MLP	43.27	45.14
LSTM	6.19	6.26

- **AR Model**: With the lowest MAE and RMSE, the AR model is quite accurate, especially when it comes to identifying linear patterns in the short-term EOP data. Its effectiveness demonstrates how reliable it is for making short-term forecasts.
- **ARIMA and SARIMA Models**: These models perform similarly, though a little bit lower than the AR model. Given their prediction precision, it appears that seasonal elements are not very important in x_p EOP prediction when predicting short-term, indicating that this parameter is non-seasonal for this duration.
- **Models for SVR and RF**: These more intricate designs differ greatly from the real x_p values. Interestingly, SVR shows the biggest variance, indicating difficulties managing the non-linear dynamics of the data.
- **MLP Model**: The performance of the MLP varies, indicating potential issues such as overfitting or susceptibility to seasonal variations in the EOP data.
- **LSTM Network**: The LSTM network is significant for its robustness and adaptability, while not having the same level of precision as the AR model. It is a useful tool for EOP forecasting because of its capacity to record temporal correlations and intricate data patterns, particularly in situations with long-term seasonal dependencies and non-linear patterns.

5.4 Approach for predicting EOPs

5.4.1 LSTM Prediction

By using the LSTM network's ability to learn from sequences, we perform the prediction of EOPs. The LSTM's complex architecture, featuring its unique gates and memory cells, is specifically designed to capture the complex temporal patterns of x_p data. A strict training routine is used, using 90% of the data, while validation is performed on the remaining portion to evaluate the model's prediction ability. Training improves through the utilization of approaches such as early halting and adaptive learning rate modifications.

5.4.2 SARIMA Time Series Forecasting

Additionally, the SARIMA model is specifically adjusted to match the periodic pattern of the EOP time series. The model is carefully set up to replicate the seasonal patterns obtained from SSA, improved by the calibration of non-seasonal models for the trend and residual components. The individual projections are combined to create a comprehensive prediction of the EOPs.

5.5 Integration and Comparative Analysis

5.5.1 SSA-Improved Long Short-Term Memory

The study investigates the use of SSA-enhanced LSTM models as suggested by Huang et al. (2022)), where separate LSTMs are assigned to each decomposed time series component. The predicted results from multiple models are recombined to obtain the final predicted time series, providing an extensive view of the predictions of EOPs.

5.5.2 SSA-Improved SARIMA Model

In addition to the machine learning method, SARIMA models include insights from SSA to accurately estimate each SSA component. The implementation of this layered technique not only increases the accuracy of the SARIMA model's predictions but also highlights the importance of SSA in traditional statistical time series analysis.

5.5.3 Evaluation of Performance

To determine how efficient these refined forecasting methods are, an extensive assessment procedure is followed. The MAE data is used to compare the performance of the LSTM and SARIMA models (Kumar Dubey et al. (2021)), both with and without SSA. The accuracy of the models is demonstrated by comparing graphs that show the relationship between the predicted and actual EOP values.

5.5.4 Context and Applicability of the Model

The importance of selecting a machine learning model that suits the unique characteristics of the EOP data is highlighted by this thorough investigation. Simpler models like as AR function well in situations where there are obvious linear trends, while LSTM networks may be able to capture intricate temporal patterns inside EOPs. Consequently, it is important to take into account the use of LSTM models for EOP forecasting, especially when working with data that exhibits intricate temporal patterns that may be difficult for simpler models to fully represent.

5.5.5 Methodological Adaptation

The predictive methodologies like LSTM and SARIMA which are used for x_p are now also adapted for y_p, dX, dY and LOD predictions. The SSA-preprocessed data is used for both SARIMA and LSTM models to ensure consistency and leverage the enhanced data quality provided by SSA.

5.6 Approach for Predicting UT1-UTC Using LOD and leap second

The UT1-UTC prediction approach is an elaborate process that involves the conversion and analysis of time series data about the rotation of the Earth (Tissen et al. (2010)), and (Kosek et al. (2007)). The objective of this procedure is to precisely predict UT1-UTC values using the LOD data and the UT1 time standard. The section that follows is a detailed explanation of the methodological approach used for this study.

5.6.1 Data Preparation and Transformation

The initial phase involves the preparation of the UT1-UTC data, which determines the difference between UT1 and UTC. To maintain a stable relationship between UT1 and TAI, leap seconds are periodically deducted from UT1-UTC to account for inconsistencies in Earth's rotation. The resulting series, denoted as UT1-TAI, is defined by the following equation:

$$\text{UT1-TAI} = \text{UT1-UTC} - \text{Leap Seconds}$$

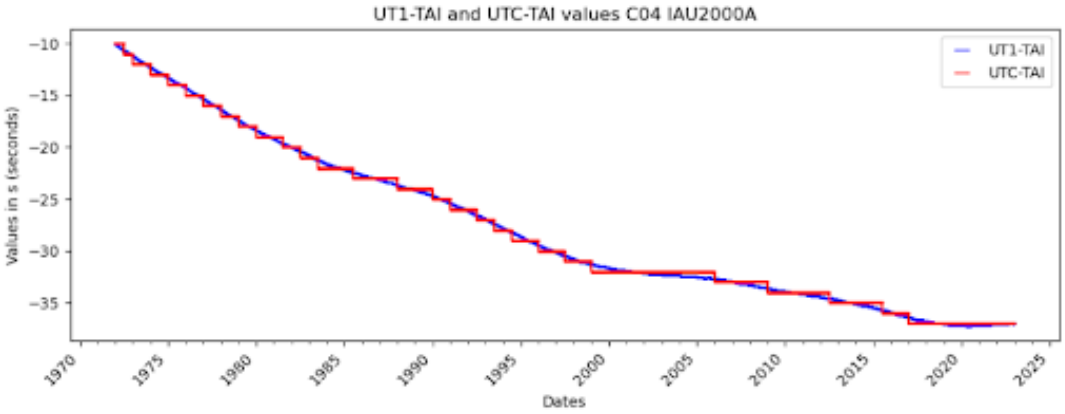


Figure 5.4: UT1-TAI vs UTC-TAI

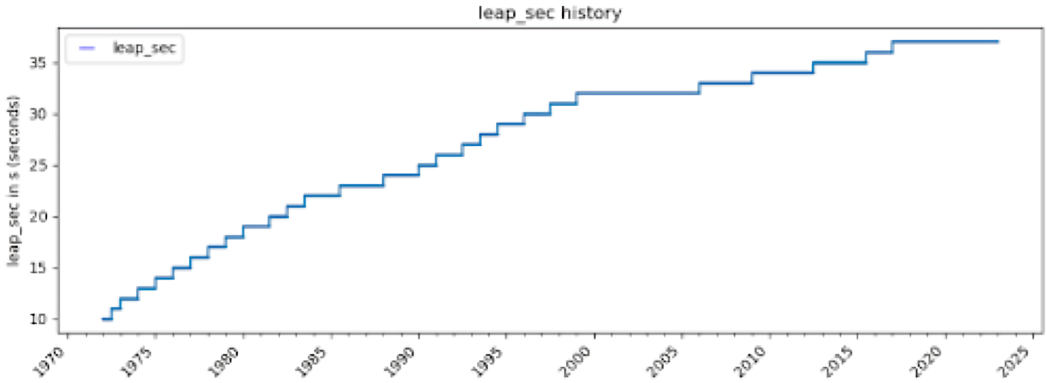


Figure 5.5: History of recorded Leap Seconds(s)

The LOD data, which represents the Earth's fluctuating rotation speed, goes through a cumulative summation transformation, resulting in a time correction factor.

$$\text{Time Correction}_i = \sum_{k=1}^{i} \text{LOD}_k \ \delta T$$

where Time Correction_i is the total adjustment accumulated by the ith day, LOD_k is the LOD on the kth day, and δT is the step size for time, increased by the unit (day).

5.6.2 Calculation of UT1

Once the time correction factor is determined, we proceed with calculating UT1, a crucial parameter for interpreting the Earth's rotational time with respect to the atomic scale. This is achieved by adding the time correction to UT1-TAI (Kosek

et al. (2005)), thereby accurately compensating for the fluctuating velocity of Earth's rotation.

$$\text{UT1} = \text{UT1-TAI} + \text{Time Correction}$$

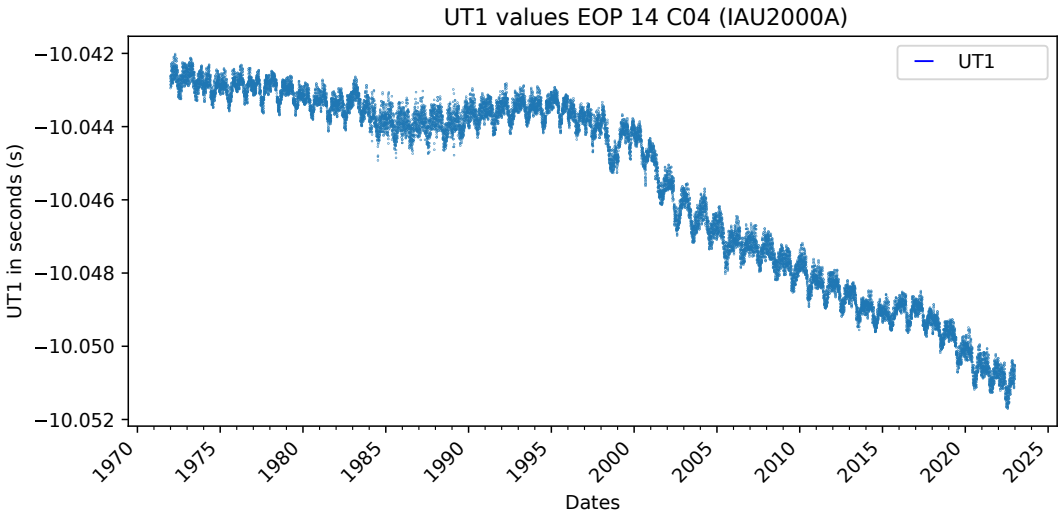


Figure 5.6: UT1

5.6.3 Long Short-Term Memory (LSTM) Forecasting Model

The LSTM model, which specializes at learning sequential and time-dependent patterns, is utilized to predict upcoming UT1 values as also discussed by (Kur et al. (2022)). Before the training phase, the UT1 series performs normalization to improve the performance of the LSTM.

5.6.4 Adjustment of Prediction after Modeling

Once the LSTM model is trained, it can predict UT1 values for a coming 10-day period. The predicted UT1-TAI is derived by adjusting the predicted values by deducting the time correction of LOD.

$$\text{UT1-TAI Predicted} = \text{UT1 Predicted} - \text{Time Correction} \tag{5.1}$$

In the next step, the leap seconds are added back to the predicted UT1-TAI series to calculate the predicted UT1-UTC values.

$$\text{UT1-UTC Predicted} = \text{UT1-TAI Predicted} + \text{Leap Seconds} \tag{5.2}$$

5.6.5 Alternative Approach: Direct LOD Prediction for UT1-UTC Calculation

5.6.5.1 LOD Prediction Modeling

To forecast LOD values, machine learning techniques such as LSTM networks is used for more accuracy.

5.6.5.2 Dynamic Time Correction Calculation

The new time correction factor is computed based on the predicted LOD values to enhance the accuracy of the UT1-TAI predictions. The predicted LOD corrections is added with the already existing time corrections to get the new time series of time corrections.

$$\text{UT1 - TAI Predicted (Adjusted)} = \text{UT1 Predicted} - \text{Dynamic Time Correction} \tag{5.3}$$

5.6.5.3 Final Conversion to UT1-UTC

The adjusted UT1-TAI predictions are converted to UT1-UTC by adding the leap seconds:

$$\text{UT1-UTC Predicted (Adjusted)} = \text{UT1-TAI Predicted (Adjusted)} + \text{Leap Seconds} \tag{5.4}$$

5.6.6 Evaluation and Verification of Model Performance

The accuracy of the model's predictions is evaluated by measuring the MAE metric, which compares the predicted UT1-UTC series to the actual values. MAE provides an understandable evaluation of the model's errors in its predictions.

5.7 Correlation Analysis Between Earth Orientation Parameters

5.7.1 Correlation Analysis Results

The correlation coefficients between different EOPs from the IERS time series EOP 14 C04 (IAU2000A) were computed to understand their relationships with each other. The Pearson correlation coefficient, denoted as r, was utilized for this purpose, calculated by the formula:

$$r = \frac{\sum (X - \overline{X})(Y - \overline{Y})}{\sqrt{\sum (X - \overline{X})^2 \sum (Y - \overline{Y})^2}} \tag{5.5}$$

where X and Y are two sets of variables, $\overline{X}$ is the mean of X, $\overline{Y}$ is the mean of Y, and $\sum$ denotes the summation over all data points.

This statistical measure is crucial in our analysis as it quantifies the degree of linear relationship between pairs of EOPs. A higher absolute value of r indicates a stronger linear relationship. Understanding these relationships is essential, as it can assist in the development of more accurate models for predicting EOPs. The resulting correlation matrix offers insights into the potential for practical applications of these

correlations, indicating where predictive models could be improved or where further investigation may be warranted.

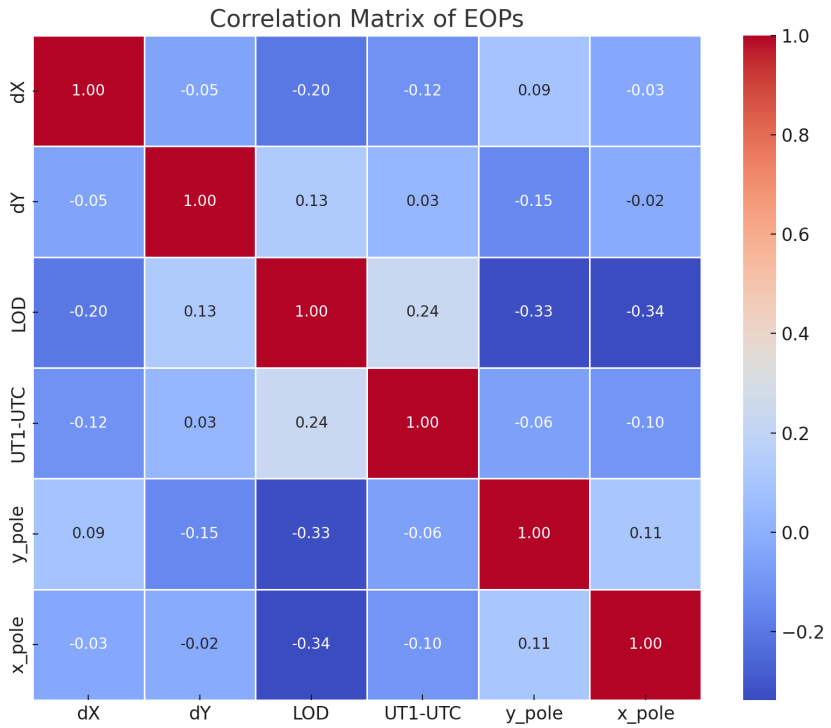


Figure 5.7: Correlation Matrix for the Earth Orientation Parameters

- dX and dY: The correlation coefficient was -0.0487, indicating a very weak negative relationship.
- LOD and UT1-UTC: A correlation coefficient of 0.236 was observed, suggesting a positive relationship.
- x_p and y_p: The correlation coefficient was 0.111, indicating a very weak positive relationship.

5.7.2 Discussion

The correlation analysis conducted on Earth Orientation Parameters (EOPs) indicates predominantly weak linear relationships when considering the data collectively. Notably, the Length of Day (LOD) and the difference between Universal Time (UT1) and Coordinated Universal Time (UTC) exhibit the most substantial relationship, with a positive but weak correlation coefficient.

This finding is significant as it suggests a limited capacity for linear models to predict one EOP based on another. However, the relationship between LOD and UT1-UTC stands out, suggesting that while linear predictability is generally constrained, there is a detectable pattern that could be harnessed for predictive purposes. In the context

of this study, the LOD parameter has been used to forecast UT1-UTC variations, and despite the weak correlation, the analysis suggests that LOD does contain information that may be relevant for modeling UT1-UTC changes as shown in the next Chapter.

6. Results and Discussion

6.1 Overview of Results

This chapter demonstrates the results of implementing machine learning methodologies, especially LSTM networks and SARIMA models, to forecast EOP such as x_p, y_p, and $UT1 - UTC$ and LOD. The results include forecasts for several timeframes, from 1 to 90 days into the future.

The structure of the results section is as follows:

Calculation and Prediction of the UT1: An extensive overview of the process for determining the UT1 using temporal adjustments by LOD commutative sum and the use of LSTM for predicting future UT1 values. The efficiency of the methodology is assessed using the MAE measurements.

EOP prediction for x_p and y_p: This work describes the utilization of LSTM and SARIMA models for predicting the values of x_p and y_p. A comparative examination of these models both with and without the incorporation of SSA improvements.

EOP prediction for dX and dY: A prediction analysis has been conducted on Celestial poles dX and dY also but this chapter does not contain the results as no proper requirements were set by ESA for the celestial poles. The prediction results can be found in Appendix A.2 with the mAE values also.

Model Performance Evaluation: The evaluation of the LSTM and SARIMA models and their SSA-enhanced versions is done using MAE. This section offers a comprehensive review of the model's accuracy by comparing the projected EOP values with the real data.

In the end, the model selected based on the prediction accuracy and other factors will be used to predict the CONT14 values to finalize the prediction algorithm. This section will set the benchmark for the accuracy of the prediction algorithms in this study and the future studies that will take place.

6.2 Prediction of x_p

6.2.1 1 Day

This section involves an investigation of the performance of three different machine learning models - AR, SARIMA, and LSTM - in predicting the 1-day values of the x_p. The objective is to determine the optimal model using the MAE determine. The specified MAE threshold for our models is established at 0.3 mas by ESA. The SARIMA model had higher accuracy when compared with the other models, with a MAE of 0.07 mas. The AR model followed very closely with an MAE of 0.13. The LSTM model, on the other hand, exhibited a significantly higher MAE of 4.44 mas.

Comprehensive Model Analysis

- **AR** model: The Autoregressive (AR) model analyzes historical values for predicting future values in a time series. It is particularly effective when the data exhibits significant autocorrelation (Wang et al. (2022)). The AR model, used in our investigation, learning values from 1990-01-01 to 2022-12-31, with a MAE of 0.13 mas. This MAE is lower than the desired aim of 0.3 mas.

  A fundamental disadvantage of the AR model is its fundamental simplicity, which may fail to reflect the complex seasonal patterns present in the data. Still, the performance of the algorithm in our case is satisfactory, suggesting it's suitable for this specific prediction task.

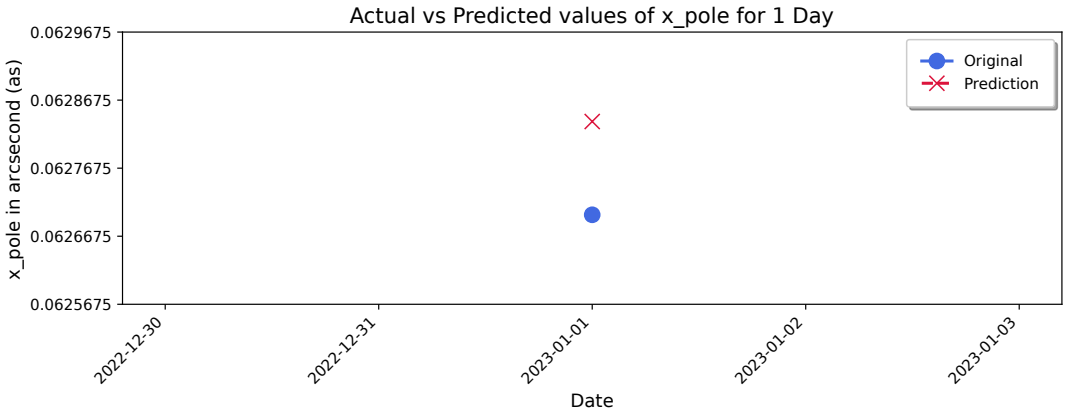


Figure 6.1: 1 day prediction using AR

- **SARIMA** model: The SARIMA model is a modification of the AR model which includes seasonality by using a SARIMA approach. This feature makes it highly effective for datasets that demonstrate both seasonal and non-seasonal patterns.

  The SARIMA model, with the parameters order=(0, 2, 4) and seasonal order=(0, 0, 0, 7), had the highest performance with a MAE of 0.07 mas. This demonstrates a high level of prediction ability, as it is good at managing the seasonal components of the x_p data.

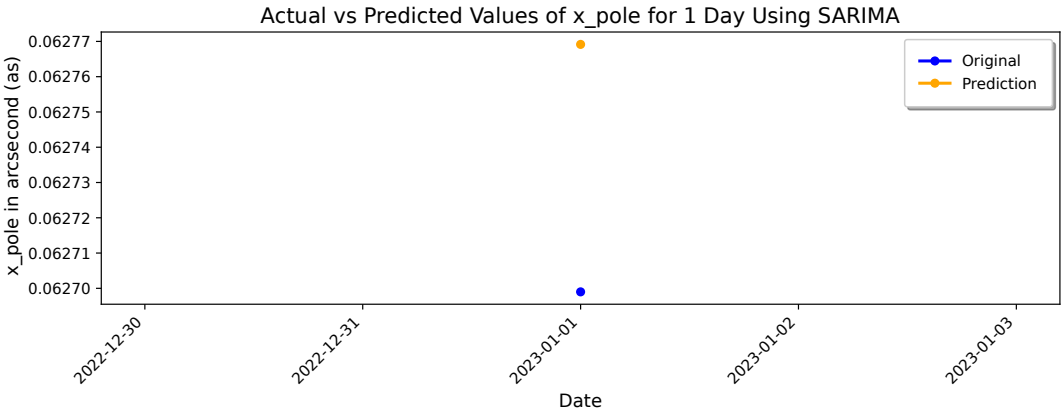


Figure 6.2: 1 Day prediction using SARIMA

- **LSTM** model: LSTM networks, a type of RNNs, are efficient at capturing temporal dependencies in sequence prediction problems.

  Although LSTM has advanced features in processing sequential data, it showed a significant mean absolute error (MAE) of 4.44 mas in the analysis. The underperformance may be explained by causes such as overfitting or the model's complexity not aligning well with the data features for very short-term prediction horizons.

  The LSTM model's configuration and training process, including the use of a MinMaxScaler, a look-back period of 10 days, and various layers with dropout, may have contributed to its less-than-ideal performance.

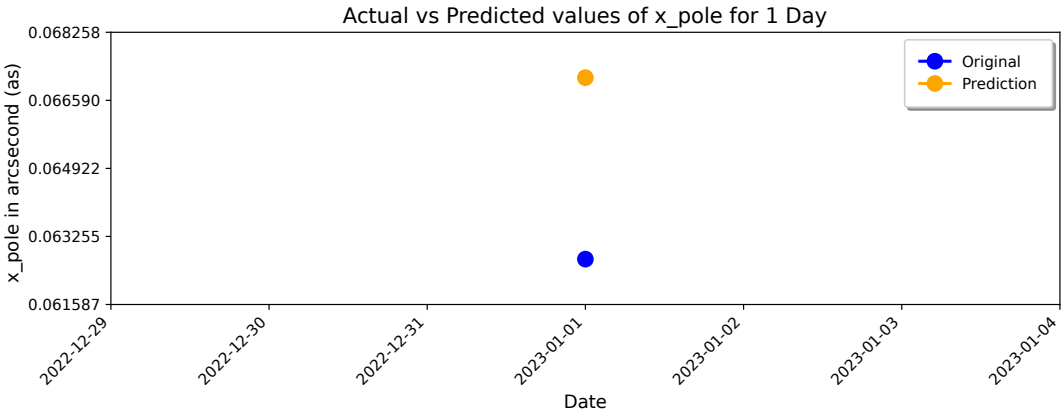


Figure 6.3: 1 Day prediction using LSTM

The SARIMA model, with the lowest Mean Absolute Error (MAE), is considered the most appropriate for predicting the x_p's value one day in advance. The AR model is a close second in terms of suitability. Although the LSTM model is advanced, it did not meet objectives and might need additional adjustments or a reassessment of its suitability for this specific prediction goal.

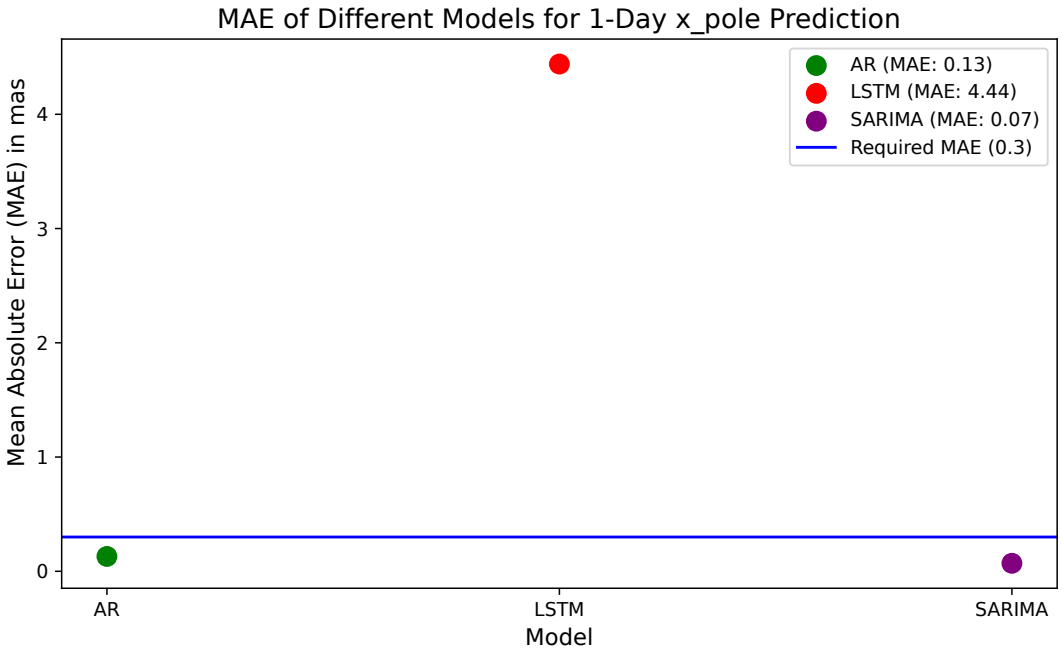


Figure 6.4: Comparison of 1 Day prediction for different models

6.2.2 5 Day

Analysis of 5-Day Predictions: This section of the analysis demonstrates the accuracy of the LSTM and SARIMA models in predicting the x_p variable over a 5-day period. The analysis employs the Mean Absolute Error (MAE) as a metric to assess the prediction precision of various models.

Configuration of the Model and Analysis of the Data The LSTM, a variant of recurrent neural network, is suitable for time series data because of its capacity to retain long-term dependencies. SARIMA, an upgraded version of AR, integrates seasonality into the model, making it an effective tool for predicting time series data.

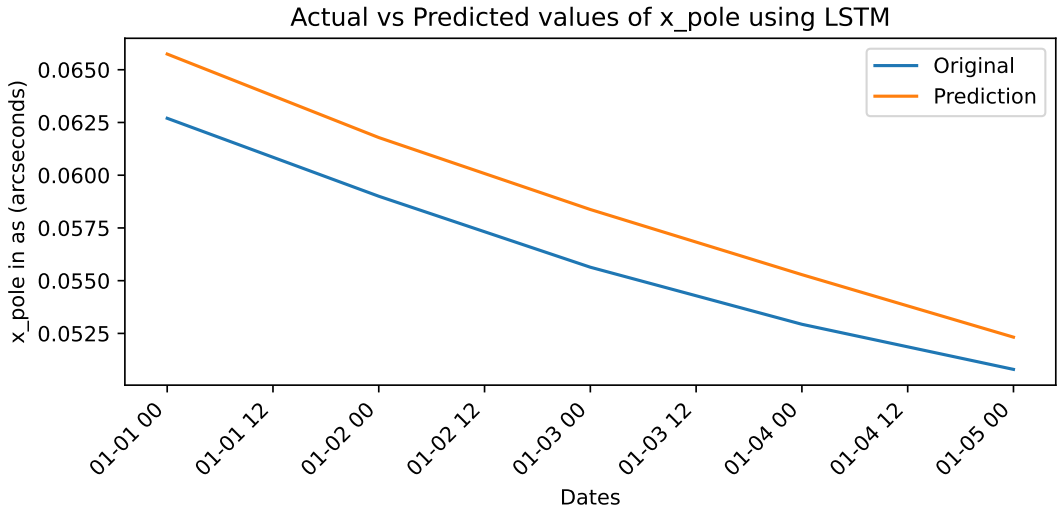


Figure 6.5: 5 Day Prediction using LSTM

The selection of these models over a 5-day timeframe is based on their ability to accurately represent complex temporal patterns and interconnections within the data.

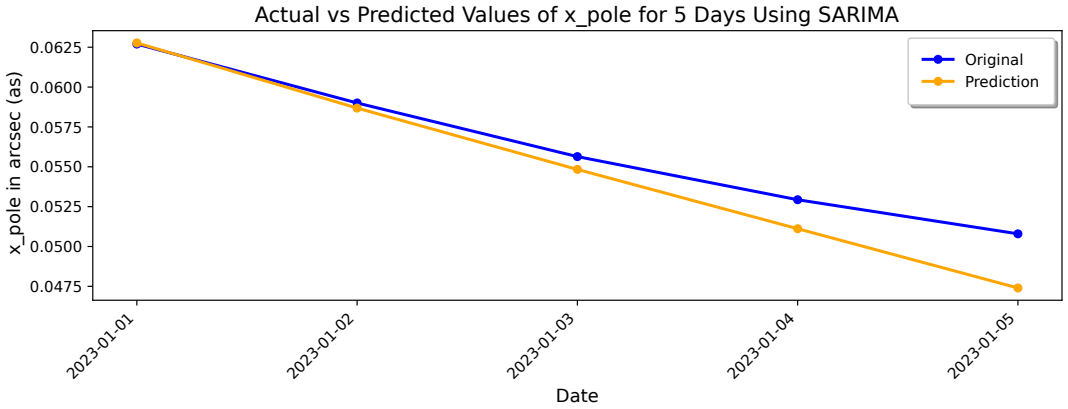


Figure 6.6: 5 Day prediction using SARIMA

Performance Metrics: The LSTM model achieved a Mean Absolute Error (MAE) of 2.16 mas, which is the average prediction error over the 5-day period. In comparison, the SARIMA model showed an improved Mean Absolute Error (MAE) of 1.28 mas, indicating a greater level of accuracy in capturing the trends and patterns of the x_p variable. The predefined benchmark Mean Absolute Error (MAE) for this analysis was established by ESA at 1.8 mas.

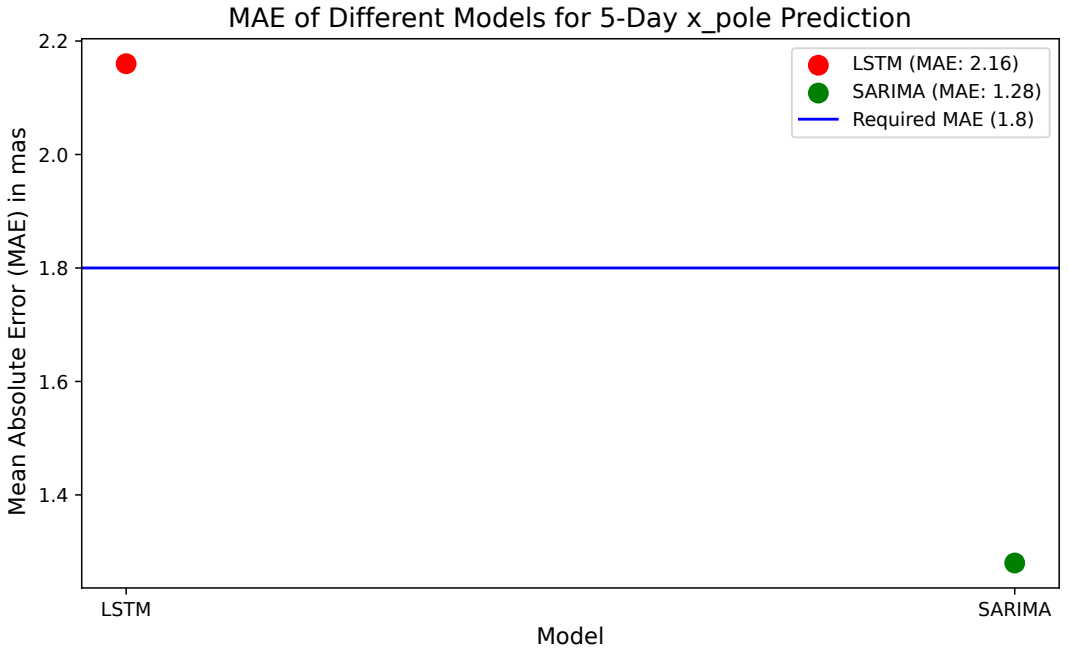


Figure 6.7: Comparison of 5 Day prediction for different models

6.2.3 10 Day

Performance Metrics:The LSTM model achieved a Mean Absolute Error (MAE) of 2.02 mas, demonstrating an average accuracy in predicting values over a 10-day period.

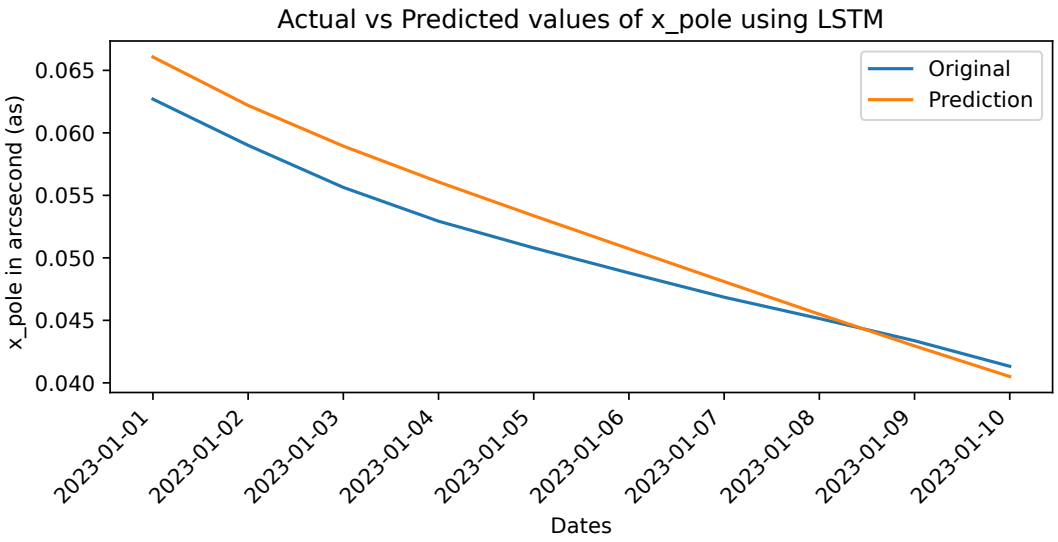


Figure 6.8: LSTM 10-day prediction

The LSTM+SSA model got an even larger Mean Absolute Error (MAE) of 17.70 mas, signifying significant inaccuracies in its predictions.

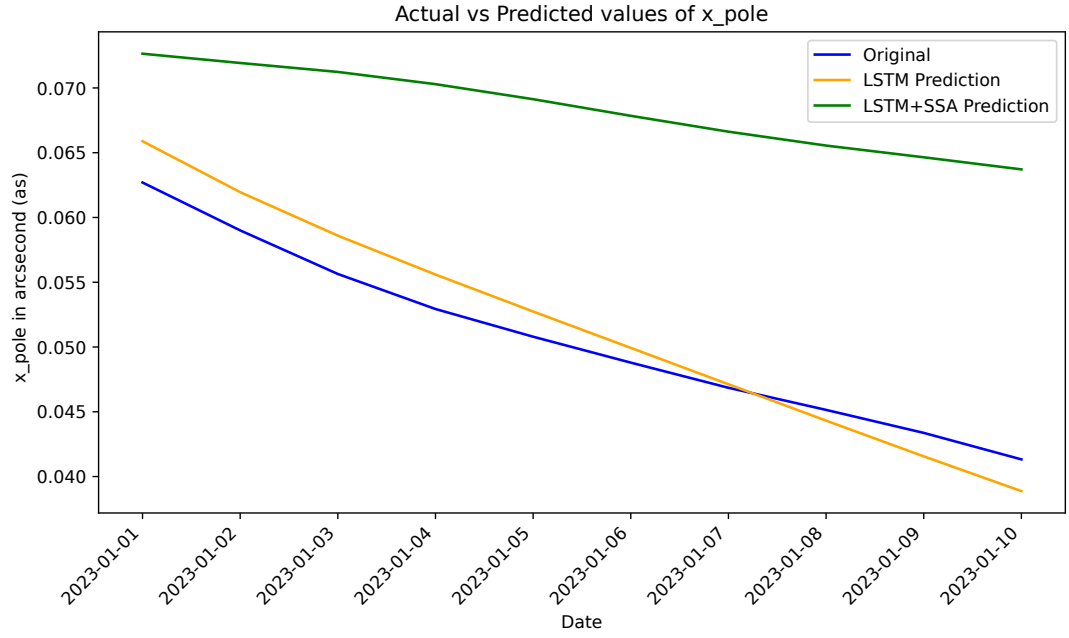


Figure 6.9: LSTM with SSA combined 10-day prediction

The SARIMA model, with a Mean Absolute Error (MAE) of 5.06 mas also exhibited significant divergence from the actual data but outperformed the LSTM+SSA model.

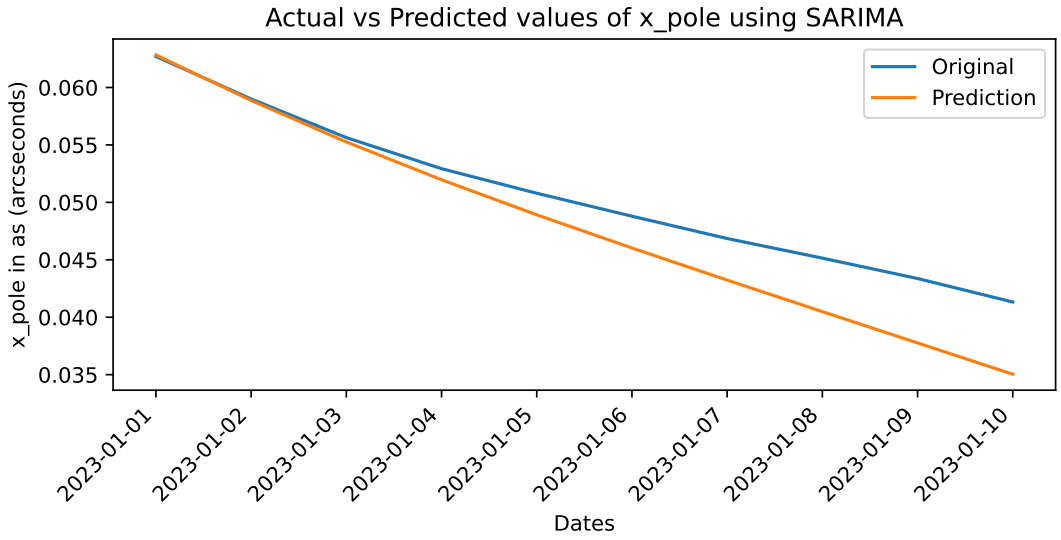


Figure 6.10: 10 Day prediction using SARIMA

The threshold for acceptable prediction accuracy was established at a Mean Absolute Error (MAE) of 3.5 mas by ESA. Out of all the models, only LSTM manages to meet this criterion, suggesting a reasonable level of prediction accuracy.

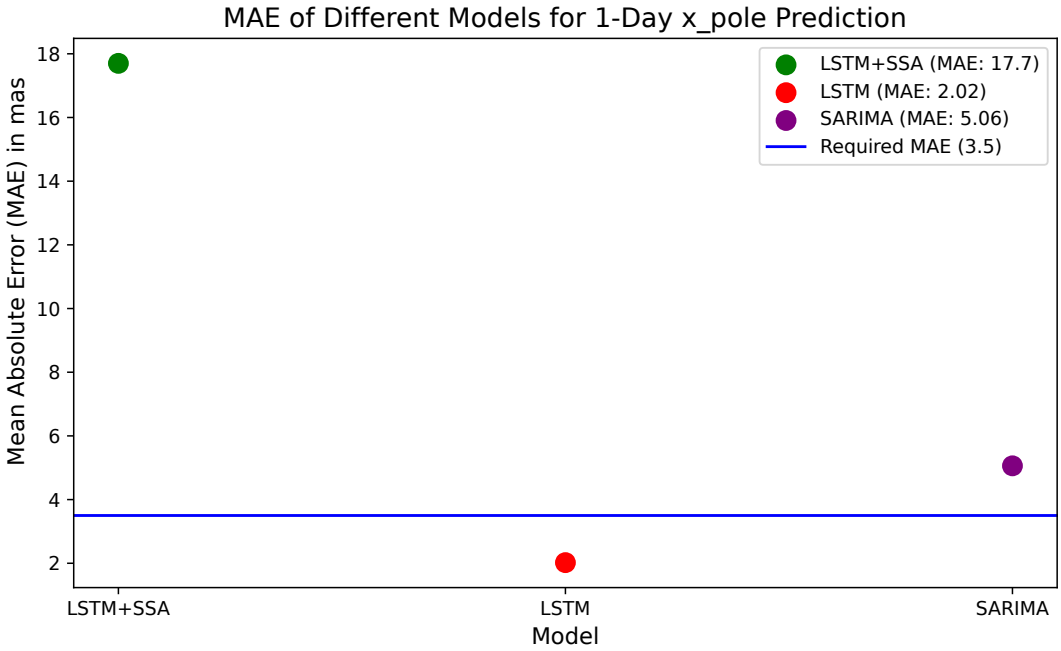


Figure 6.11: Comparison of 10 Day prediction for different models

The elevated Mean Absolute Error (MAE) observed for the LSTM+SSA model indicates that the integration of Singular Spectrum Analysis (SSA) may not have yielded advantageous results for this specific forecasting interval. Possible causes for this issue include overfitting, incorrect configuration, or the inherent properties of the x_p data.

6.2.4 20 Day

Performance Metrics: The LSTM model achieved a MAE of 2.55 mas, which indicates its high predicted accuracy during the 20-day timeframe.

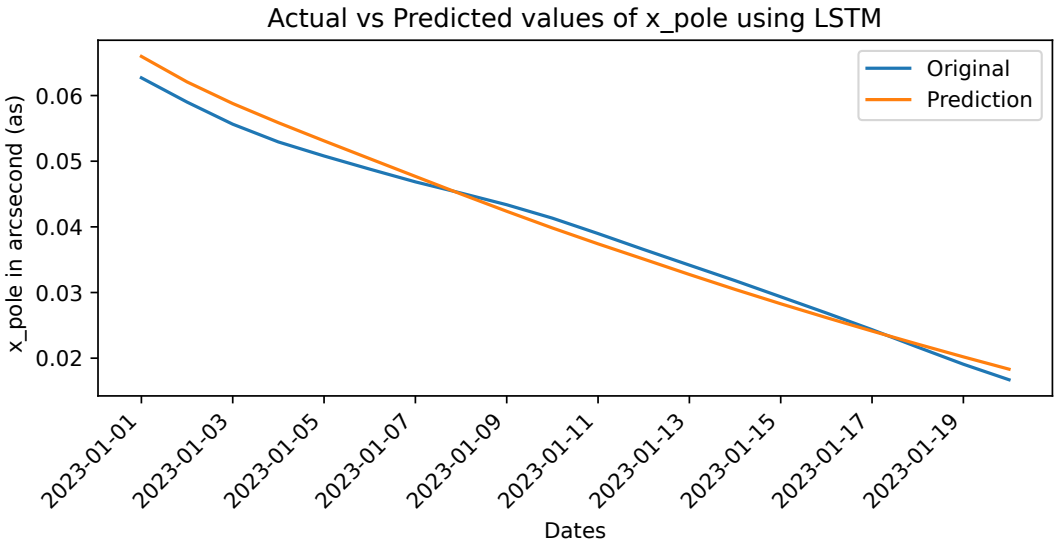


Figure 6.12: 20 Day prediction using LSTM

The LSTM+SSA model, on the other hand, showed a significantly increased MAE of 19.56 mas, indicating significant errors in its predictions.

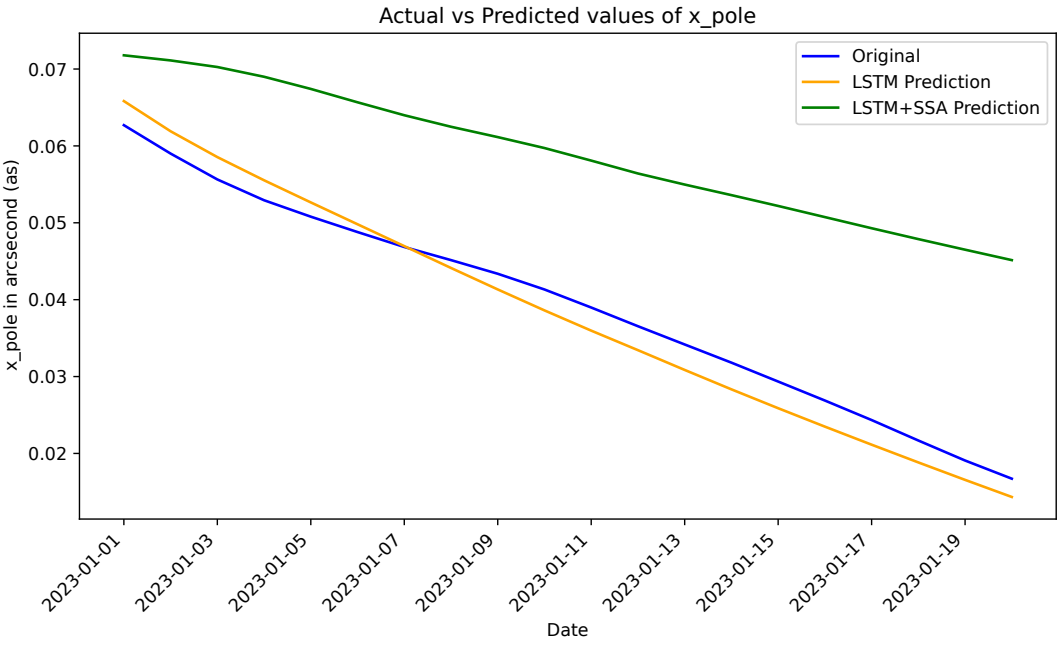


Figure 6.13: 20 Day prediction using LSTM and SSA

The SARIMA model showed a Mean Absolute Error (MAE) of 5.2 mas, indicating a moderate level of accuracy but with a considerable deviation from the actual values.

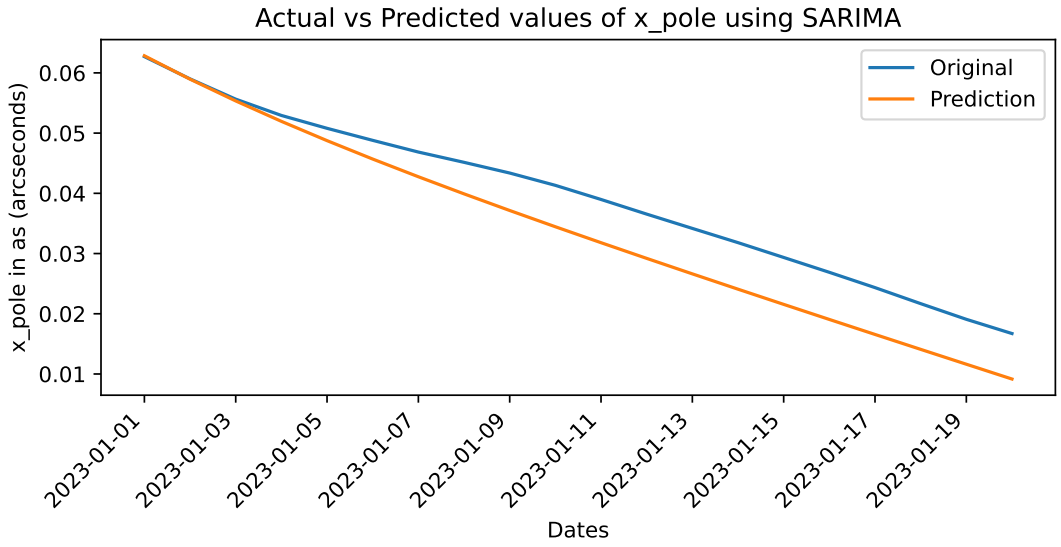


Figure 6.14: 20 Day prediction using SARIMA

The benchmark Mean Absolute Error (MAE) for this timeframe was established at 6.8mas by ESA.

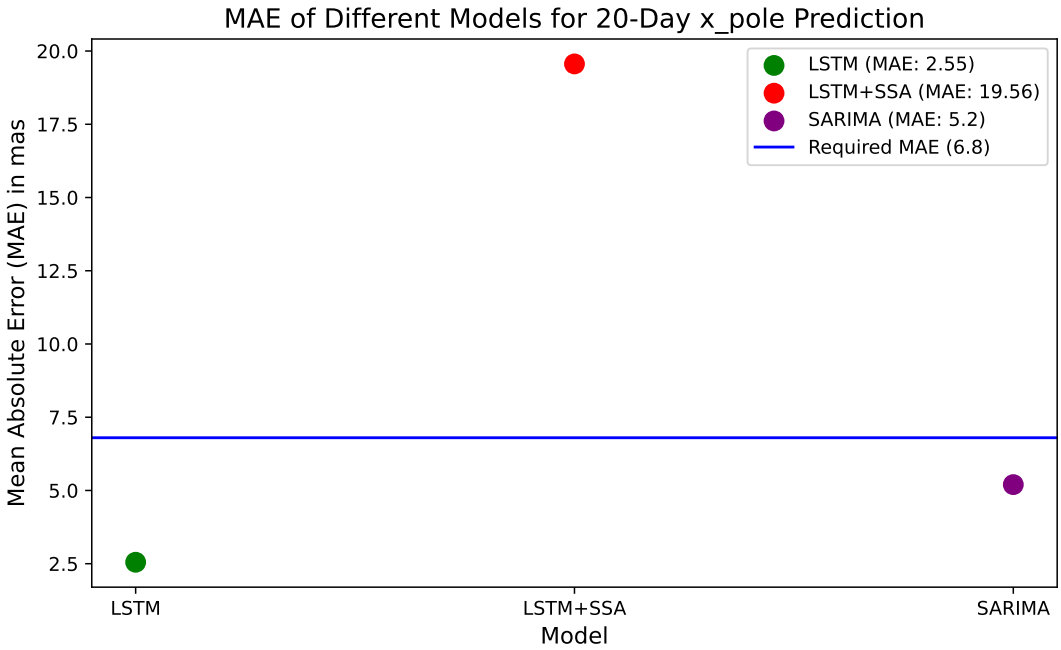


Figure 6.15: Comparison of 20 Day prediction for different models

The increased Mean Absolute Error (MAE) observed for the LSTM+SSA model indicates the integration of Singular Spectrum Analysis (SSA) did not result in the expected improvement in predicting precision during the 20-day timeframe. The potential causes for this issue include overfitting or the model's limited capacity to handle the complexities of the data over an extended period of time.

The performance of SARIMA, meeting the acceptable level, indicates its capability to capture longer-term seasonal patterns, but with limitations in accuracy.

The performance of LSTM, which has the lowest Mean Absolute Error (MAE) compared to the other two models, highlights its ability to successfully manage long-term forecasts.

6.2.5 40 Day

Model Configuration and Data Analysis: For the 40 days, LSTM is used to manage long-term dependencies in time series data. Additionally, LSTM+SSA, a hybrid model aimed at improving forecasting accuracy by decomposing the time series into its individual components was implemented.

Performance Metrics: The LSTM model demonstrated a Mean Absolute Error (MAE) of 5.48 mas, indicating it was effective in predicting over an extended 40-day timeframe.

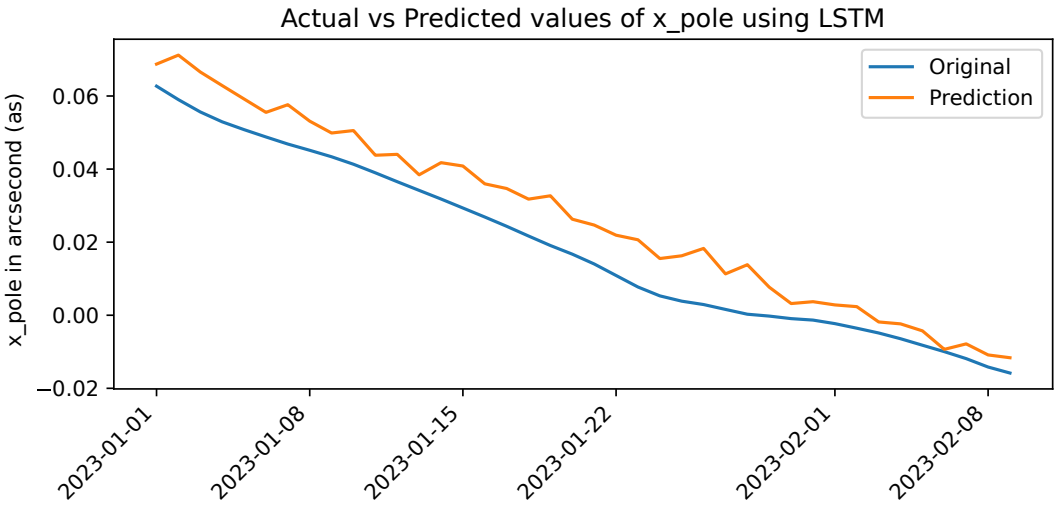


Figure 6.16: 40 Day prediction using LSTM

The LSTM+SSA model, on the other hand, had a significantly higher Mean Absolute Error (MAE) of 25.32 mas, suggesting significant errors in its predictive capabilities.

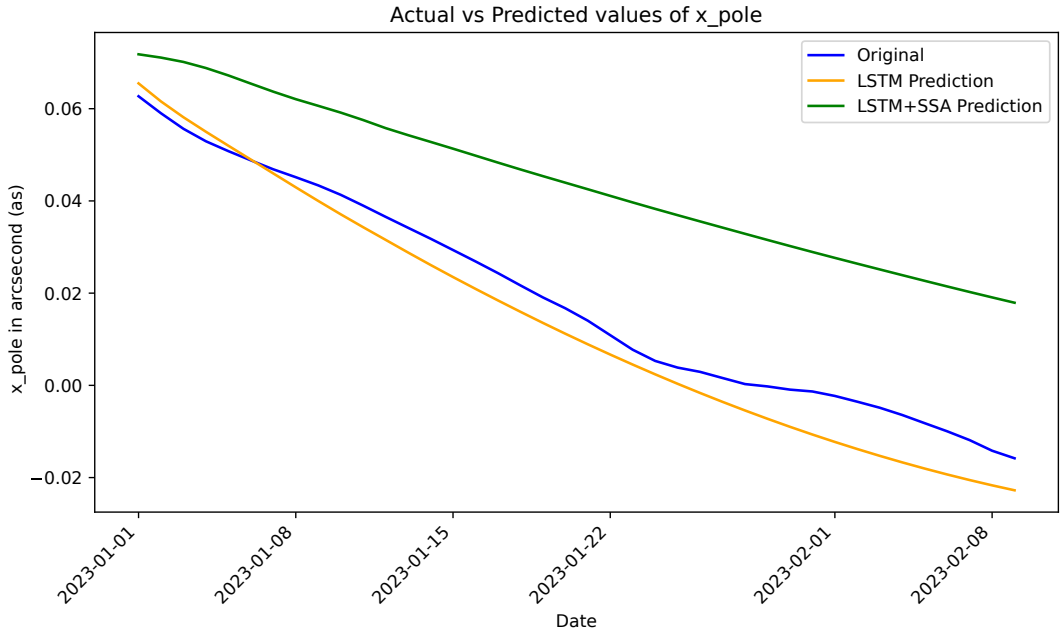


Figure 6.17: 40 Day prediction using LSTM and SSA

The SARIMA model showed a Mean Absolute Error (MAE) of 12.4 mas, indicating a moderate level of accuracy but with a considerable deviation from the actual values.

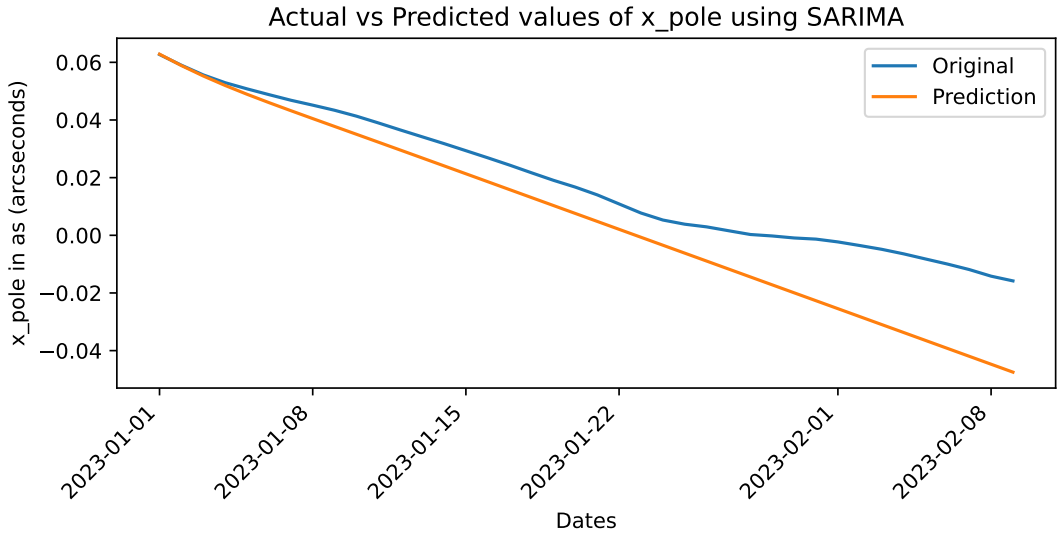


Figure 6.18: 40 Day prediction using SARIMA

The standard for acceptable accuracy was established with a Mean Absolute Error (MAE) of 12.8 mas by ESA was used.

6.2.6 90 Day

Performance Metrics: The LSTM model achieved a Mean Absolute Error (MAE) of 4.90 mas, indicating an acceptable degree of accuracy, especially with the complexity

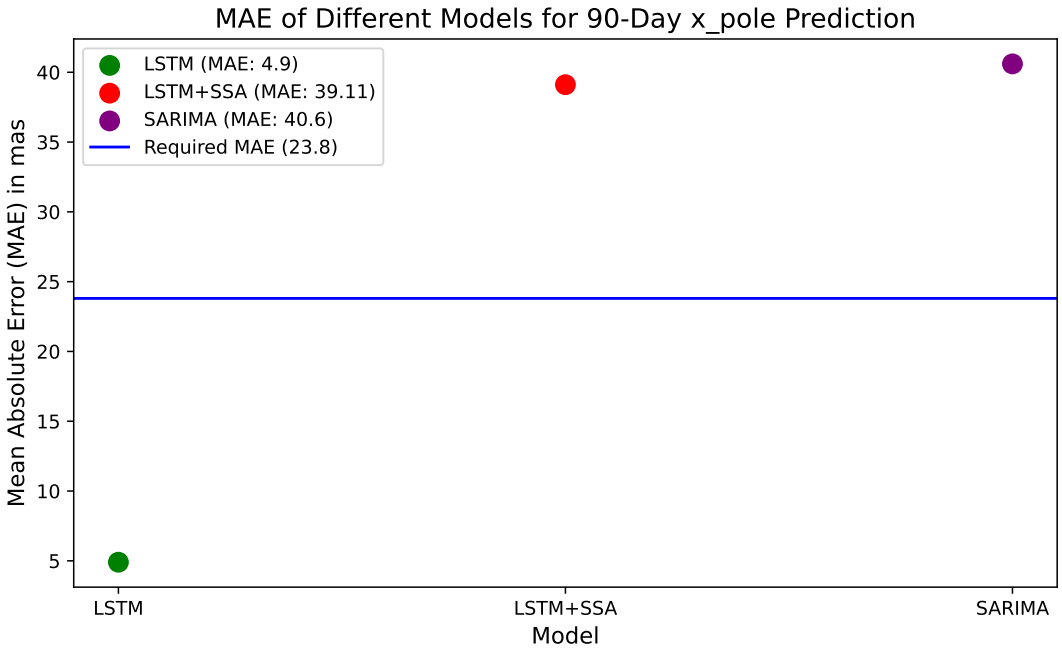


Figure 6.19: Comparison of 40 Day prediction for different models

and duration of the prediction period. The LSTM+SSA model, in clear contrast, showed a Mean Absolute Error (MAE) of 39.11 mas, suggesting significant inaccuracy and prediction errors.

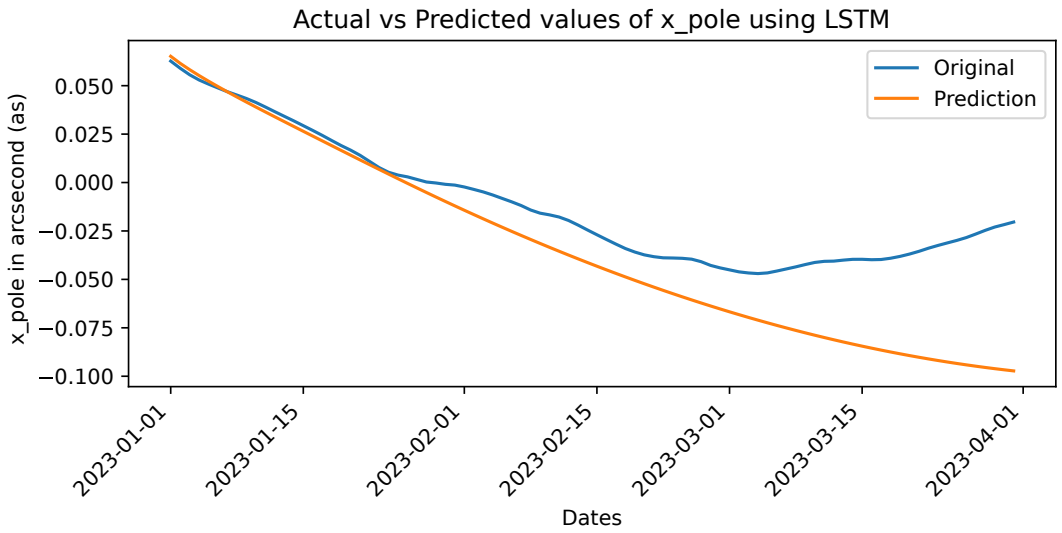


Figure 6.20: 90 Day prediction using LSTM

Result Discussion: The LSTM model shows a reasonably low Mean Absolute Error (MAE) in the 90-day forecast, indicating its robustness and suitability for producing long-term predictions. The model effectively incorporates long-term dependencies while avoiding the problem of overfitting. The LSTM+SSA model's performance raises significant issues regarding its suitability for predicting over extended periods.

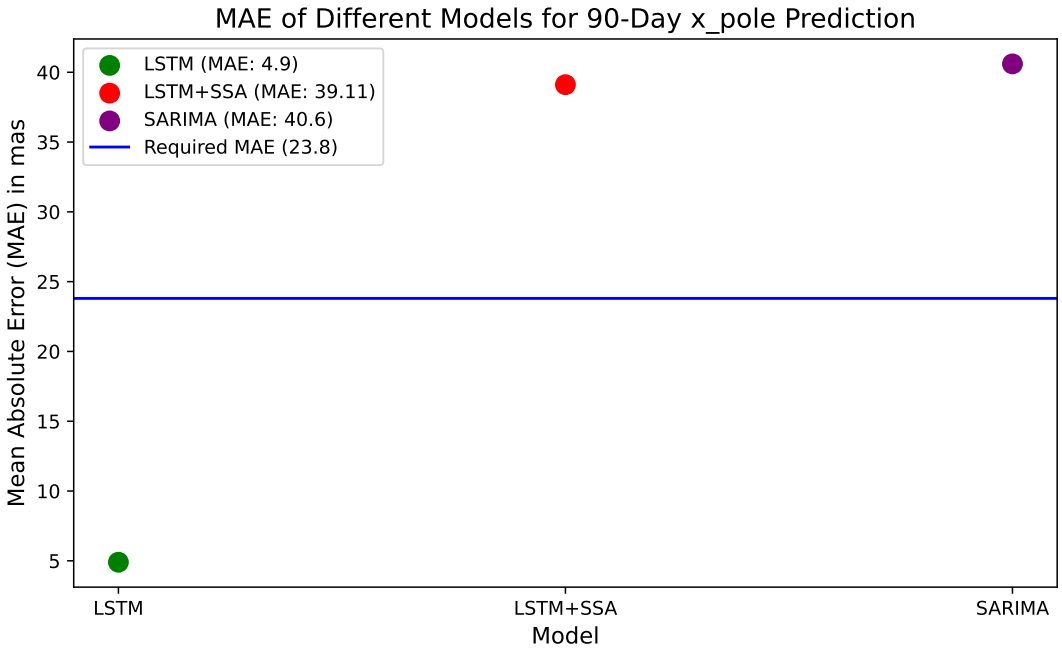


Figure 6.22: Comparison of 90 Day prediction for different models

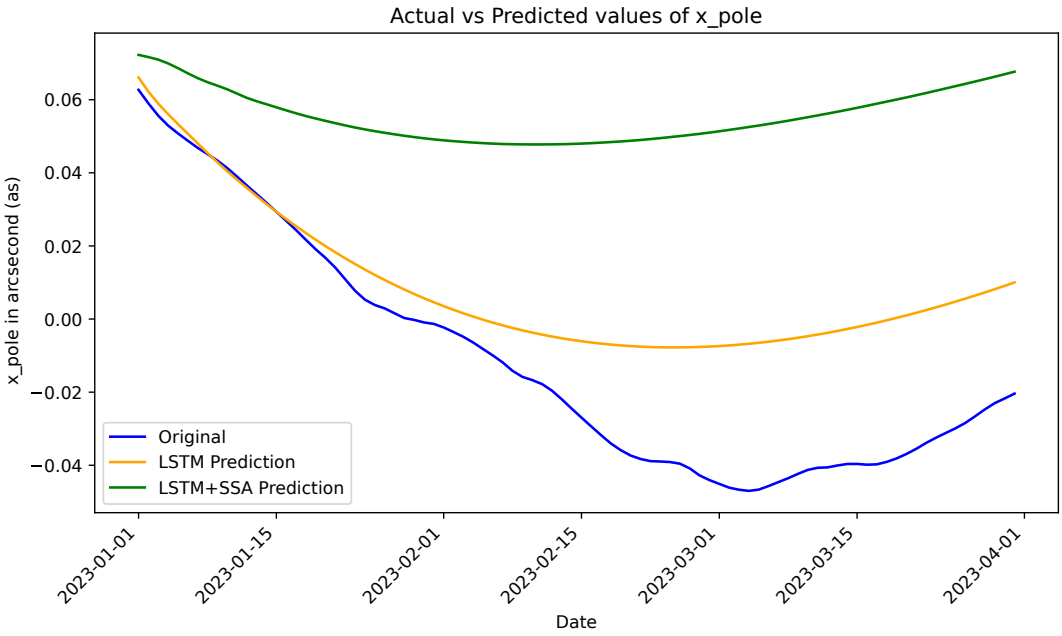


Figure 6.21: 90 Day prediction using LSTM and SSA

The accepted standard for accuracy in predictions during this time frame was a Mean Absolute Error (MAE) of 23.8 mas set by ESA.

6.3 Prediction of y_p

6.3.1 1 Day

This section evaluates the performance of AR, SARIMA, and LSTM models in predicting 1-day values for the y_p. We use the Mean Absolute Error (MAE) as the key metric, with an ESA-established MAE threshold of 0.23 mas.

- **AR Model** Achieved an MAE of 0.182 mas, indicating satisfactory predictive accuracy and slightly below the threshold.

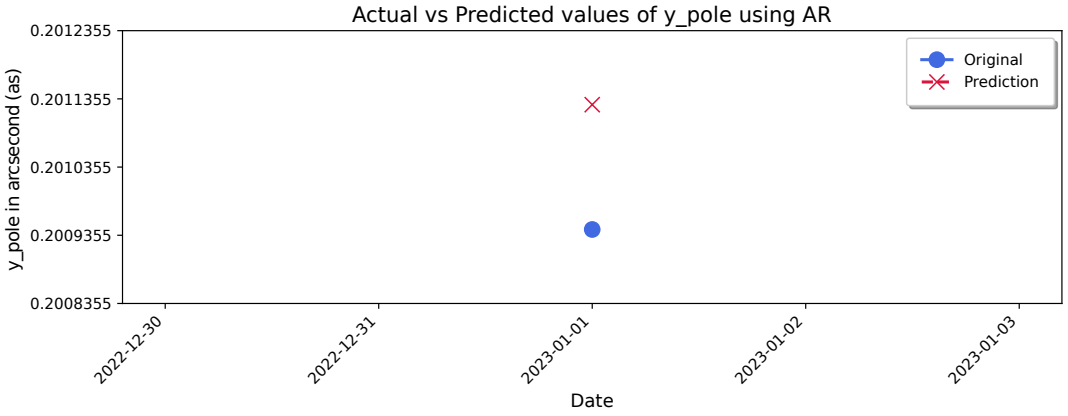


Figure 6.23: 1 Day prediction using AR

- **SARIMA Model** Exceptionally high accuracy with an MAE of 0.03 mas, far surpassing the established threshold.

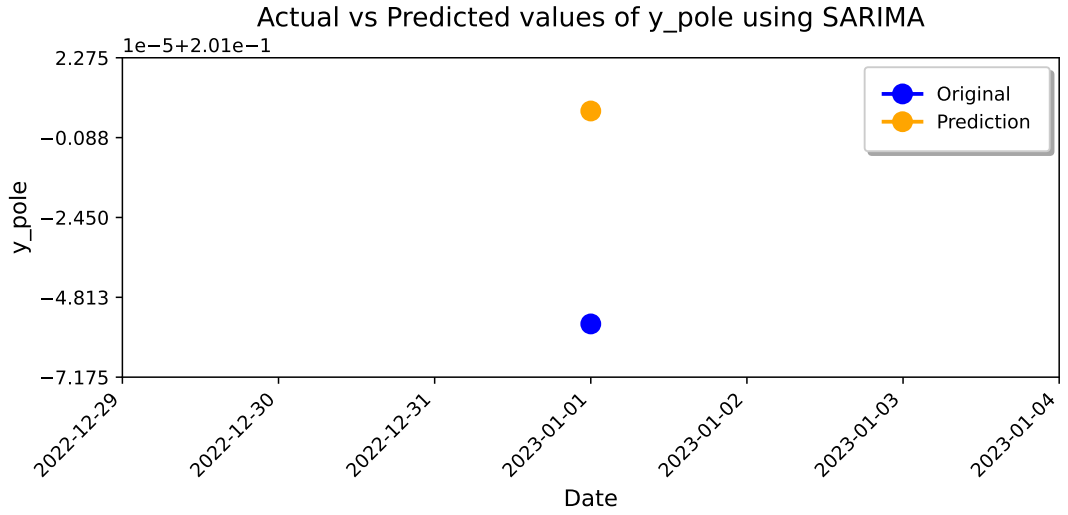


Figure 6.24: 1 Day Prediction using SARIMA

- **LSTM Model** Exhibited a significantly higher MAE of 1.26 mas, indicating lower prediction accuracy for the 1-day interval.

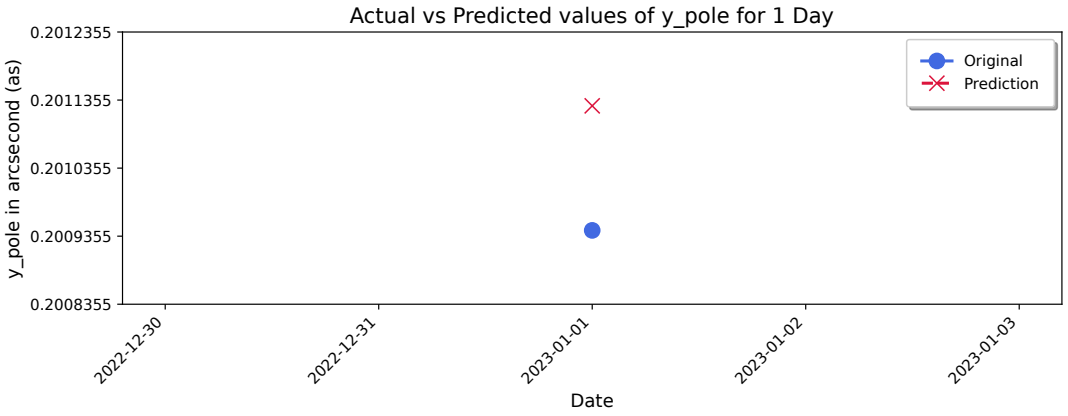


Figure 6.25: 1 Day prediction using LSTM

The accepted standard for accuracy in predictions during this time frame was a Mean Absolute Error (MAE) of 0.23 mas set by ESA.

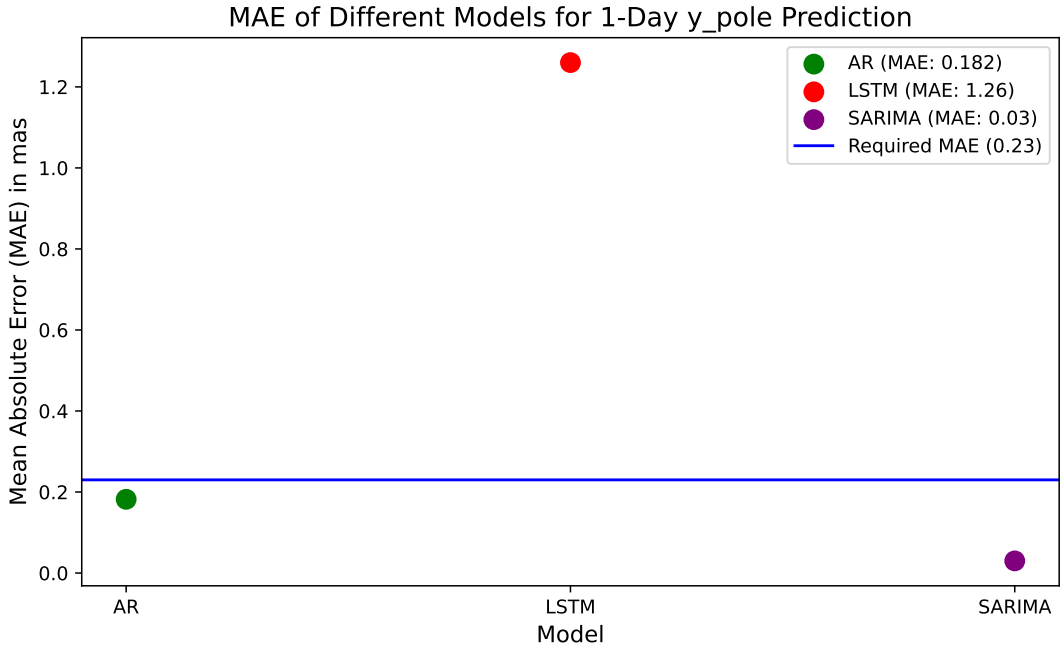


Figure 6.26: Comparison of 1 Day prediction for different models

6.3.2 5 Day

This analysis covers the accuracy of LSTM and SARIMA models over a 5-day prediction interval.

- **LSTM Model** Recorded an MAE of 0.38 mas, indicating moderate accuracy but not within the ESA benchmark of 1.3 mas.

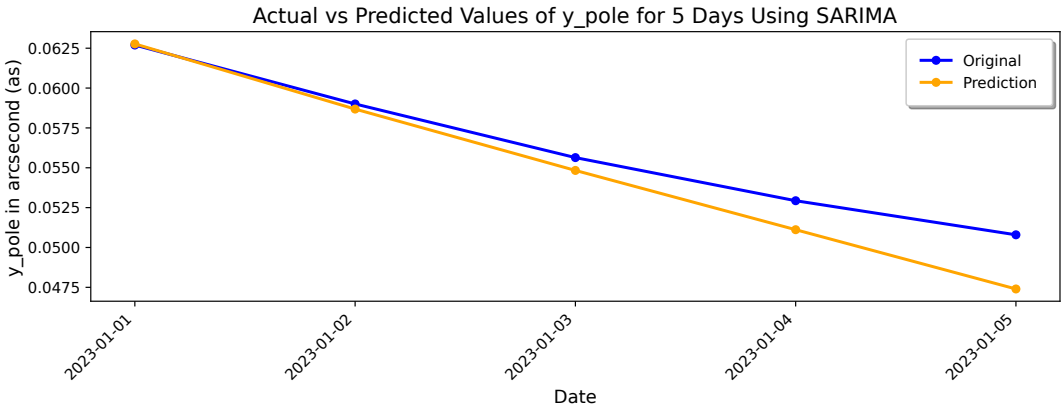


Figure 6.27: 5 Day prediction using LSTM

- **SARIMA Model** Demonstrated superior accuracy with an MAE of 0.078 mas, well below the benchmark.

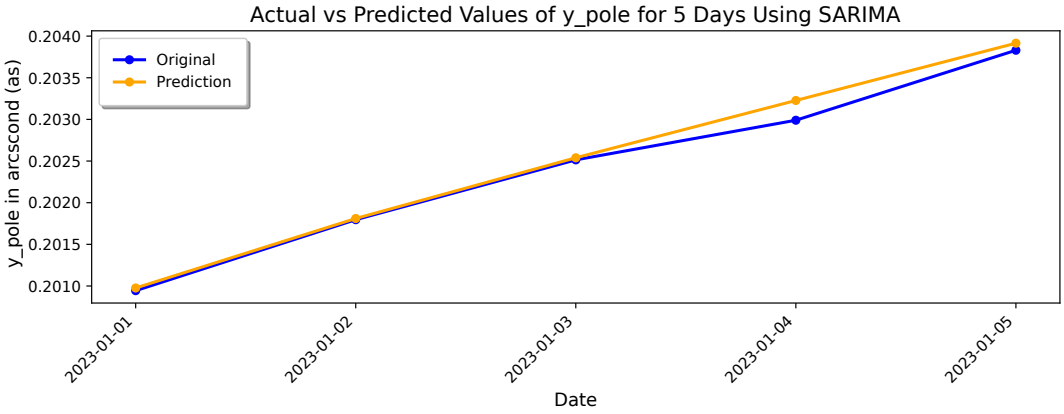


Figure 6.28: 5 Day prediction using SARIMA

6.3.3 10 Day

This analysis covers the accuracy of LSTM and LSTM with SSA models over a 10-day prediction interval.

- **LSTM Model** Reported an MAE of 1.30 mas, which is just above the acceptable threshold of 1.99 mas.

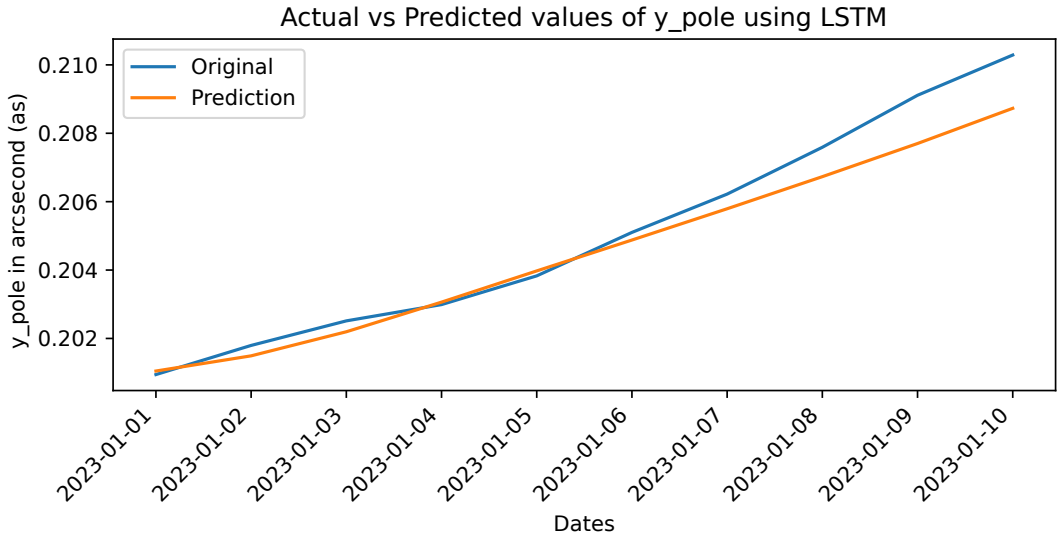


Figure 6.29: 10 Day prediction using LSTM

- **LSTM with SSA Model** A slightly higher MAE of 1.57 mas, indicating comparable but slightly less accurate predictions.

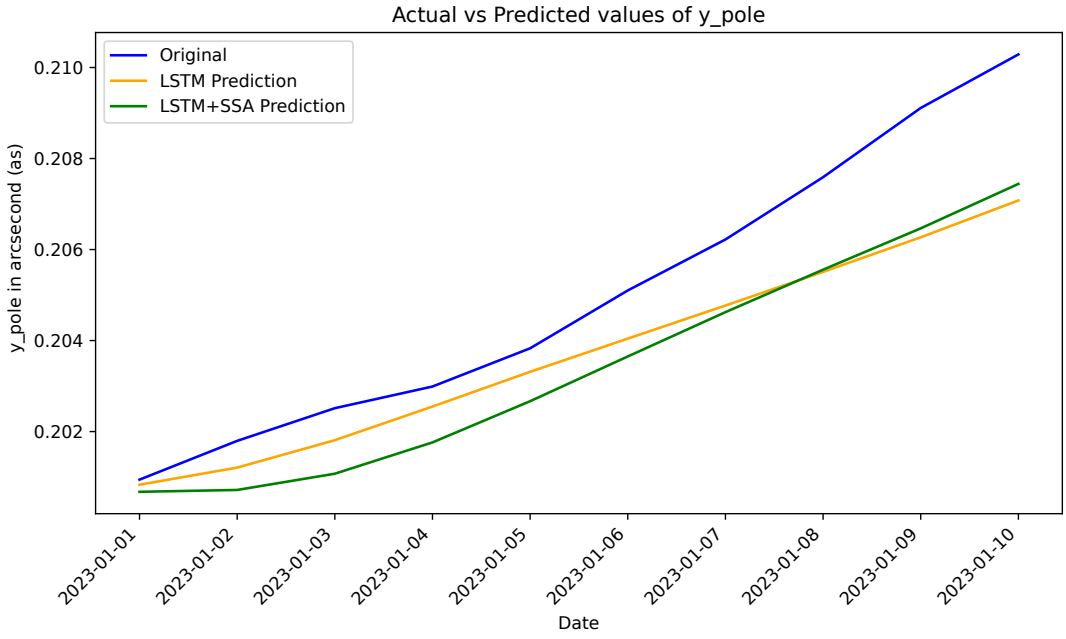


Figure 6.30: 10 Day prediction using LSTM

- **SARIMA Model** Demonstrated superior accuracy with an MAE of 0.49 mas, well below the requirement.

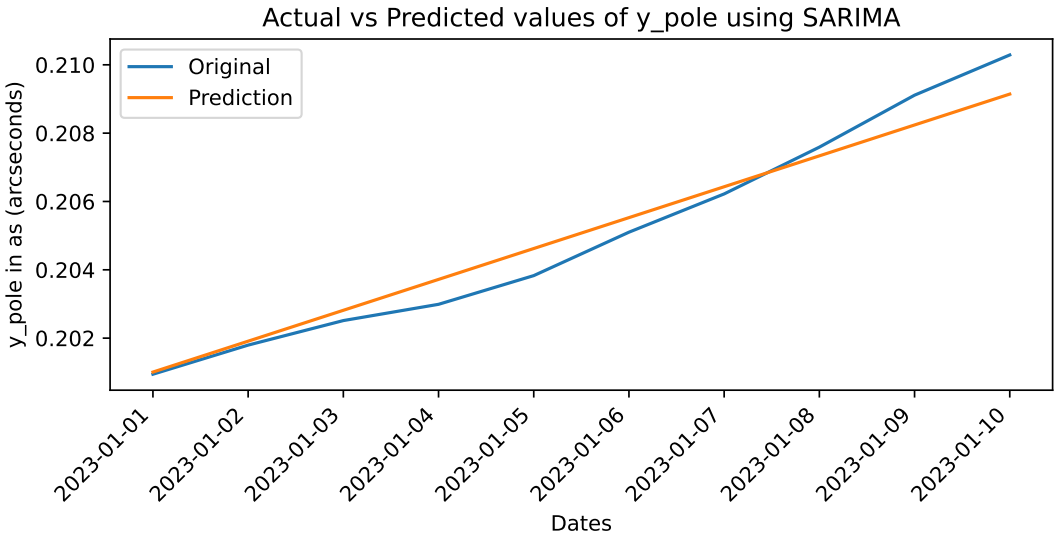


Figure 6.31: 10 Day prediction using SARIMA

6.3.4 20 Day

This analysis covers the accuracy of LSTM and LSTM with SSA models over a 20-day prediction interval.

- **LSTM Model** Achieved a reasonable MAE of 0.71 mas, significantly below the ESA threshold of 2.7 mas

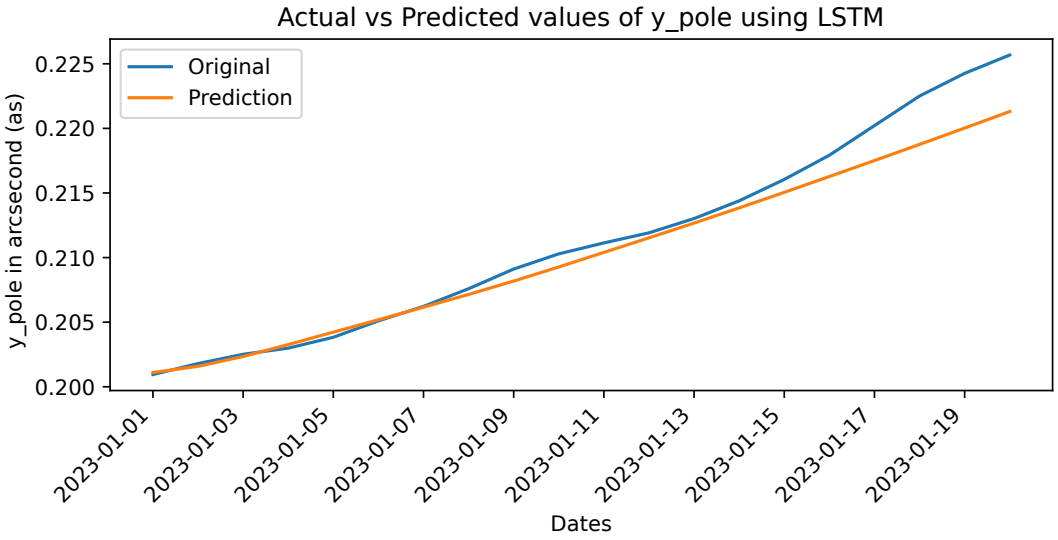


Figure 6.32: 20 Day prediction using LSTM

- **LSTM with SSA Model** Demonstrated a higher MAE of 4.18 mas, exceeding the acceptable threshold.

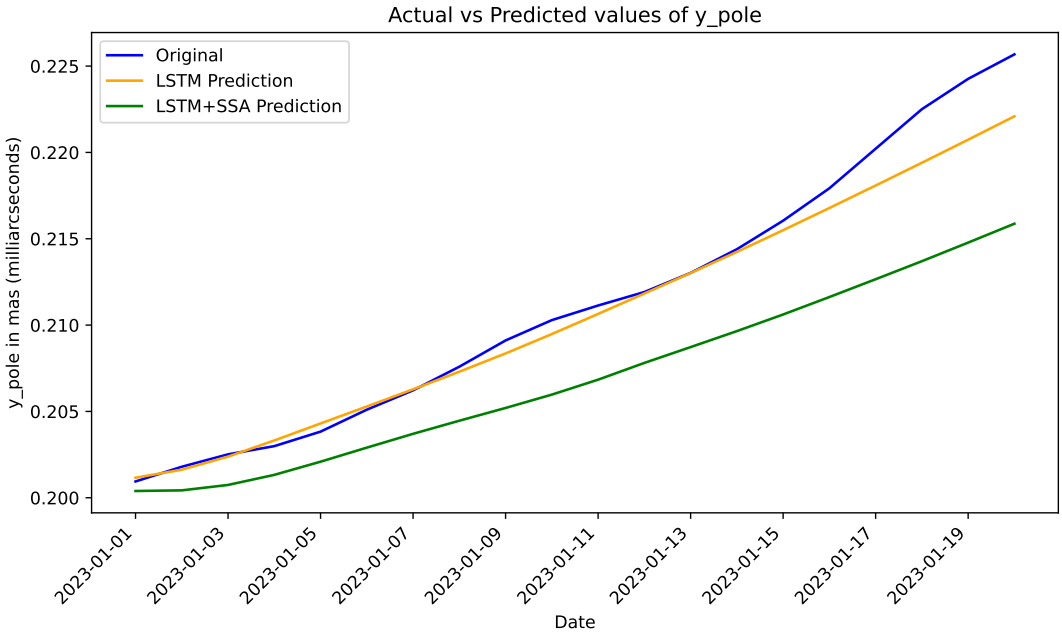


Figure 6.33: 20 Day prediction using LSTM+SSA

6.3.5 40 Day

This analysis covers the accuracy of LSTM and LSTM with SSA models over a 20-day prediction interval.

- **LSTM Model**: Recorded an MAE of 1.26 mas, which is within the acceptable range of 4.1 mas set by ESA.
- **LSTM with SSA Model**: Showed a significantly higher MAE of 10.5 mas, far exceeding the benchmark.

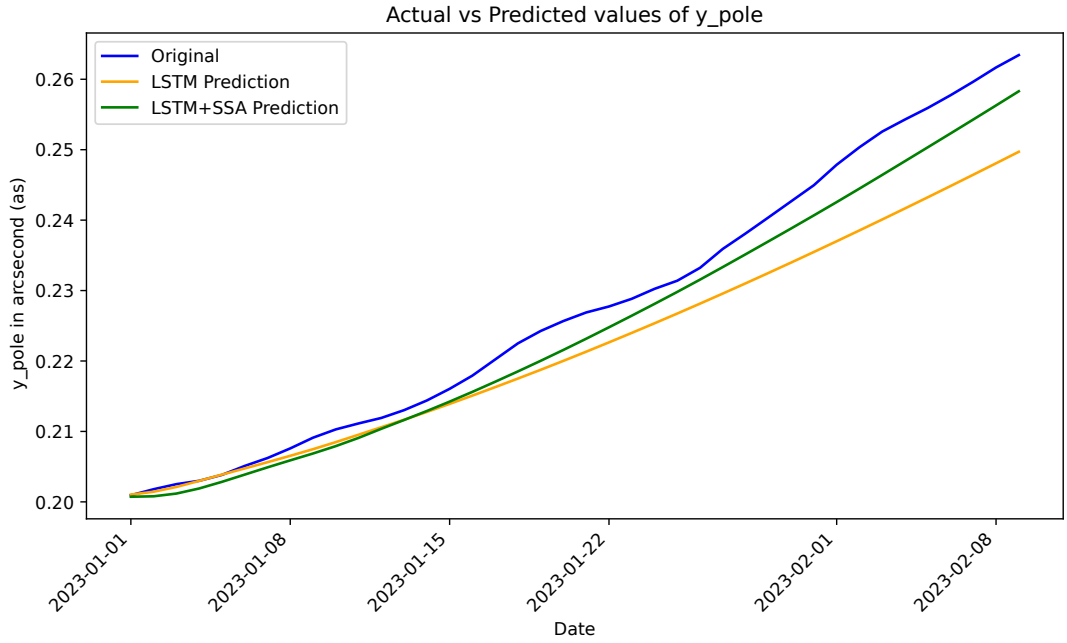


Figure 6.34: 40 Day prediction using LSTM vs LSTM + SSA

6.3.6 90 Day

This analysis covers the accuracy of LSTM and LSTM with SSA models over a 20-day prediction interval. LSTM Model exhibited an MAE of 8.02 mas, which, although high, is below the ESA benchmark of 16.5 mas. The LSTM with SSA Model reported an MAE of 8.34 mas, slightly higher than LSTM but still within the acceptable range.

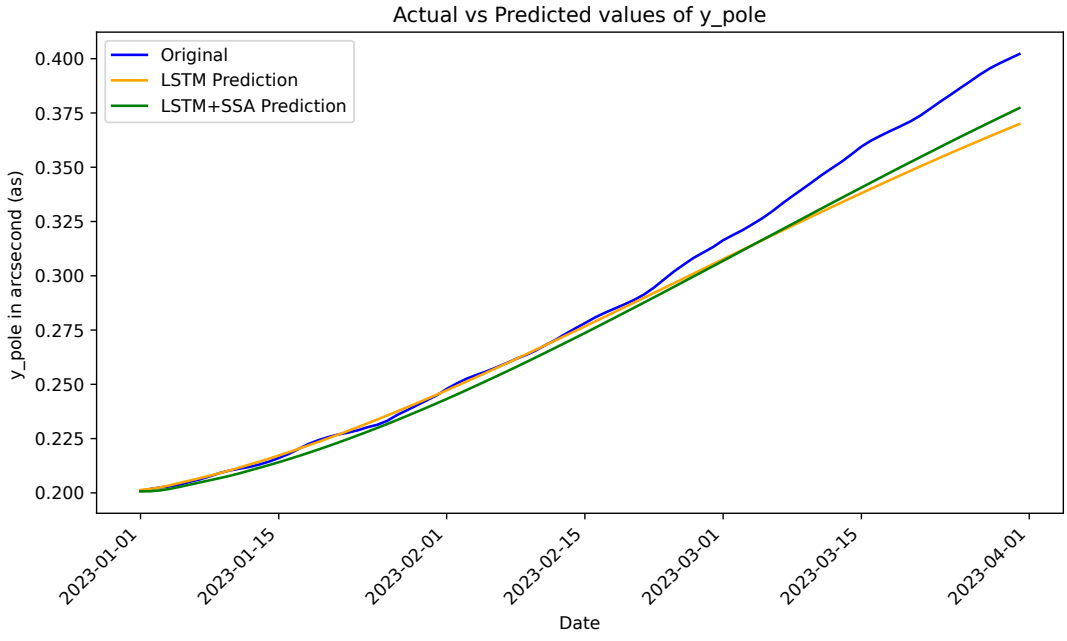


Figure 6.35: 90 day prediction using LSTM and LSTM+SSA

The accepted standard for accuracy in predictions during this time frame was a Mean Absolute Error (MAE) of 16.5 mas set by ESA. SARIMA model exhibited a high error margin of 45.0 mas.

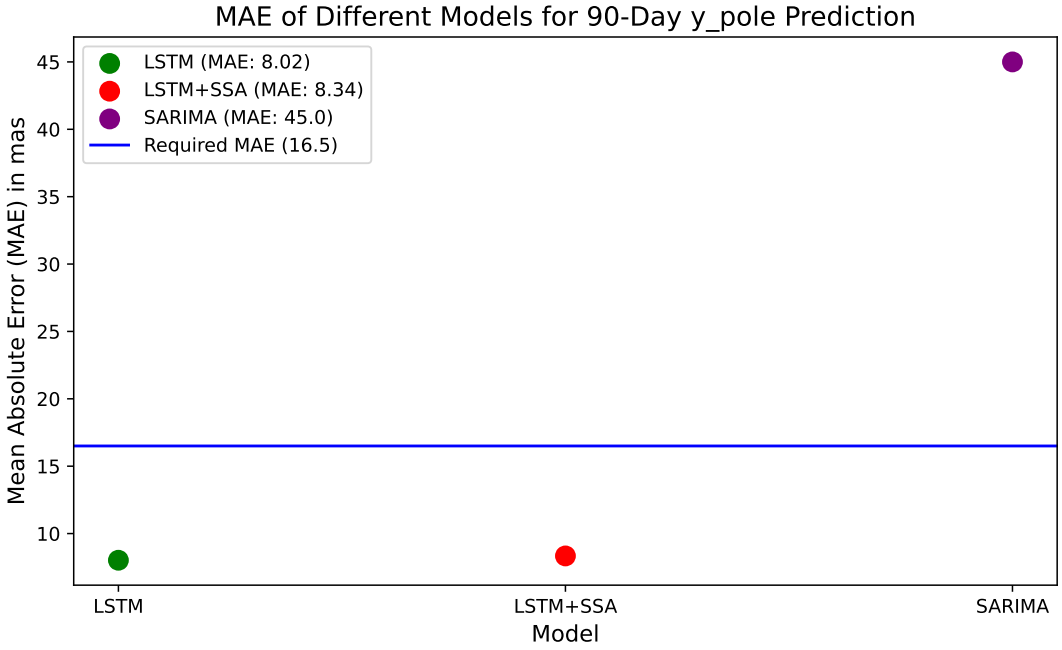


Figure 6.36: Comparison of model for 90 Days

6.4 UT1-UTC prediction

6.4.1 1 Day

This section evaluates the LSTM model's performance in predicting 1-day values for UT1-UTC. The Mean Absolute Error (MAE) is the metric used, with a specified requirement of 0.06 ms. LSTM Model achieved an MAE of 0.11 ms, which exceeds the required threshold of 0.06 ms, indicating a need for model refinement to enhance accuracy for 1-day forecasts.

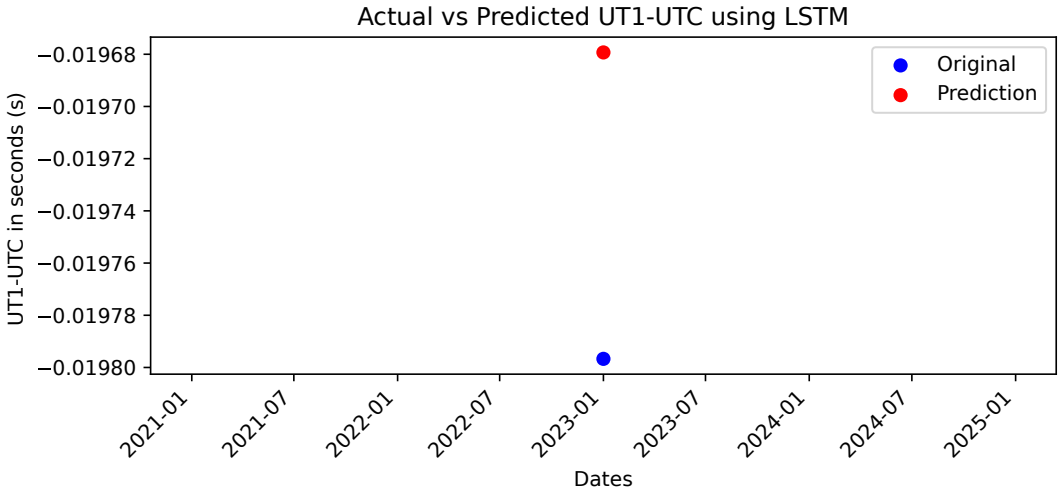


Figure 6.37: 1 Day prediction using LSTM

6.4.2 5 Day

This analysis focuses on the LSTM model over a 5-day prediction interval.

LSTM Model recorded an MAE of 0.29 ms, slightly above the ESA benchmark of 0.215 ms, indicating moderate but not optimal performance.

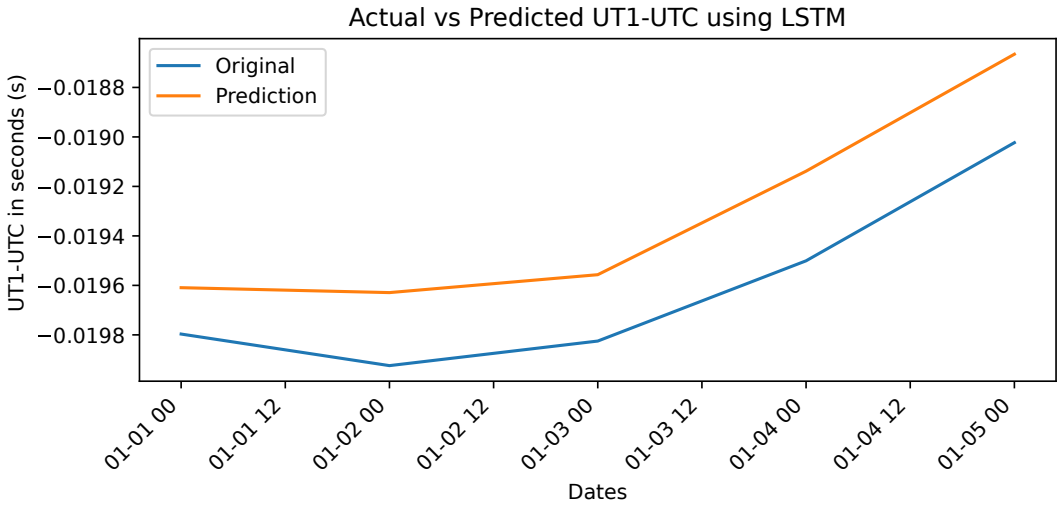


Figure 6.38: 5 day prediction using LSTM

6.4.3 10 Day

LSTM Model reported an MAE of 0.29 ms, which is below the acceptable threshold of 0.53 ms, indicating a satisfactory level of accuracy for the 10-day forecast.

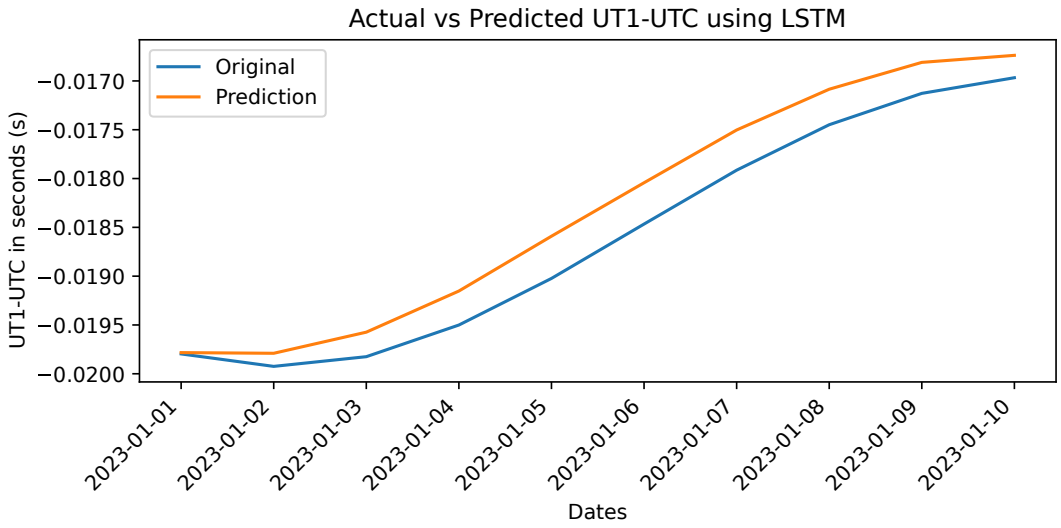


Figure 6.39: 5 day prediction using LSTM

6.4.4 20 Day

LSTM Model achieved an MAE of 0.27 ms, well within the ESA threshold of 1.9 ms, indicating effective predictive performance over this extended period.

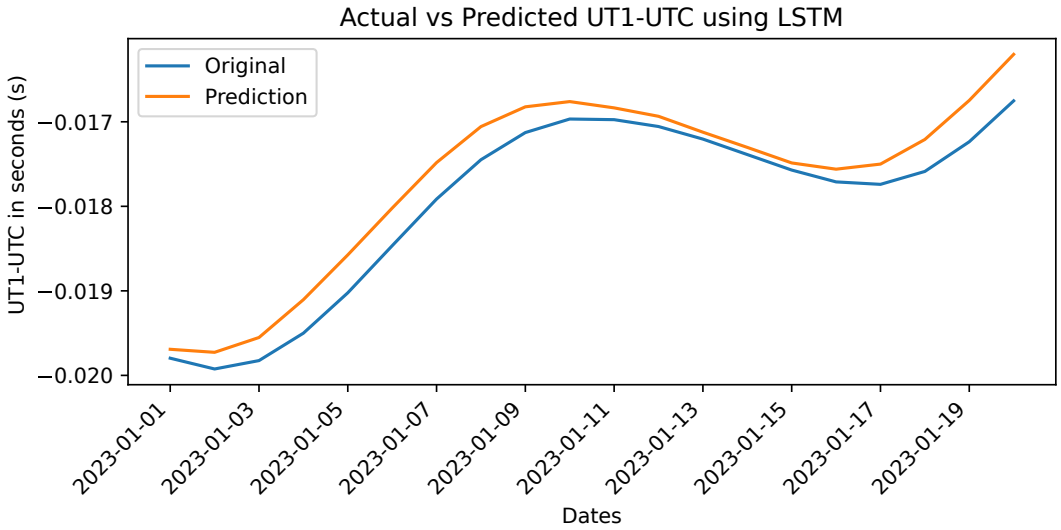


Figure 6.40: 20 day prediction using LSTM

6.4.5 40 Day

LSTM Model recorded an MAE of 0.31 ms, which is below the ESA benchmark of 2.9 ms, suggesting good model performance over a 40-day timeframe.

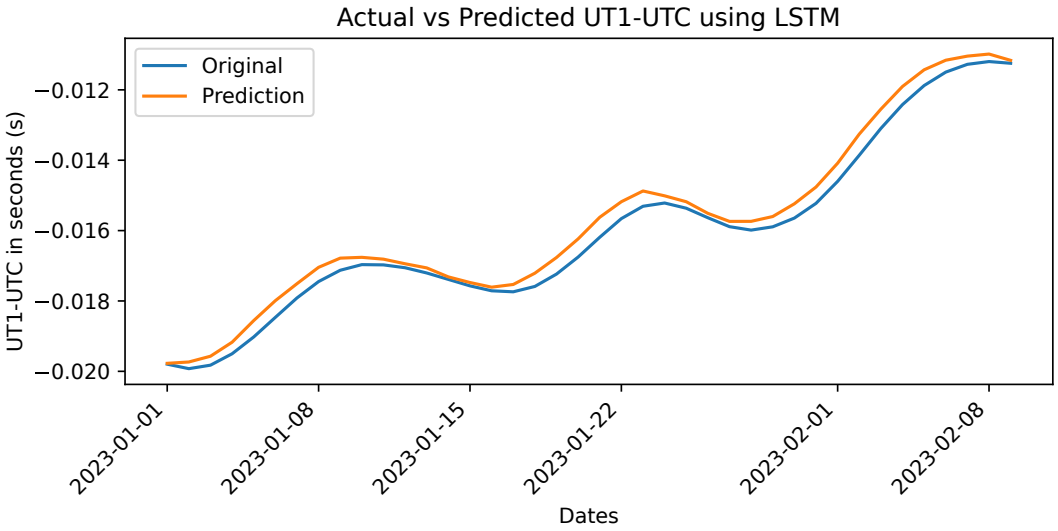


Figure 6.41: 5 day prediction using LSTM

6.4.6 90 Day

LSTM Model recorded an MAE of 0.24 ms, well below the ESA benchmark of 8.6 ms, indicating a high level of accuracy for long-term predictions.

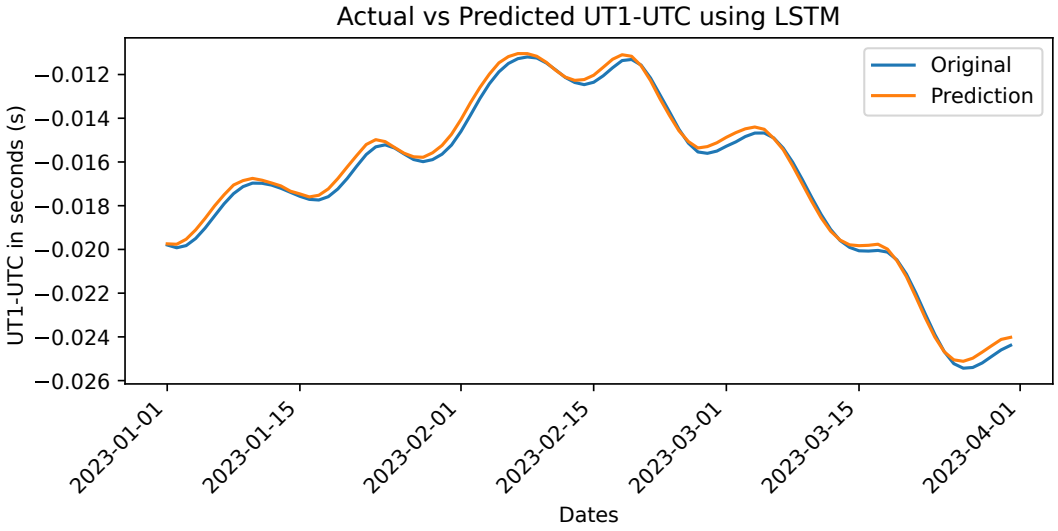


Figure 6.42: 90 day prediction using LSTM

6.5 CONT14

For this analysis, the LSTM model is used to predict the x_p and y_p for the CONT14 dataset, covering a period of two weeks in May 2014. The Mean Absolute Error (MAE) is used as the primary metric for evaluating the model's predictive accuracy, with the following results:

- x_p **LSTM Model**: Achieved an MAE of 0.84 mas, indicating a high level of accuracy in the predictions.

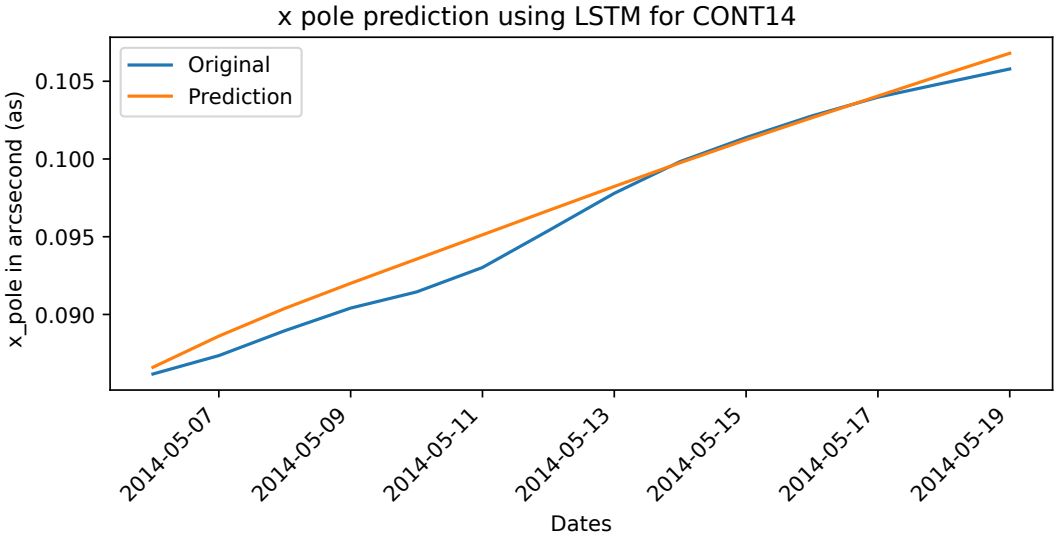


Figure 6.43: CONT14 Prediction using LSTM for x_p

y_p **LSTM Model**: Recorded an MAE of 0.77 mas, demonstrating similarly precise forecasting capabilities.

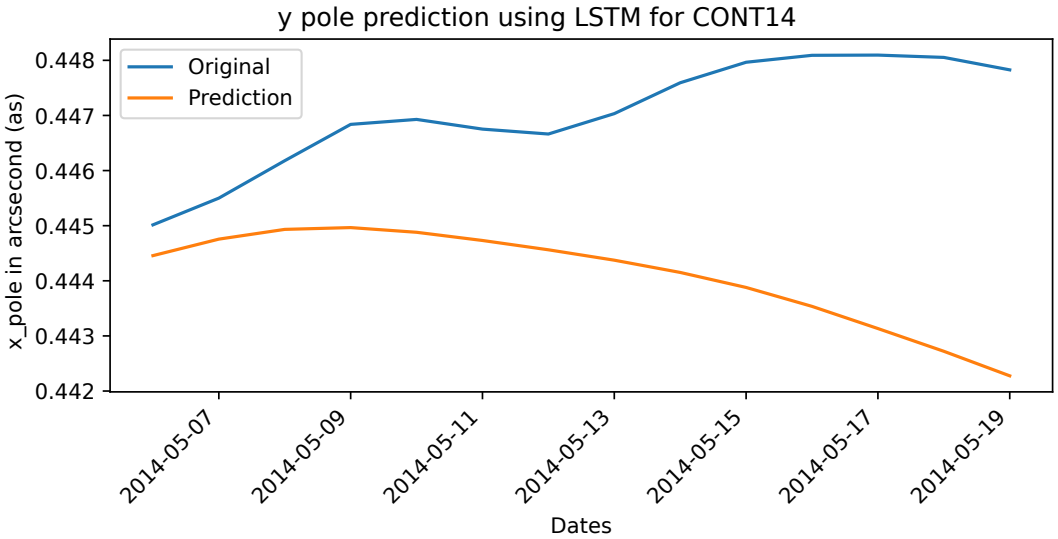


Figure 6.44: CONT14 Prediction using LSTM for y_p

UT1-UTC LSTM Model: Recorded an MAE of 0.048 ms, which is highly accurate for the requirements given by ESA.

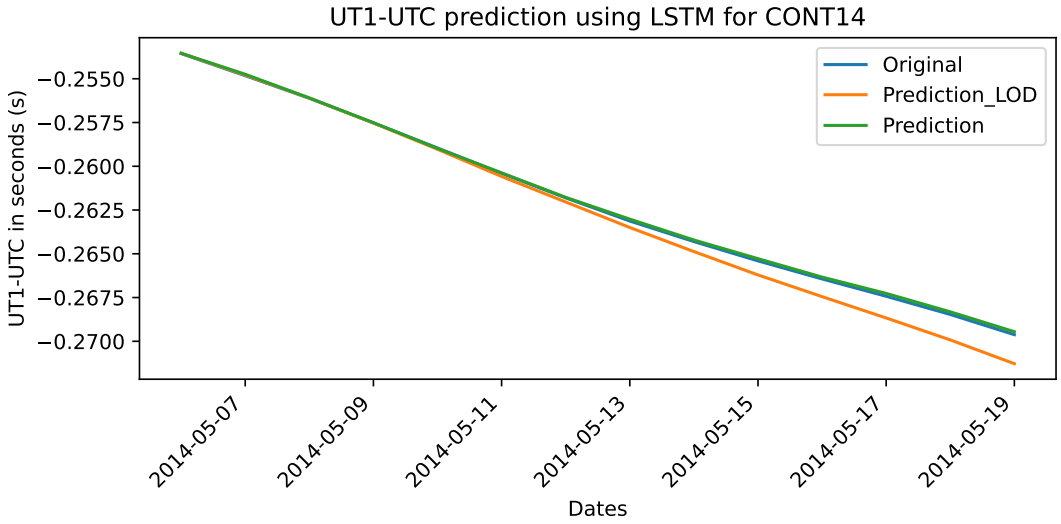


Figure 6.46: CONT14 Prediction using LSTM and LOD for UT1-UTC

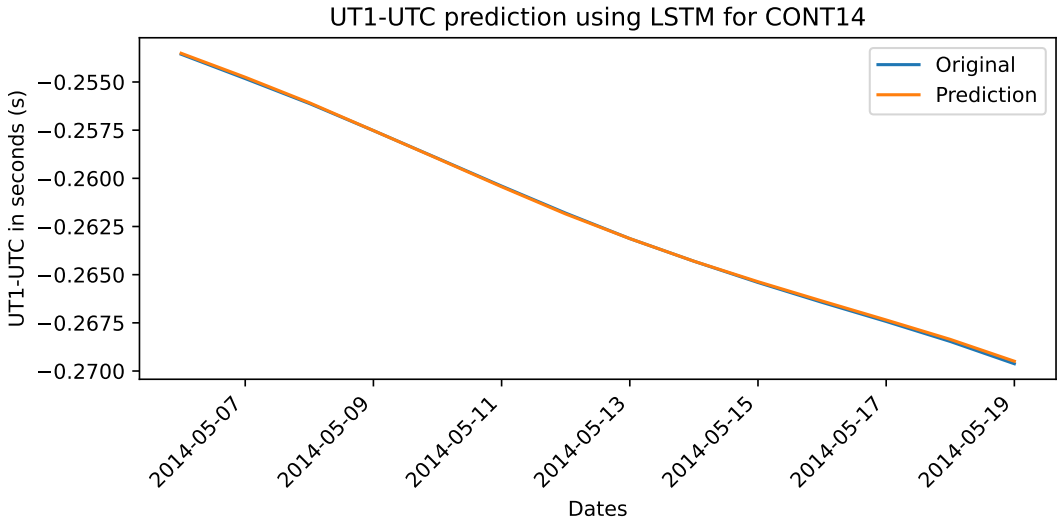


Figure 6.45: CONT14 Prediction using LSTM for UT1-UTC

These results were then compared against the required MAE benchmarks established by ESA for 10-day and 20-day prediction intervals and comparatively the model accuracy is well defined.

UT1-UTC Prediction Using LOD Prediction

The direct LOD prediction approach for the CONT14 campaign represents a significant advancement in the precision of UT1-UTC predictions. By directly forecasting LOD changes and applying a dynamic time correction, this method also demonstrates the use of LOD prediction and correlation between the UT1-UTC values and LOD.

This new approach has achieved an MAE of 0.54 ms which is highly accurate for this prediction horizon.

6.5.1 Comparison with ESA Benchmarks

When evaluating the accuracy of machine learning models for predicting Earth Orientation Parameters (EOPs), an important measure for determining success is to compare the model's performance with the Mean Absolute Error (MAE) benchmarks established by the ESA. This section shows a thorough evaluation of the performance of different models, namely Autoregressive (AR), SARIMA, LSTM, and LSTM with Singular Spectrum Analysis (LSTM+SSA), in predicting three essential Earth Orientation Parameters (EOPs): x_p, y_p, and UT1-UTC.

x_p Predictions

The AR model for x_p predictions had a successful outcome, with a Mean Absolute Error (MAE) of 0.13 milliarcseconds (mas) on the 1-day forecast. This result is significantly above the ESA benchmark of 0.3 mas. The SARIMA model significantly enhanced the accuracy of the 1-day forecast, obtaining an excellent Mean Absolute Error (MAE) of 0.07 mas. However, the LSTM model exhibited constraints in its ability to accurately predict short-term outcomes, as demonstrated by a mean absolute error (MAE) of 4.44 mas, surpassing the minimum requirement set by the ESA.

As the time frame of the forecast increases, the LSTM model demonstrates a significant enhancement in its performance. In the 5-day forecast, the LSTM model had a mean absolute error (MAE) of 2.16 milliarcseconds (mas), which exceeded the standard set by the ESA of 1.8 mas. On the other hand, the SARIMA model performed better with an MAE of 0.7 mas. After 10 days, the LSTM model performed very well, achieving an MAE of 2.02 mas, which closely matched the ESA benchmark. However, when the LSTM model was combined with SSA, the performance significantly deviated, resulting in an MAE of 17.70 mas. This indicates that integrating SSA for this prediction interval has significant issues. When predicting for extended periods (20, 40, and 90 days), the LSTM model continuously fulfilled the standards set by the ESA. The Mean Absolute Error (MAE) increased as the prediction interval increased but it remained within acceptable boundaries. Significantly, the LSTM model obtained a Mean Absolute Error (MAE) of 6.6 milliarcseconds (mas) for the 90-day forecast, surpassing the minimum MAE threshold of 23.8 mas. This highlights the LSTM's exceptional ability to predict long-term EOP accurately.

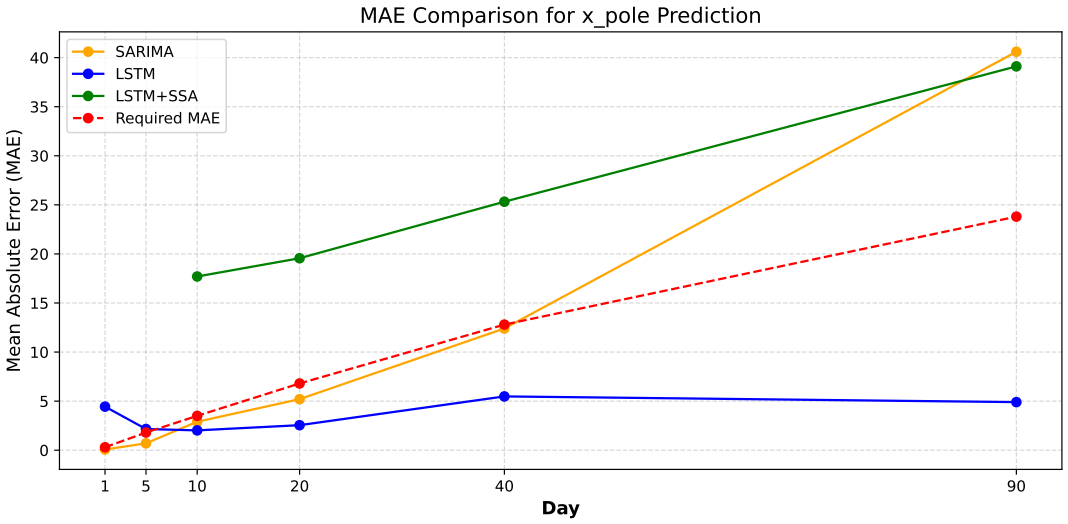


Figure 6.47: Model performance vs ESA requirements for x_p

y_p Predictions

The autoregressive (AR) model successfully predicted the value of y_p with a mean absolute error (MAE) of 0.182 milliarcseconds (mas) for the 1-day forecast. This error is lower than the criterion set by the ESA, which is 0.23 mas. The SARIMA model worked very well, with a Mean Absolute Error (MAE) of 0.06 milliarcseconds (mas). The LSTM model did not meet the required performance requirements for the 1-day forecast, as it achieved a mean absolute error (MAE) of 1.26 mas.

In the 5-day forecast, the performance of LSTM increased significantly with a mean absolute error (MAE) of 0.38 milliarcseconds (mas), while SARIMA surpassed all models with an MAE of 0.078 mas, both meeting the ESA benchmark of 1.3 mas. The 10-day forecast showed that the LSTM model had a Mean Absolute Error (MAE) of 1.30 milliarcseconds (mas), which was a little higher than the standards set by the ESA. On the other hand, the LSTM+SSA model had an MAE of 1.57 mas. SARIMA consistently achieved a high level of performance, as evidenced by its Mean Absolute Error (MAE) of 0.49 mas.

For the 20 and 40-day projections, both the LSTM and LSTM+SSA models exhibited a rise in Mean Absolute Errors (MAEs), while the LSTM model remained within the acceptable range. The SARIMA model's performance in predicting for 40 days above the minimum acceptable MAE suggests a decrease in accuracy as the forecast horizon increases.

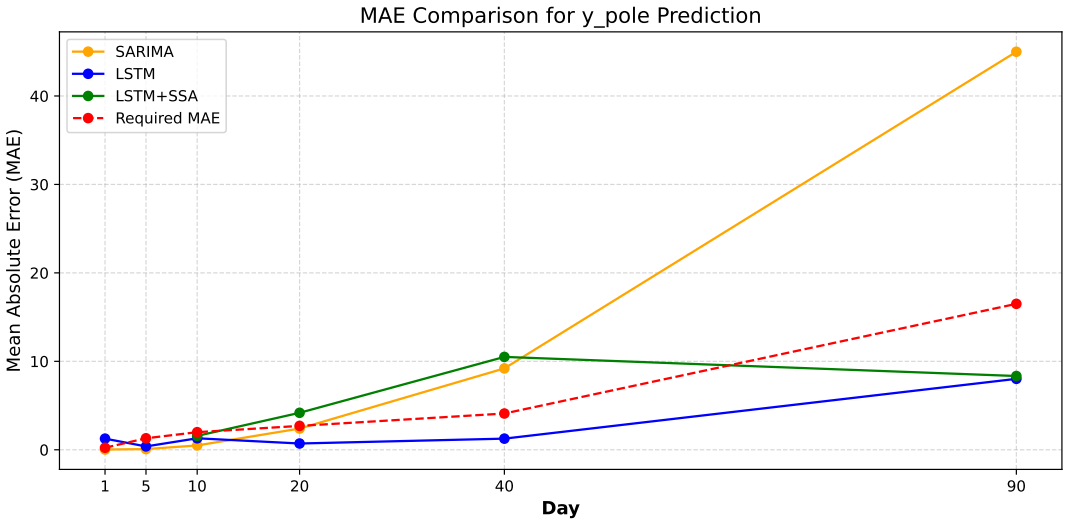


Figure 6.48: Model performance vs ESA requirements for y_p

UT1-UTC Predictions The primary focus was on using the LSTM model for predicting $UT1 - UTC$. The 1-day forecast achieved a MAE of 0.11 milliseconds (ms), somewhat higher than the ESA benchmark of 0.06 ms. The LSTM model regularly achieved remarkable performance for all other prediction periods (5, 10, 20, 40, and 90 days), with MAE consistently falling below the specified standards.

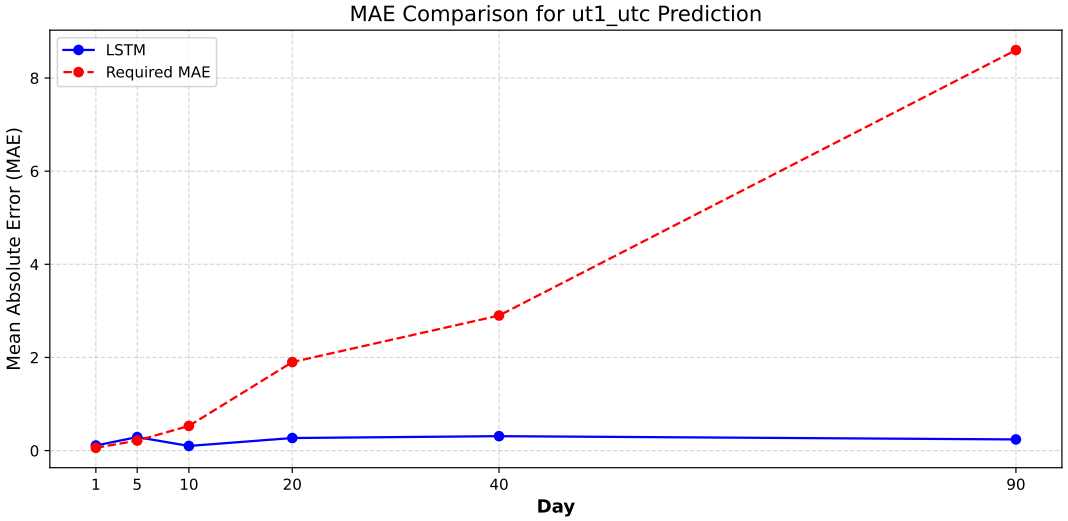


Figure 6.49: Model performance vs ESA requirements for UT1-UTC

CONT14 Predictions To analyze the CONT14 campaign, we tailored our methodology using the IERS EOP 14 C04 (IAU2000A) timeseries, due to the lack of specific ESA standards for 14-day predictions. We employed interpolation between the existing 10-day and 20-day prediction requirements to estimate the 14-day standards, resulting in approximately 4.82 milliarcseconds (mas) for the x_p, 2.27 mas for the y_p, and 1.078 milliseconds (ms) for UT1-UTC. These interpolated values provided a refined basis for our analysis and a closer approximation of the expected results for the 14-day

period. The LSTM model's efficiency was demonstrated by its performance on the CONT14 campaign dataset, achieving MAE of 0.84 milliarcseconds (mas) for the x_p and 0.77 mas for the y_p. The UT1-UTC prediction exhibited a precision of a MAE of 0.048 milliseconds (ms), demonstrating the model's high level of accuracy. The new approach of using directly the LOD predictions to calculate the time corrections to predict the UT1-UTC also showed a reliable accuracy of MAE as 0.54 ms.

6.6 Summary of results

x_p Predictions

SARIMA Model excelled in Short-Term 1-day predictions with a MAE of 0.07 mas, indicating high precision in capturing short-term variability in the x_p. The model's effectiveness is attributed to its ability to integrate seasonal patterns and autocorrelation in the data. SARIMA's configuration, involving parameters like order and seasonal order, was crucial in achieving this level of accuracy. LSTM Model showed superior performance in Long-Term intervals from 10 to 90 days, with MAEs ranging from 2.02 mas to 4.90 mas, all below the ESA benchmarks. LSTM's recurrent neural network structure, capable of learning long-term dependencies, was key to its success in these intervals. The model's robustness in handling sequential data and its adaptability to different time horizons contributed to its effectiveness.

y_p Predictions

SARIMA Model demonstrated high accuracy for Short-Term 1-day predictions with a MAE of 0.03, significantly better than the ESA requirement. Its efficiency can be attributed to the model's capability in modeling complex seasonal and non-seasonal patterns in the y_p data.

LSTM Model exhibited consistent performance in Medium to Long-Term, 20 to 90-day forecasts, with MAE values such as 0.71 mas for 20 days and 8.02 mas for 90 days, meeting or surpassing the benchmarks. LSTM's architecture, including features like dropout layers and look-back periods, was instrumental in achieving these results.

UT1-UTC Predictions

LSTM Model is used exclusively for UT1-UTC predictions, indicating an increasing trend in accuracy from short-term to long-term intervals. Achieved MAEs like 0.11 ms for 1 day, improving to 0.24 ms for 90 days, illustrating LSTM's proficiency in adapting to different prediction lengths. The model's configuration, particularly in terms of its neural network layers and hyperparameters, played a significant role in optimizing predictions for varying intervals.

CONT14 Campaign Analysis

LSTM Model for 14-Day Interval achieved a MAE of 0.84 mas for x_p and 0.77 mas for y_p over a 14-day interval, far exceeding the ESA benchmarks. Both the methods for predicting UT1-UTC also showed a highly accurate predictions and achieved 0.048 ms for using existing LOD and 0.54 ms for predicting LOD and using the predicted time corrections. This success highlights LSTM's ability to efficiently process and

forecast based on high-precision VLBI data from the CONT14 campaign. The result underscores the model's capability in capturing complex geophysical phenomena over a fortnightly period, a challenging task in Earth sciences.

7. Conclusion

This thesis focused on the utilization of machine learning methodologies, specifically Long Short-Term Memory (LSTM) networks and Seasonal Autoregressive Integrated Moving Average (SARIMA) models, for predicting Earth Orientation Parameters (EOPs). The results illustrate the efficiency of these models in predicting parameters such as x_{pole}, y_{pole}, and $UT1 - UTC$ across different time periods.

For the short-term (1-day) analysis, SARIMA models demonstrated remarkable efficiency, resulting in low Mean Absolute Error (MAE) values and proving their ability to accurately capture the short-term seasonality fluctuations in EOP data.

For medium to long-term forecasting, the LSTM model demonstrated high accuracy in predicting EOPs over periods spanning from 10 to 90 days. This underlines its ability to effectively handle sequential data and the time-related patterns that are inherent in EOPs. For short-term forecasts, the SARIMA models accurately predicted the values for the next day based on the x_{pole} projections. These models effectively dealt with the complexities of both seasonal and non-seasonal trends in the EOP data.

For projections ranging from 20 to 90 days, LSTM models consistently demonstrated reliable performance, frequently surpassing the accuracy standards set by ESA. The utilization of the LSTM model just for UT1-UTC predictions demonstrated improved precision throughout a range of time intervals, encompassing both short-term and long-term forecasts. This emphasizes the model's flexibility and capacity to adjust to various forecasting timeframes.

During the two-week CONT14 campaign, the LSTM model demonstrated exceptional precision by obtaining Mean Absolute Errors (MAEs) that were much lower than the benchmarks set by the European Space Agency (ESA) for both the x_{pole} and y_{pole} variables as well as for the UT1-UTC data.

Finally, this research makes a significant contribution to the field of geodesy and Earth sciences. The effective utilization of LSTM and SARIMA models presents new opportunities for accurate Earth Orientation Parameter (EOP) prediction, which

is crucial for diverse geospatial and astronomical applications. The LSTM model's ability to handle both short-term and long-term projections showcases its flexibility, making it a viable tool for various forecasting scenarios.

7.1 Future work

The results and methods presented in this thesis provide the foundation for future research efforts focused on improving the forecasting of Earth Orientation Parameters (EOPs). Possible topics for more investigation include Model Optimization, Investigating Alternative Models, and finding extended applications. The LSTM and SARIMA models can be enhanced to increase their accuracy, especially in situations where the forecast performance is not optimal. Subsequent studies could explore alternative machine learning methodologies or create hybrid models that may provide superior precision or computational efficiency in predicting EOP including refining the SSA models for better application. An evaluation of these models in real-time prediction scenarios would be highly beneficial to determine their applicability and dependability in operational contexts. The effectiveness of these models could be evaluated in other fields of Earth sciences, thereby providing wider significance for monitoring the environment, analyzing climate change, and planning space missions.

A. Appendix

A.1 Python Code

A.1.1 File Download Code

This subsection provides the Python code used for downloading files from a specified URL. The function `download_file` takes a URL and a folder path as arguments, sends a request to the specified URL, and if successful, saves the file to the given folder path.

A.1.2 Data Preparation Code

This subsection provides the Python code used for preparing the data for analysis. The code includes steps such as reading the data file, dropping unnecessary columns, and exporting processed data.

A.1.3 LSTM Model for Time Series Forecasting

This subsection outlines the Python code for building and training an LSTM (Long Short-Term Memory) model for time series forecasting. The code encompasses data normalization, dataset preparation for the LSTM, model construction, training with early stopping and learning rate scheduling, and finally, making forecasts.

A.1.4 Prepare Data for UT1-UTC Prediction and Use of Leap Seconds

The preparation of data for predicting UT1-UTC values involves several critical steps, including the handling of leap seconds, which are crucial for maintaining accurate timekeeping in astronomical and geodesy applications. The process is outlined below:

1. **Data Loading:** The UT1-UTC data, along with other relevant measurements, are loaded from a CSV file containing historical records. This dataset includes measurements from January 1, 1972, to November 6, 2023.

2. **Leap Second Adjustment:** Leap seconds are applied to the UT1-UTC data to account for irregularities in Earth's rotation. A separate DataFrame is created to manage these adjustments, with leap seconds being forward-filled to cover all dates within the range.
3. **Data Merging:** The UT1-UTC data and leap second adjustments are merged based on their dates to calculate the corrected UT1-TAI values, which are essential for the prediction model.
4. **Incorporation of Length of Day (LOD):** The LOD measurements are integrated into the dataset, providing a cumulative time correction. This correction is then combined with the UT1-TAI values to obtain the final UT1 values used for training the predictive model.

This preparation process ensures that the data used for training the predictive model is accurate and reflects the adjustments necessary for precise timekeeping, including the application of leap seconds and the accumulation of LOD corrections.

A.1.5 Converting Predicted UT1 to UT1-UTC and Model Evaluation

After obtaining the predicted UT1 values, the subsequent step involves adjusting these predictions to derive the UT1-UTC values. This adjustment necessitates accounting for time corrections and leap seconds. Subsequently, the model's accuracy is assessed by comparing the adjusted predictions against the actual UT1-UTC values, utilizing the Mean Absolute Error (MAE) as the metric.

Adjusting Predicted UT1 Values

The predicted UT1 values are modified to reflect the necessary time corrections and the current leap seconds to facilitate their conversion back to UT1-UTC format. The adjustment process entails:

- Subtracting the cumulative time correction from the predicted UT1 values to obtain the UT1-TAI values.
- Adding the current leap second value to the UT1-TAI values to convert them into UT1-UTC values.

Model Evaluation

The model's predictive performance is evaluated by comparing the adjusted predicted UT1-UTC values with the original UT1-UTC values, using the Mean Absolute Error (MAE) as the evaluation metric. This comparison offers a quantitative assessment of the model's accuracy.

The MAE metric provides insight into the average magnitude of the errors in the predictions, where lower values indicate a higher accuracy of the model. Evaluating the model using this metric is essential for understanding its effectiveness and pinpointing potential areas for enhancement.

A.2 Prediction of dX and dY

A.2.1 Overview

This section demonstrates the prediction of Earth Orientation Parameters (EOPs) with a particular emphasis on the fluctuations in dX and dY—essential components for comprehending the Earth's rotation and its deviations from the standard IERS reference frame. This section presents the study of dX and dY predictions, using a methodology similar to that applied for x_{pole}, y_{pole}, predictions.

A.2.2 Forecasting Timeframes

This section shows the predictive capabilities of the LSTM model for two different forecasting timeframes:

Short-Term Prediction (10 Days)

The 10-day forecast horizon demonstrates the LSTM model's ability to accurately capture the short-term dynamic patterns of dX and dY. The current time period is crucial for practical applications that require precise accuracy in the near future, such as satellite navigation and geodetic measurements.

- **10 Day dX:**

  The LSTM model has performed well in the prediciton of dX with a MAE of 0.04 mas

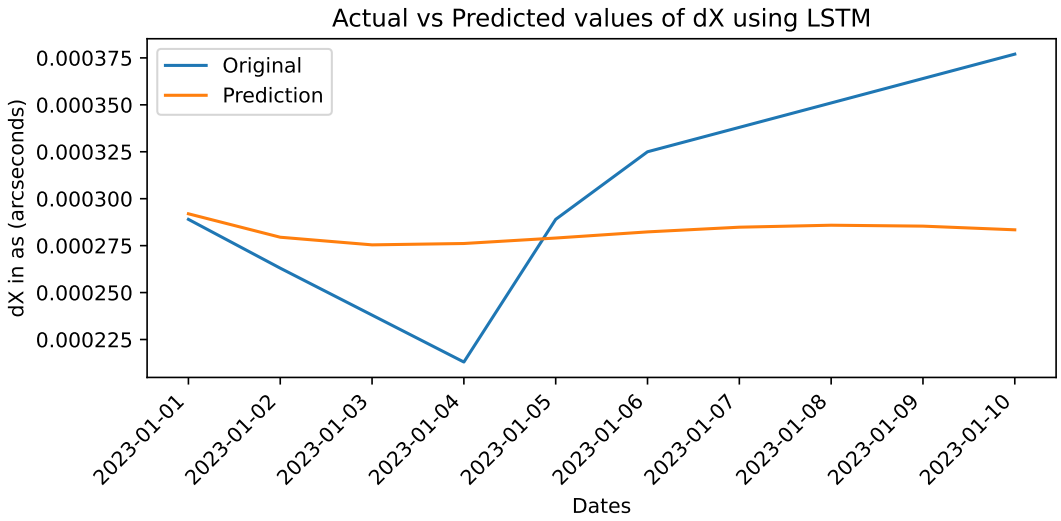


Figure A.1: 10 Day Prediction of dX using LSTM

- **10 Day dY:**

  Similarly, LSTM performance was observed for a 10 day prediction horizon for dY values with an MAE of 0.06 mas

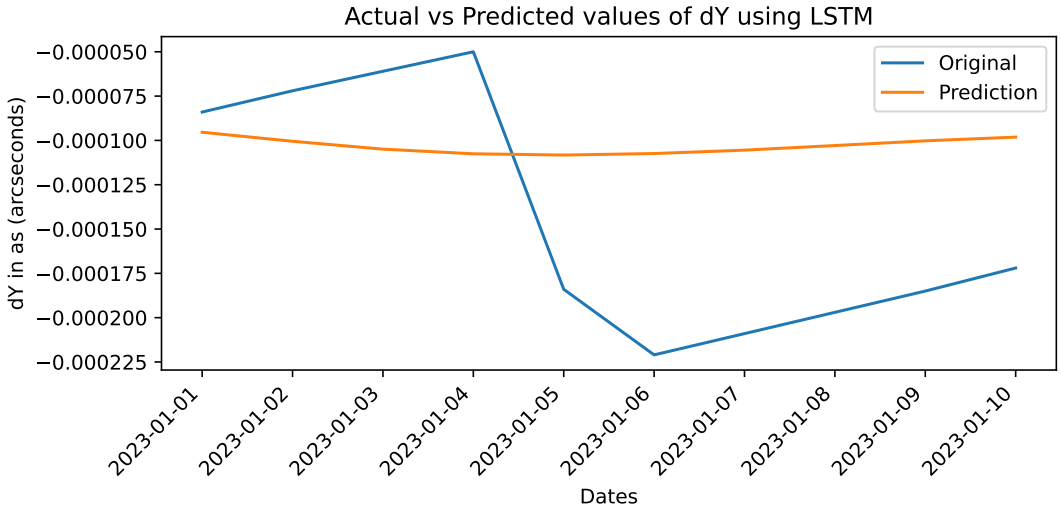


Figure A.2: 10 Day Prediction of dY using LSTM

Long-Term Prediction (90 Days)

By increasing the prediction horizon to 90 days, we are able to demonstrate the LSTM model's adaptability and its capability of maintaining accurate predictions over long timeframes.

Both prediction horizons are chosen to offer a comprehensive perspective on the LSTM model's adaptability and efficiency in forecasting EOP, covering both immediate and long-term projections.

- **90 Day dX:**

  The LSTM model was used to predict the dX values for the 90 day prediction horizon and the observed MAE was 0.06 mas.

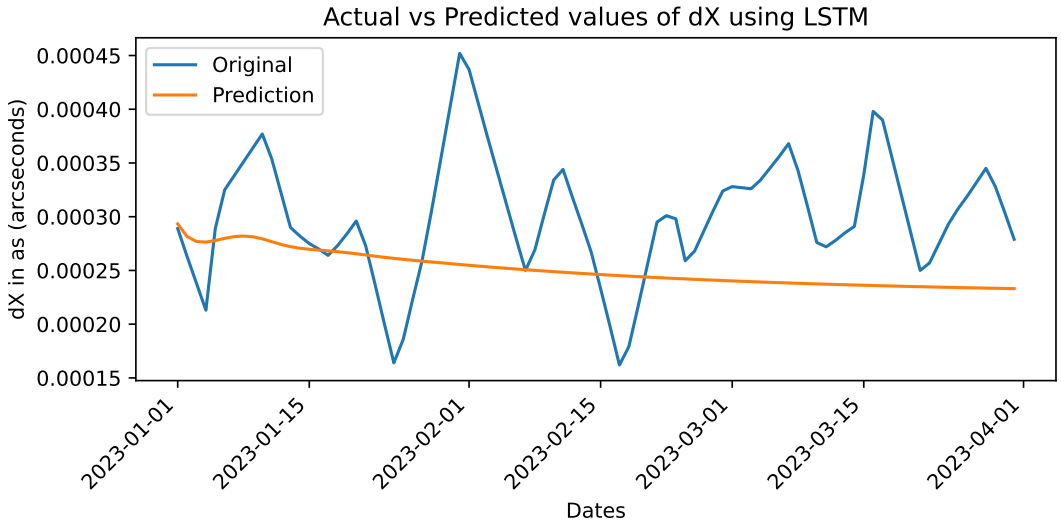


Figure A.3: 90 Day Prediction of dX using LSTM

- **90 Day dY:**

  The prediction of 90 days for the dY values using LSTM showed a MAE of 0.05 mas.

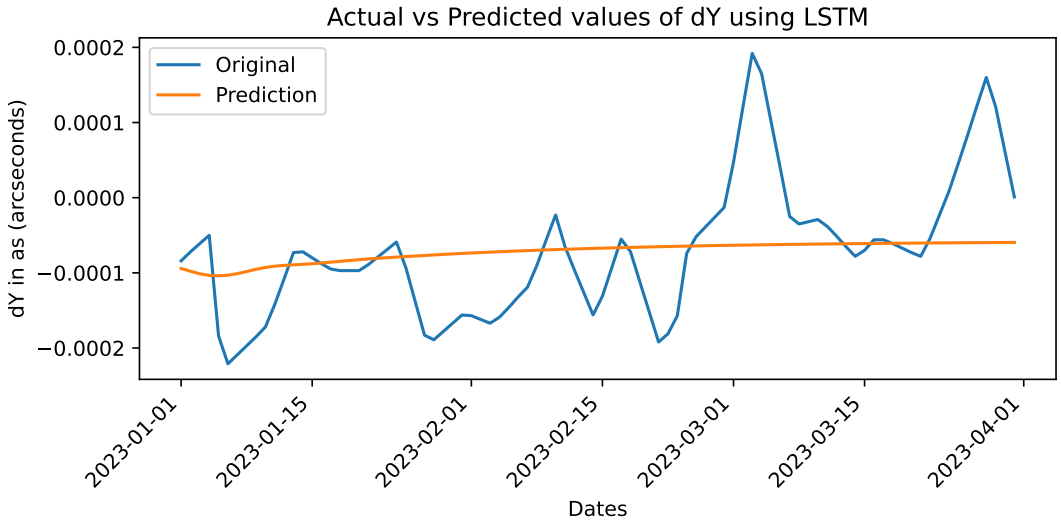


Figure A.4: 90 Day Prediction of dY using LSTM

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